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2018-19-20

Tax Withholding and Estimated

Capital Gains and Losses

Contents

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Exemptions
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MODULE 1: INTERCORPORATE INVESTMENTS

LEARNING OUTCOMES

- 1.a.** Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS) for 1) investments in financial assets, 2) investments in associates, 3) joint ventures, 4) business combinations, and 5) special purpose and variable interest entities
- 1.b.** Compare and contrast IFRS and US GAAP in their classification, measurement, and disclosure of investments in financial assets, investments in associates, joint ventures, business combinations, and special purpose and variable interest entities
- 1.c.** Analyze how different methods used to account for intercorporate investments affect financial statements and ratios

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

1.

Introduction to intercorporate investments

Intercorporate investments can have a significant impact on an investing company's financial performance and position.

Why invest in other companies?

1

Diversify
their asset
base

2

Enter new
markets

3

Obtain
competitive
advantages

4

Deploy
excess cash

5

Achieve
additional
profitability

What instrument can investors invest in?

Debt securities

Commercial paper

Corporate/government bonds & notes

Redeemable preferred stock

Asset-backed securities (ABS)

Equity securities

Common stock

Non-redeemable preferred stock

What does the percentage of equity ownership depend on?

Resources available

Ability to acquire the
shares

Desired level of
influence or control

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

2.

Overview of basic corporate investment categories

Categories of investments in marketable debt and equity securities

Investments in
financial assets
(3.)

Investments
in associates
(4.)

Joint ventures
(4.)

Business
combinations
(5.)

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

2.

Overview of basic corporate investment categories

Summary of Accounting Treatments for Investments

Features	Financial assets	Associate	Joint Venture	Business combination
Ownership	Usually <20%	Usually 20% - 50%	Shared control	Usually > 50%
Degree of Influence(*)	<i>No significant influence</i>	<i>Significant influence</i>	<i>Joint control</i>	<i>Control</i>
Accounting treatment	Amortized cost or fair value	Equity method	Equity method (US GAAP & IFRS) Proportionate consolidation (rare)	Acquisition method

(*) *The distinction between investments in financial assets, investments in associates, and business combinations is **based on the degree of influence or control** rather than purely on the ownership percentage held.*

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

2.

Overview of basic corporate investment categories



Example

Fill in the blank.

For financial reporting purposes, these companies' ownership interest are most likely considered as:

Features	Company A	Company B	Company C
Ownership	15%	49%	51%
Degree of Influence	Significant influence	Control	Significant influence
Classification	...	...	...
Accounting treatment	...	...	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

2.

Overview of basic corporate investment categories



Example

Fill in the blank.

For financial reporting purposes, these companies' ownership interest are most likely considered as:

Features	Company A	Company B	Company C
Ownership	15%	49%	51%
Degree of Influence	<i>Significant influence</i>	<i>Control</i>	<i>Significant influence</i>
Classification	Investments in associates	Business combinations	Investments in associates
Accounting treatment	Equity method	Acquisition method	Equity method

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

Both IASB and FASB developed revised standards for financial investments.

Key changes in IFRS 9

Classification terminology

Previous standard

Held-to-maturity

Held for trading

Available for sale

IFRS 9

Amortized cost

Fair value through P&L

Fair value through OCI

Approach to loan impairment

Companies are required to migrate from an incurred loss model to an **expected credit loss model**.

→ Earlier recognition of impairment (12-month expected losses for performing loans and lifetime expected losses for nonperforming loans).

→ Companies evaluating not only historical and current information about loan performance, but also forward-looking information.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

All financial assets are measured at fair value when initially acquired.

In subsequent periods, financial assets are measured at either fair value or amortized cost.

3.1. Amortized Cost (*for Debt Securities Only*)

Criteria for amortized cost accounting

Business model test

Debt securities are being held to collect contractual cash flows.

Cash flow characteristic test

The contractual cash flows are solely payments of principal and interest on principal.

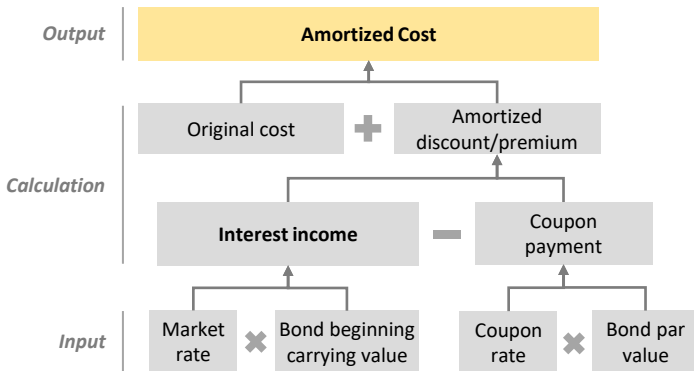
Debt securities are reported on the balance sheet at **amortized cost**

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

3.1. Amortized Cost (for Debt Securities Only)



Balance sheet

Amortized Cost

Income Statement

Dividends, interest income received

Realized gain/loss

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

3.2. Fair Value Through Profit or Loss (for Debt and Equity Securities)

Debt Securities

Debt Securities held for trading *may be* classified as FVPL

If accounting for debt securities at amortized cost results in an accounting mismatch^(*)
→ classified as FVPL

Equity Securities

Equity securities held for trading *must be* classified as FVPL

Other equity securities *may be* classified as either FVPL or FVOCI

Once classified, the choice is **irrevocable**

(*) Accounting mismatch: *Inconsistency resulting from different measurement bases for assets and liabilities*

Derivatives that are not used for hedging are always carried at FVPL (except for hedging instruments).

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.
Investments in financial assets

3.2. Fair Value Through Profit or Loss (for Debt and Equity Securities)

How financial assets in FVPL are reported on balance sheet and income statement?

Balance sheet

Fair value of the financial assets

Income Statement

Dividends, interest income received

Realized gain/loss

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.
Investments in financial assets

3.3. Fair Value Through OCI (for Debt and Equity Securities)

What securities can be classified as FVOCI?

Can be **debt or equity** securities designated at FVOCI.

How financial assets in FVOCI are reported?

Balance sheet

Fair value of the financial assets

Unrealized gain/loss (reported in OCI)

Income Statement

Dividends, interest income received

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

Summary of Classifications of Financial Assets

	Amortized cost	FVPL	FVOCI
Balance Sheet	Amortized cost	Fair value	Fair value, with G/L recognized in equity (OCI)
Income Statement	Interest (including amortization) Realized G/L	Interest Dividend G/L	Interest Dividend

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

Classification and measurement



Illustration

At the beginning of the year, Midland Corporation purchased a 9% bond with a face value of \$100,000 for \$96,209 to yield 10%. The coupon payments are made annually at year-end. Suppose that the fair value of the bond at the end of the year is \$98,500.

Determine the impact on Midland's balance sheet and income statement if the bond investment is classified as

- 1) *amortized cost*
- 2) *fair value through profit or loss, and*
- 3) *fair value through OCI*

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

Classification and measurement



Illustration

	Amortized cost	Fair value	
		FVPL	FVOCI
Balance sheet (opening)	Original cost	Fair value	Fair value
	=	=	=
Income statement	Interest income	Interest income	Interest income
	=	=	=
Balance sheet (ending)	Amortized cost	Fair value	Fair value
	=	=	=
Realized/unrealized gain			
	=	=	=

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

Classification and measurement



Illustration

	Amortized cost	Fair value	
		FVPL	FVOCI
Balance sheet (opening)	Original cost $= FV_0 = 96,209$	Fair value $= FV_0 = 96,209$	Fair value $= FV_0 = 96,209$
Income statement	Interest income $= 96,209 \times 10\%$ $= 9,621$	Interest income $= 96,209 \times 10\%$ $= 9,621$	Interest income $= 96,209 \times 10\%$ $= 9,621$
Balance sheet (ending)	Amortized cost = $96,209 + 9,621 -$ $100,000 \times 9\% =$ $96,830$	Fair value $= 98,500$	Fair value $= 98,500$
Realized/ unrealized gain	N/A	Unrealized gain = $98,500 - 96,209 -$ $621 = 1,670$ in P&L	Unrealized gain = $98,500 - 96,209 -$ $621 = 1,670$ in OCI

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.
Investments in financial assets

Classification and measurement



Illustration

Now let's imagine that the bonds are called on the first day of the next year for \$101,000.

Calculate the gain or loss recognition for each classification.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.
Investments in financial assets

Classification and measurement



Illustration

Amortized cost			Fair value		
			FVPL	FVOCI	
Balance sheet value	Amortized cost		Fair value	Fair value	
	=		=	=	
Call price	=		=	=	
Realized gain	=		=	=	
Unrealized gain	=		=	=	

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.

Investments in financial assets

Classification and measurement



Illustration

	Amortized cost	Fair value	
		FVPL	FVOCI
Balance sheet value	Amortized cost = 96,830	Fair value = 98,500	Fair value = 98,500
Call price	101,000	101,000	101,000
Realized gain	101,000 – 96,830 = 4,170 in P&L	101,000 – 98,500 = 2,500 in P&L	101,000 – 98,500 = 2,500 in P&L
Unrealized gain	N/A	N/A	Unrealized gain 1,670 is removed from equity

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

3.
Investments in financial assets

Reclassification of Investments



Reclassification of **equity instruments is not permitted**



Reclassification of **debt instruments** is only permitted if the business model for the financial assets has changed in a way that significantly affects operations

No restatement of prior periods at the reclassification date

How should analyst evaluate?

- Analysts typically evaluate performance separately for operating and investing activities. Analysis of operations should exclude:
 - Interest income
 - Dividend income
 - Gain/loss on investments
- Analysts should exclude non-operating assets in the determination of return on net operating assets.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

**A
company
(investor)**

Holds 20 to 50% of the voting rights of an associate (investee)

Has (or can exercise) **significant influence**, but not control

***Associate or
joint
ventures***

**Significant
influence may
be evidenced
by**



Representation on the board of directors



Participation in the *policy-making* process



Material transactions between the investor and the investee



Interchange of managerial personnel



Technological dependency

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

Joint ventures

- Ventures undertaken and controlled by two or more parties
- Can be a convenient way to enter foreign markets, conduct specialized activities, and engage in risky projects.



Common characteristics

A contractual arrangement exists between two or more venturers.

The contractual arrangement establishes joint control.

Accounting methods for joint ventures

Both IFRS and US GAAP9 require the **equity method**.

Allowed to use **proportionate consolidation** under rare circumstances.

Shown on a **single line item** with similar items under its sole control

Shown on a **line-by-line basis** in income statement and balance sheet

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

**Carrying amount
of equity
investment**

=

Initial
investment
(cost to acquire)

+

% of
investee's
profit

—

Dividends or
other
distributions



Balance sheet

Assets

Current assets

↓ Cash

↑ Cash from dividends

Non-current assets

↑ Equity investment
(less dividend)

Liabilities

↑ **Equity**



Income statement

Earning

+ Proportionate share of
the investee's **profit**

With investee's loss:

Carrying amount = Initial investment — % of investee's loss — Dividend or other distributions

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

**Carrying amount
of equity
investment**

=

Initial
investment
(cost to acquire)

+

% of
investee's
profit

-

Dividends or
other
distributions

Example:

\$208,000

=

\$200,000

+

\$10,000

-

\$2,000



Balance sheet

Assets

Current assets

↓ Cash: \$200,000

↑ Cash: \$2,000

Non-current assets

↑ Equity
investment:
\$208,000

Liabilities

↑ **Equity**
\$10,000



Income statement

Earning

+ Proportionate share of
the investee's **profit**
\$10,000



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

Example: Equity Method: Balance in Investment Account

Branch (a fictitious company) purchases a 20% interest in Williams (a fictitious company) for €200,000 on 1 January 2016. Williams reports income and dividends as follows:

	Income	Dividend
2016	€200,000	€50,000
2017	300,000	100,000
2018	400,000	200,000
	€900,000	€350,000

Calculate the investment in Williams that appears on Branch's balance sheet as of the end of 2018.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

 **Example: Equity Method: Balance in Investment Account**

Answer: Investment in Williams at 31 December 2018:

Calculation				
Initial cost				€200,000
+ Equity income 2016	=	...	=	...
− Dividends received 2016	=	...	=	...
+ Equity income 2017	=	...	=	...
− Dividends received 2017	=	...	=	...
+ Equity income 2018	=	...	=	...
− Dividends received 2018	=	...	=	...
Balance-Equity Investment			=	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

Example: Equity Method: Balance in Investment Account

Answer: Investment in Williams at 31 December 2018:

Calculation

Initial cost		€200,000
+ Equity income 2016	= $20\% \times €200,000$	= €40,000
– Dividends received 2016	= $20\% \times €50,000$	= (€10,000)
+ Equity income 2017	= $20\% \times €300,000$	= €60,000
– Dividends received 2017	= $20\% \times €100,000$	= (€20,000)
+ Equity income 2018	= $20\% \times €400,000$	= €80,000
– Dividends received 2018	= $20\% \times €200,000$	= (€40,000)
Balance-Equity Investment		= €310,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

 Example: Equity Method: Balance in Investment Account

Answer:

We can also calculate the investment in Williams at 31 December 2018 as:

= Initial investment + % acquired x (Total income over the 3 years – Total dividend over the 3 years)

= [€200,000 + 20% x (€900,000 – €350,000)] = €310,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

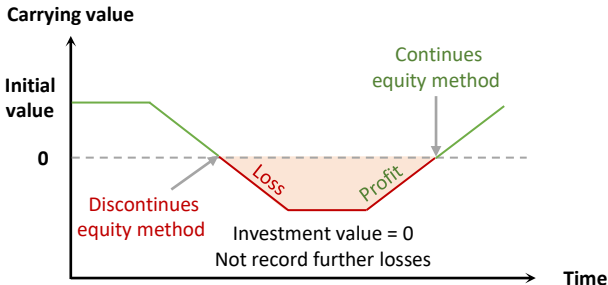
Investments in associates and joint ventures

4.1. Equity Method of Accounting: Basic Principles

The recorded investment value can decline as a result of:

- Investee losses
- A permanent decline in the investee's market value

What does the investor company do when there is a decline in investment value?



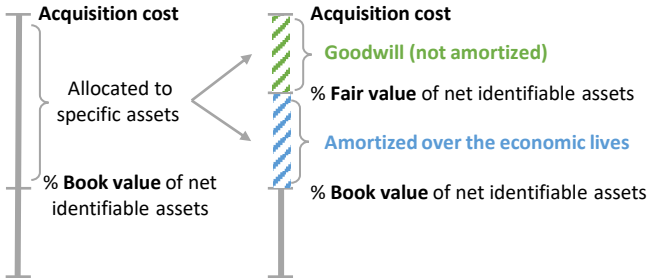
[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

- The cost (purchase price) to acquire shares of an investee is often greater than the book value of those shares.
- IFRS and US GAAP both treat the difference between the cost of the acquisition and investor's share of the fair value of the net identifiable assets as **goodwill**.



Over time, as the differences are amortized, the balance in the investment account will ***come closer to representing the ownership percentage*** of the book value of the net assets of the associate.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Illustration: Equity Method Investment in Excess of Book Value

Blake Co. and Brown Co. are two hypothetical companies. Assume that Blake Co. acquires 30% of the outstanding shares of Brown Co. At the acquisition date, book values and fair values of Brown's recorded assets and liabilities are as follows:

	Book value	Fair value
Current assets	€10,000	€10,000
Plant and equipment	190,000	220,000
Land	120,000	140,000
	€320,000	€370,000
Liabilities	100,000	100,000
Net assets	€220,000	€270,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Illustration: Equity Method Investment in Excess of Book Value

Blake Co. believes the value of Brown Co. is higher than the book value of its identifiable net assets.

They offer €100,000 for a 30% interest in Brown, which represents a €34,000 excess purchase price.

Calculate goodwill.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Illustration: Equity Method Investment in Excess of Book Value

Answer:

Step 1: Calculate excess purchase price.

Excess purchase price = Purchase price – Proportionate share of the investee's book value

Purchase price	...
----------------	-----

– 30% of book value of Brown ($30\% \times \text{€}220,000$)	...
--	-----

Excess purchase price	...
------------------------------	-----

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Illustration: Equity Method Investment in Excess of Book Value

Answer:

Step 1: Calculate excess purchase price.

Excess purchase price = Purchase price – Proportionate share of the investee's book value

Purchase price	€100,000
– 30% of book value of Brown ($30\% \times €220,000$)	66,000
Excess purchase price	€34,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Illustration: Equity Method Investment in Excess of Book Value

Answer:

Step 2: Calculate excess purchase price attributable to net assets

Excess purchase price attributable to net assets = Fair value of net assets –
Book value of net assets

Excess purchase price = ...



Goodwill = ...

Attributable to net assets = ...

Comes from
differences in

Plant and equipment

= ...

Land = ...

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

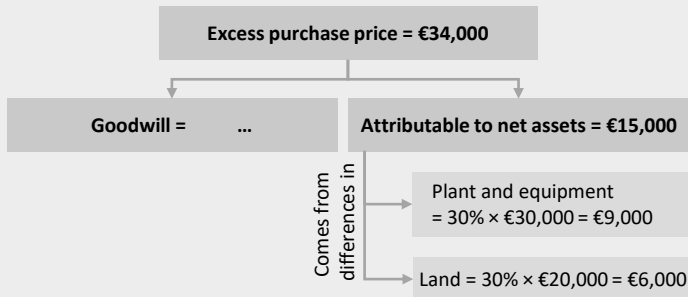


Illustration: Equity Method Investment in Excess of Book Value

Answer:

Step 2: Calculate excess purchase price attributable to net assets

Excess purchase price attributable to net assets = % Fair value of net assets
– % Book value of net assets



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

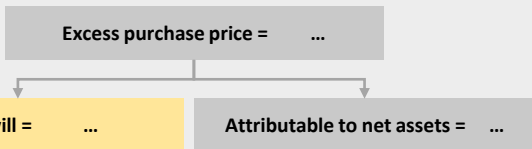


Illustration: Equity Method Investment in Excess of Book Value

Answer:

Step 3: Calculate goodwill.

Goodwill = Excess purchase price – Total attributable to net assets



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Illustration: Equity Method Investment in Excess of Book Value

Answer:

Step 3: Calculate goodwill.

Goodwill = Excess purchase price – Total attributable to net assets

Excess purchase price = €34,000

Goodwill = €19,000

Attributable to net assets = €15,000

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

Property, plant, and equipment measurement

IFRS

Historical cost

Fair value

(less accumulated depreciation)

US GAAP

Historical cost

(less accumulated depreciation)

Are amortized

Effect on

Balance sheet

Income statement

Assets

↓ Equity investment

Liabilities

Equity

Earning

↓ Proportionate share of the investee's profit

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

Amounts allocated to	Treatment
Inventory	Expensed
Depreciable or amortizable assets	Capitalized and subsequently <i>depreciated or amortized</i> over an appropriate period of time.
Land and other assets or liabilities that are <i>not amortized</i>	Reported at fair value

Carrying amount of equity investment

- = Initial investment $\pm$ % investee's profit or loss
- Dividend or other distribution – Annual amortization of excess purchase price attributable to net assets

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

Using the **illustration on slide 29** and assuming a 10-year useful life for plant, property, and equipment, using straight-line depreciation, the annual amortization is as follows:

Account	Excess Price (€)	Useful Life	Amortization /Year (€)
Plant and equipment	9,000	10 years	900
Land	6,000	Indefinite	0
Goodwill	19,000	Indefinite	0

Annual amortization would reduce the investor's share of the investee's reported income (equity income) and the balance in the investment account by €900 for each year over the 10-year period.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

On 1 January 2018, Parker Company acquired 30% of Prince Inc. common shares for the cash price of €500,000 (both companies are fictitious). It is determined that Parker has the ability to exert significant influence on Prince's financial and operating decisions. The following information concerning Prince's assets and liabilities on 1 January 2018 is provided:

Prince, Inc.

	Book Value	Fair Value	Difference
Current assets	€100,000	€100,000	€0
Plant and equipment	1,900,000	2,200,000	300,000
	€2,000,000	€2,300,000	€300,000
Liabilities	800,000	800,000	0
Net assets	€1,200,000	€1,500,000	€300,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

The plant and equipment are depreciated on a straight-line basis and have 10 years of remaining life. Prince reports net income for 2018 of €100,000 and pays dividends of €50,000.

Calculate the following:

1. *Goodwill included in the purchase price.*
2. *Investment in associate (Prince) at the end of 2018.*

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

Example: Equity Method Investments with Goodwill

Answer to 1: Goodwill included in the purchase price.

Step 1: Calculate excess purchase price.

Excess purchase price = Purchase price – Proportionate share of the investee's book value

Purchase price	...
– 30% of book value of Prince's net assets	...
Excess purchase price	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

Answer to 1: Goodwill included in the purchase price.

Step 1: Calculate excess purchase price.

Purchase price	€500,000
– 30% of book value of Prince's net assets (30% × €1,200,000)	360,000
Excess purchase price	€140,000

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

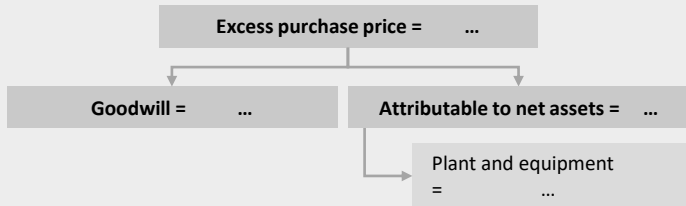
Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

 Example: Equity Method Investments with Goodwill

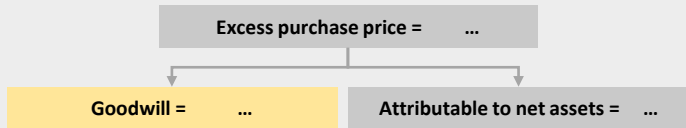
Answer to 1: Goodwill included in the purchase price.

Step 2 : Calculate excess purchase price attributable to net assets.



Step 3: Calculate goodwill.

Goodwill = Excess purchase price – Total attributable to net assets



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

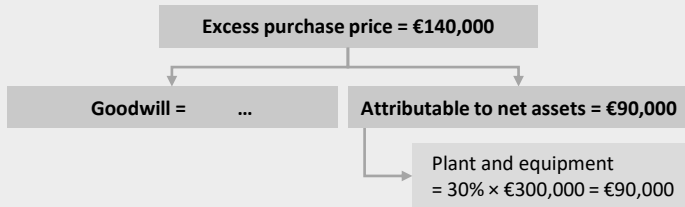
Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price

 Example: Equity Method Investments with Goodwill

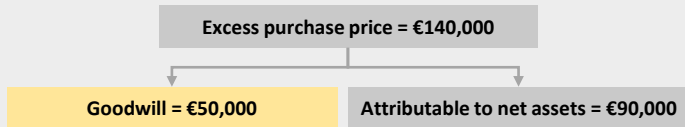
Answer to 1: Goodwill included in the purchase price.

Step 2 : Calculate excess purchase price attributable to net assets.



Step 3: Calculate goodwill.

Goodwill = Excess purchase price – Total attributable to net assets



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

Answer to 2: Investment in associate (Prince) at the end of 2018.

Carrying amount of equity investment

= Initial investment $\pm$ % of investee's profit or loss – Dividend or other distribution – Annual amortization of excess purchase price attributable to net assets

Step 1: Calculate % of investee's profit or loss

Parker's share of Prince's net income = ...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

Answer to 2: Investment in associate (Prince) at the end of 2018.

Carrying amount of equity investment

= Initial investment $\pm$ % investee's profit or loss – Dividend or other distribution – Annual amortization of excess purchase price attributable to net assets

Step 1: Calculate % of investee's profit or loss

Parker's share of Prince's net income = $(30\% \times \text{€}100,000) = 30,000$

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

Answer to 2: Investment in associate (Prince) at the end of 2018.

Step 2: Calculate dividend or other distribution received

Dividends received = ...

Step 3: Calculate annual amortization of excess purchase price attributable to net assets

Amortization of excess purchase price attributable to plant and equipment:

= ...

Step 4: Calculate ending carrying amount of equity investment:

= Initial investment $\pm$ % investee's profit or loss – Dividend – Annual amortization of excess purchase price attributable to net assets

= ...

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.2. Amortization of Excess Purchase Price



Example: Equity Method Investments with Goodwill

Answer to 2: Investment in associate (Prince) at the end of 2018.

Step 2: Calculate dividend or other distribution received

$$\text{Dividends received} = (30\% \times €50,000) = 15,000$$

Step 3: Calculate annual amortization of excess purchase price attributable to net assets

Amortization of excess purchase price attributable to plant and equipment:

$$= €90,000 / 10 \text{ years} = 9,000$$

Step 4: Calculate ending carrying amount of equity investment:

= Initial investment $\pm$ % investee's profit or loss – Dividend – Annual amortization of excess purchase price attributable to net assets

$$= €500,000 + €30,000 - €15,000 - €9,000 = €506,000$$

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.3. Fair Value Option

Definition

Option to account for their equity method investment at fair value.

Standards

US GAAP

Available to all entities

IFRS

Restricted to venture capital,
mutual funds, unit trusts,
insurance funds

Both standards require that the election to use the fair value option occur at the time of ***initial recognition*** and is ***irrevocable***

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.3. Fair Value Option

Accounting treatment for fair value option

- Investment account on investor's balance sheet is reported at **fair value** (not reflect the investor's proportionate share in the investee's earnings, dividends, or other distributions).
- Unrealized G/L arising from changes in fair value → **Income statement**
- Interest and dividends received → **Income statement**
- Excess of cost over the fair value of the investee's identifiable net assets → **not amortized**
- Goodwill is **not created**

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.4. Impairment

Both IFRS and US GAAP require **periodic reviews** of equity method investments for impairment.

IFRS

When can the impairment be recorded?

- There is objective evidence of impairment as a result of **one or more (loss) events** occurred after the initial recognition of the investment.
- That loss event has an impact on the investment's future cash flows.

How is the equity investment tested for impairment?

By comparing:

1. Value in use and net selling price
→ Recoverable amount = Higher value
2. Carrying amount and recoverable amount
→ Use the lower value

Value in use

vs.

Net selling price
= Fair value – cost to sell

Carrying amount

vs.

Recoverable amount

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.4. Impairment

IFRS

Comparison #1

Value in use

<

Net selling price
= Fair value –
cost to sell

Comparison #2

Carrying amount

>

Recoverable amount

Impaired

Carrying amount

<

Recoverable amount

Not impaired

After comparison 2, only the smaller value is reported on the balance sheet.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.4. Impairment

IFRS

Goodwill is included in the carrying amount of the investment and not separately tested for impairment.

How is the investment impairment recorded?

- Carrying amount is written down to **recoverable amount** directly or through the use of an allowance account.
- **Impairment loss** is recognized on the *Income statement*.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.4. Impairment

US GAAP

When can the impairment be recorded?

- Fair value of the investment declines below its carrying value.
- The decline is **determined to be permanent**.

How is the equity investment tested for impairment?

By comparing: Fair value of the investment and its carrying value.

Carrying value

>

Fair value



Impaired

Carrying value

<

Fair value



Not impaired

After the comparison, only the smaller value is reported on the balance sheet.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.4. Impairment

US GAAP

How is the impairment recorded?

- Carrying amount is written down to **fair value**.
- **Impairment loss** is recognized on the *Income statement*.

Reversal of impairment

US GAAP

Prohibit reversal of impairment losses

IFRS

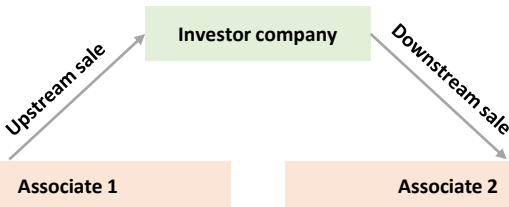
Permit reversal of impairment losses

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Profit from upstream/downstream transactions can only be realized until confirmed through:

- 1. Use**
- 2. Sale to third party**

The remaining is called Unconfirmed profit.

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Associate 1

Associate's IS

+ Profit from upstream transaction

Investor company's IS

+ (Profit from upstream transaction – Unconfirmed profit from upstream transaction) × % ownership

Associate 2

Associate's IS

No record

Investor company's IS

+ Profit from downstream transaction
– (Unconfirmed profit from downstream transaction × % ownership)

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Upstream Sale

On 1 January 2018, Wicker Company acquired a 25% interest in Foxworth Company (both companies are fictitious) for €1,000,000 and used the equity method to account for its investment.

The book value of Foxworth's net assets on that date was €3,800,000. An analysis of fair values revealed that all fair values of assets and liabilities were equal to book values except for a building. The building was undervalued by €40,000 and has a 20-year remaining life. The company used straight-line depreciation for the building.

Foxworth paid €3,200 in dividends in 2018. During 2018, Foxworth reported net income of €20,000.

During the year, Foxworth sold inventory to Wicker. At the end of the year, there was €8,000 profit from the upstream sale in Foxworth's net income. The inventory sold to Wicker by Foxworth had not been sold to an outside party.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Upstream Sale

1. *Calculate the equity income to be reported as a line item on Wicker's 2018 income statement.*
2. *Calculate the balance in the investment in Foxworth to be reported on the 31 December 2018 balance sheet.*

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates


Illustration: Equity Method with Sale of Inventory: Upstream Sale

Answer to 1: Equity income reported on IS

Equity income of investee company on investor's IS

= Wicker's share of Foxworth's reported income – Amortization attributable to net assets – Unconfirmed profit from upstream transaction

Wicker's share of Foxworth's reported income	...
--	-----

Amortization of excess purchase price attributable to building	...
--	-----

Unconfirm profit	...
------------------	-----

Equity income 2018	...
---------------------------	------------

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates


Illustration: Equity Method with Sale of Inventory: Upstream Sale

Answer to 1: Equity income reported on IS

Equity income of investee company on investor's IS

= Wicker's share of Foxworth's reported income – Amortization attributable to net assets – Unconfirmed profit from upstream transaction

Wicker's share of Foxworth's reported income (25% × €20,000)	€5,000
---	--------

Amortization of excess purchase price attributable to building (25% × €40,000 / 20)	(500)
---	-------

Unconfirm profit (25% × €8,000)	(2,000)
---------------------------------	---------

Equity income 2018	€2,500
---------------------------	---------------

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates


Illustration: Equity Method with Sale of Inventory: Upstream Sale

Answer to 2: Investment in Foxworth reported on BS

Investment in Foxworth reported on BS

= Initial investment + Equity income – Dividends received

Purchase price	...
Equity income 2018	...
Dividends received (25% × €3,200)	...
Investment in Foxworth, 31 Dec 2018	...

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Upstream Sale

Answer to 2: Investment in Foxworth reported on BS

Investment in Foxworth reported on BS

= Initial investment + Equity income – Dividends received

Purchase price	€1,000,000
Equity income 2018	2,500
Dividends received (25% × €3,200)	(800)
Investment in Foxworth, 31 Dec 2018	€1,001,700

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Downstream Sale

Jones Company owns 25% of Jason Company (both fictitious companies) and appropriately applies the equity method of accounting.

Amortization of excess purchase price, related to undervalued assets at the time of the investment, is €8,000 per year.

During 2017 Jones sold €96,000 of inventory to Jason for €160,000. Jason resold €120,000 of this inventory during 2017. The remainder was sold in 2018.

Jason reports income from its operations of €800,000 in 2017 and €820,000 in 2018.

1. *Calculate the equity income to be reported as a line item on Jones's 2017 income statement.*
2. *Calculate the equity income to be reported as a line item on Jones's 2018 income statement.*

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 1: Equity income reported on IS in 2017 (remove unconfirmed profit from downstream transaction)

Equity income reported on Jone's IS in 2017

= Jones's share of Jason's reported income – Amortization attributable to net assets – Unconfirmed profit from downstream transaction

We first need to calculate the Unconfirmed profit from downstream transaction.

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 1: Equity income reported on IS in 2017 (remove unconfirmed profit from downstream transaction)

Calculate unconfirmed profit from downstream transaction:

Output

Step 4: Jones's share of the unrealized profit

= ...

Calculation

% ownership

= ...



Step 3: Unrealized profit

= ...

Input

Step 1: Jones's profit on the sale to Jason

= ...



Step 2: % of goods is unsold

= ...

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

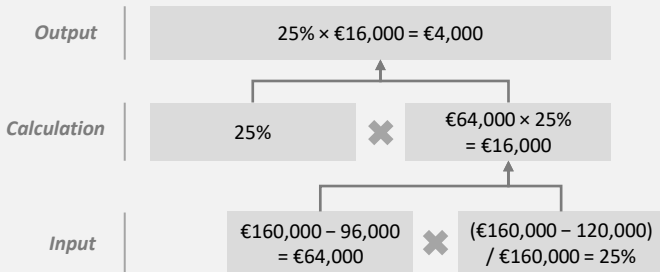
Investments in associates and joint ventures

4.5. Transaction with associates


Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 1: Equity income reported on IS in 2017 (remove unconfirmed profit from downstream transaction)

Calculate unconfirmed profit from downstream transaction:



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 1: Equity income reported on IS in 2017 (remove unconfirmed profit)

Equity income reported on Jone's IS in 2017

= Jones's share of Jason's reported income – Amortization attributable to net assets – Unconfirmed profit from downstream transaction

Jones's share of Jason's reported income	...
--	-----

Amortization of excess purchase price	...
---------------------------------------	-----

Unrealized profit	...
-------------------	-----

Equity income 2017	...
---------------------------	-----

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates


Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 1: Equity income reported on IS in 2017 (remove unconfirmed profit)

Equity income reported on Jone's IS in 2017

= Jones's share of Jason's reported income – Amortization attributable to net assets – Unconfirmed profit from downstream transaction

Jones's share of Jason's reported income ($25\% \times \text{€}800,000$)	€200,000
Amortization of excess purchase price	(8,000)
Unrealized profit ($25\% \times \text{€}16,000$)	(4,000)
Equity income 2017	€188,000

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 2: Equity income reported on IS in 2018 (add back realized profit)

Equity income reported on Jone's IS in 2018

= Jones's share of Jason's reported income – Amortization attributable to net assets + Realized profit

Jones's share of Jason's reported income

...

Amortization of excess purchase price

...

Realized profit

...

Equity income 2018

...

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Transaction with associates



Illustration: Equity Method with Sale of Inventory: Downstream Sale

Answer to 2: Equity income reported on IS in 2018 (add back realized profit)

Equity income reported on Jone's IS in 2018

= Jones's share of Jason's reported income – Amortization attributable to net assets + Realized profit

Jones's share of Jason's reported income ($25\% \times \text{€}820,000$)	€205,000
Amortization of excess purchase price	(8,000)
Realized profit ($25\% \times \text{€}16,000$)	4,000
Equity income 2018	€201,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Issues for Analysts

1 *Is the equity method appropriate?*

Investor company owns 19% of
investee company



Investor company has significant
influence on investee company



Investor company wants to avoid
reporting associate losses

Investor company may avoid using
equity method



Not appropriate

Investor company owns 25% of
investee company



Investor company does not have
significant influence on investee



Investor company wants to
capture associate income

Investor company may prefer
using equity method



Not appropriate

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.5. Issues for Analysts

2 *Issue in investment account on BS and investment income on IS*

The investment account is reported as **single line item** on investor BS

Only investment income of the associate is reported on IS

There can be significant assets and liabilities of the investee that are not reflected on the investor's balance sheet

Proportionate share is included in net income but no proportionate share of sales is reflected

Affect debt ratios

Net profit margin ratios could be overstated

3 *Quality of the equity method earnings*

Equity income earnings are **non-cash earnings**

→ Analysts should consider potential restrictions on dividend cash flows

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

An investor company acquires 40% of the stock of an investee company at its book value \$1,200 ($\$3,000$ book value of stockholders' equity $\times$ 40%) by issuing shares. Compare the equity and proportionate consolidation methods showed on balance sheet, income statement and ratios effects.

Continue next slide →

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

	Pre-acquisition balance sheets		Post-acquisition balance sheet of investor company	
	Investor company	Investee company	Equity method	Proportionate consolidation method
Current assets	5,000	1,000	...	...
Investment			...	...
Other assets	10,000	4,000	...	...
Total assets	15,000	5,000	...	...
Liabilities	5,000	2,000	...	...
Common stock	7,000	2,000	...	...
Retained earnings	3,000	1,000	...	...
Total equity and liabilities	15,000	5,000	...	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

	Pre-acquisition balance sheets		Post-acquisition balance sheet of investor company	
	Investor company	Investee company	Equity method	Proportionate consolidation method
Current assets	5,000	1,000	5,000	$5,000 + 1,000 \times 40\% = 5,400$
Investment			1,200	0
Other assets	10,000	4,000	10,000	$10,000 + 4,000 \times 40\% = 11,600$
Total assets	15,000	5,000	16,200	17,000
Liabilities	5,000	2,000	5,000	$5,000 + 2,000 \times 40\% = 5,800$
Common stock	7,000	2,000	$7,000 + 1,200 = 8,200$	$7,000 + 1,200 = 8,200$
Retained earnings	3,000	1,000	3,000	3,000
Total equity and liabilities	15,000	5,000	16,200	17,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

	Pre-acquisition income statements		Post-acquisition income statement of investor company	
	Investor company	Investee company	Equity method	Proportionate consolidation method
Sales	20,000	3,500	...	...
Expenses	(16,000)	(1,800)	...	...
Equity income			...	...
Profit	4,000	1,700	...	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

	Pre-acquisition income statements		Post-acquisition income statement of investor company	
	Investor company	Investee company	Equity method	Proportionate consolidation method
Sales	20,000	3,500	20,000	$20,000 + 3,500 \times 40\% = 21,400$
Expenses	(16,000)	(1,800)	(16,000)	$(16,000) + (1,800) \times 40\% = (16,720)$
Equity income			680	0
Profit	4,000	1,700	4,680	4,680

Net income will be the same under both methods.

- *Under equity method*, investor will report equity income = 40% of the net income of the investee company, sales and expenses will be unaffected.
- *Under proportionate consolidation method*, the investor's sales and expenses will be increased by 40% of the investee company sales and expenses, yielding the same net income.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.
Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

The effect on return on equity and its components is:

	Equity method (EM)	Proportionate consolidation method (PC)	Ratios effects (lower/higher/ equal)
Net profit margin	...	...	...
Total asset turnover	...	...	...
ROA	...	...	...
Leverage	...	...	...
Return on equity	...	...	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

The effect on return on equity and its components is:

	Equity method (EM)	Proportionate consolidation method (PC)	Ratios effects (lower/higher/ equal)
Net profit margin	$4,680/20,000 = 0.234$	$4,680/21,400 = 0.219$	EM higher
Total asset turnover	$20,000/16,200 = 1.235$	$21,400/17,000 = 1.259$	Indeterminate
ROA	$4,680/16,200 = 0.289$	$4,680/17,000 = 0.275$	EM higher
Leverage	$16,200/11,200 = 1.446$	$17,000/11,200 = 1.518$	PC higher
Return on equity	$4,680/11,200 = 0.418$	$4,680/11,200 = 0.418$	Same

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

4.

Investments in associates and joint ventures

4.6. Proportionate consolidation method



Illustration: Proportionate consolidation method

The effect on return on equity and its components is:

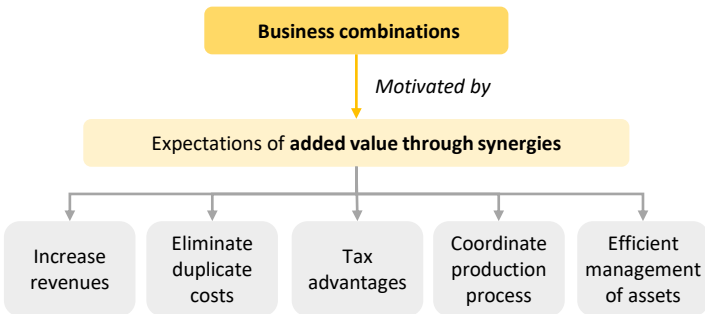
	Equity method (EM)	Proportionate consolidation method (PC)	EM ratios
Net profit margin	Same net income Fewer sales	Same net income	Higher net profit margin
ROA	Same net income Fewer assets	Same net income	Higher ROA
Return on equity	Same net income Same equity	Same net income Same equity	Same
Leverage	Same equity Fewer assets	Same equity	Lower leverage
Total assets turnover	Both the numerator and denominator change	Both the numerator and denominator change	Indeterminate

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

Why combine businesses?



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

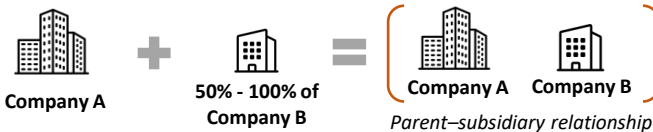
Business combinations

5.1. Types of Business Combinations

Merger



Acquisition



Consolidation



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

IFRS and US GAAP require the acquisition method of accounting for business combinations.

- Fair value (price paid) of the acquisition includes fair value of any contingent consideration.
- **Direct costs** of the business combination (professional and legal fees, valuation experts, and consultants) are ***expensed as incurred***.

Acquisition method solves **three major accounting issues** in business combinations

Recognition and measurement of the **assets and liabilities**

Initial recognition and subsequent accounting for **goodwill**

Recognition and measurement of **non-controlling interest**

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #1: Recognition and measurement of the assets and liabilities

1

Recognition and Measurement of Identifiable Assets and Liabilities

Both IFRS and US GAAP require

Measure the identifiable tangible and intangible assets and liabilities
at fair value

Recognize any assets and liabilities that acquiree had not previously recognized (such as ***identifiable intangible assets***: brand names, patents, technology)

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #1: Recognition and measurement of the assets and liabilities

2

Recognition and Measurement of Contingent Liabilities



Must recognize any contingent liability if

1. It is a present obligation that arises from past events
2. It can be measured reliably



Not recognize costs that the acquirer expects (but is not obliged) to incur

Recognizes these costs in future periods as they are incurred (e.g. restructuring costs)

Treatment of contingent liabilities

IFRS

Record if their fair values can be *reliably measured*

US GAAP

Record if they are *probable* and can be *reasonably estimated*

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #1: Recognition and measurement of the assets and liabilities

3

Recognition and Measurement of Indemnification Assets

Indemnification

Compensation for harm or loss



Indemnification assets arise when the seller (acquiree) indemnifies the buyer (acquirer) for the outcome of a contingency

Buyer recognizes an indemnification asset at the same time it recognizes the indemnified liability

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #1: Recognition and measurement of the assets and liabilities

3

Recognition and Measurement of Indemnification Assets



Illustration: Recognition and Measurement of Indemnification Assets

An acquiree is exposed to a contingent liability. Based on probabilistic estimation, the FV of the contingent liability is \$100,000. The former owners provide a contractual guarantee to indemnify the acquirer of the loss.

→ In the consolidated balance sheet, the acquirer recognized:

Assets

Indemnification asset + \$100,000

Liabilities

Contingent liabilities +\$100,000

Equity

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #1: Recognition and measurement of the assets and liabilities

4

Recognition and Measurement of Financial Assets and Liabilities

Financial assets and liabilities will be **reclassified** from acquiree to acquirer's business model based on

Contractual terms

Economic conditions

**Acquirer's operating
or accounting policies**

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5. Business combinations

5.2. Acquisition Method

Issue #2: Recognition and measurement of goodwill

IFRS allows 2 options for recognizing goodwill

Full goodwill

Full goodwill

= FV of the entity as a whole – FV
of all identifiable assets

US GAAP requires full goodwill

Partial goodwill

Partial goodwill

= FV of the acquisition – % acquire
× FV of all identifiable assets

Note:

- The goodwill option is on a transaction-by-transaction basis (IFRS).
- Goodwill is not amortized but is tested for impairment **at least annually** or if events indicate that goodwill might be impaired.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #2: Recognition and measurement of goodwill

Illustration: Recognition and Measurement of Goodwill

Acquirer contributes \$800,000 for an 80% interest in Acquiree. The identifiable net assets have a fair value of \$900,000. The fair value of the entire entity is determined to be \$1 million.

Calculate full goodwill and partial goodwill.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #2: Recognition and measurement of goodwill

Illustration: Recognition and Measurement of Goodwill

Answer:

Full goodwill

= FV of the entity as a whole
– FV of all identifiable assets

= \$1,000,000 - \$900,000
= **\$100,000**

**Used under both IFRS and US
GAAP**

Partial goodwill

= FV of the acquisition – % acquire
x FV of all identifiable assets

= \$800,000 – 80% x \$900,000
= **\$80,000**

Only used under IFRS

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Issue #2: Recognition and measurement of goodwill

*Occasionally, a company faces adverse circumstances such that its market value **drops below** the fair value of its net assets.*

Market value < FV of net assets

Acquisition price < FV of net assets

“bargain purchase” acquisition

FV of net assets – Acquisition price
 = **Gain on IS**

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.**Business combinations**

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Franklin Company, a hypothetical company, acquired 100% of the outstanding shares of Jefferson, Inc. (another fictitious company) by issuing 1,000,000 shares of its €1 par common stock (€15 market value). Immediately before the transaction, the two companies compiled the following information:

Continue next slide →

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

	Franklin Book Value (,000)	Jefferson Book Value (,000)	Jefferson Fair Value (,000)
Cash and receivables	10,000	300	300
Inventory	12,000	1,700	3,000
PP&E (net)	27,000	2,500	4,500
	€49,000	€4,500	€7,800
Current payables	8,000	600	600
Long-term debt	16,000	2,000	1,800
	24,000	2,600	2,400
Net assets	€25,000	€1,900	€5,400

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Shareholders' equity:	Franklin Book Value (,000)	Jefferson Book Value (,000)
Capital stock (€1 par)	€5,000	€400
Additional paid in capital	6,000	700
Retained earnings	€14,000	€800

Jefferson has no identifiable intangible assets.

Show the balances in the post-combination balance sheet using the acquisition method.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Answer:

Step 1: Calculate goodwill

Because Franklin acquired 100% of the outstanding shares of Jefferson, the proper goodwill recognition option is **full goodwill**.

Goodwill = Fair value of the stock issued – Fair value allocated to identifiable net assets

= ...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Answer:

Step 1: Calculate goodwill

Because Franklin acquired 100% of the outstanding shares of Jefferson, the proper goodwill recognition option is **full goodwill**.

Goodwill = Fair value of the stock issued – Fair value allocated to identifiable net assets

$$= €15,000,000 - €5,400,000 = €9,600,000$$

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Step 2: Calculate post-combination balance sheet *using FV of the acquiree*

	Franklin Book Value (,000)	Jefferson Fair Value (,000)	Franklin Consolidated (,000)
Cash and receivables	10,000	300	...
Inventory	12,000	3,000	...
PP&E (net)	27,000	4,500	...
Goodwill			...
Total assets	€49,000	€7,800	...
Current payables	8,000	600	...
Long-term debt	16,000	1,800	...
Total liabilities	€24,000	€2,400	...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Step 2: Calculate post-combination balance sheet using FV of the acquiree

	Franklin Book Value (,000)	Jefferson Fair Value (,000)	Franklin Consolidated (,000)
Cash and receivables	10,000	300	10,300
Inventory	12,000	3,000	15,000
PP&E (net)	27,000	4,500	31,500
Goodwill			9,600
Total assets	€49,000	€7,800	€66,400
Current payables	8,000	600	8,600
Long-term debt	16,000	1,800	17,800
Total liabilities	€24,000	€2,400	€26,400

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.
Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Step 2: Calculate post-combination balance sheet using FV of the acquiree

Equity	Calculation (,000)	Franklin Consolidated (,000)
Capital stock (€1 par)	...	...
Additional paid in capital	...	...
Retained earnings	...	...
Total stockholders' equity		...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Step 2: Calculate post-combination balance sheet using FV of the acquiree

Equity	Calculation (,000)	Franklin Consolidated (,000)
Capital stock (€1 par)	€1 x (5,000 + 1,000)	€6,000
Additional paid in capital	6,000 + €15 x 1,000 – 1,000	€20,000
Retained earnings	14,000	€14,000 ^(*)
Total stockholders' equity		€40,000

() At the acquisition date, only the acquirer's retained earnings are carried to the combined entity. Earnings of the target are included on the consolidated income statement and retained earnings only in post-acquisition periods.*

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

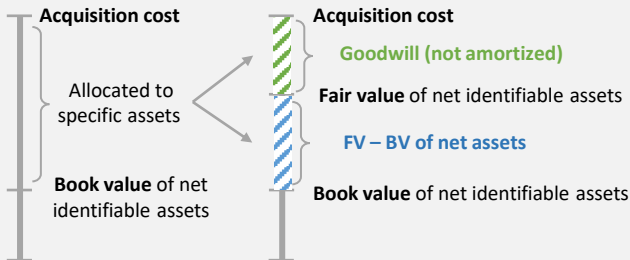
Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Additional question: How would the excess purchase price be allocated?

$$\begin{aligned} \text{Excess purchase price} &= \text{Purchase price} - \text{Book value of net assets} \\ &= \text{FV of Franklin's stocks issued to acquire} - \text{BV of Jefferson's net assets} \\ &= \dots \end{aligned}$$



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

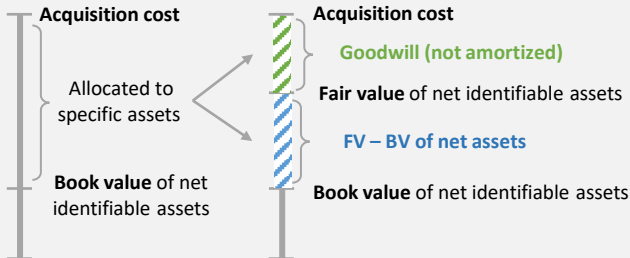
Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Additional question: How would the excess purchase price be allocated?

$$\begin{aligned} \text{Excess purchase price} &= \text{Purchase price} - \text{Book value of net assets} \\ &= \text{FV of Franklin's stocks issued to acquire} - \text{BV of Jefferson's net assets} \\ &= €15,000,000 - €1,900,000 = €13,100,000 \end{aligned}$$



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

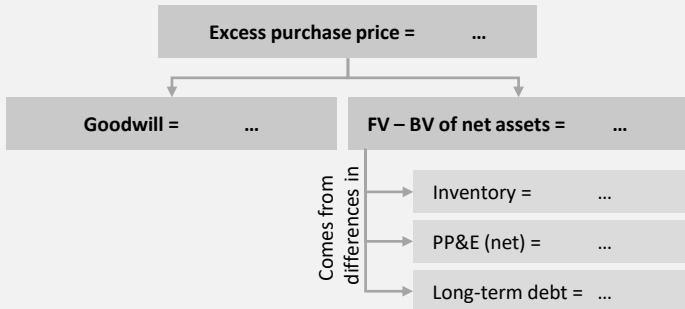
5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Additional question: How would the excess purchase price be allocated?



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business combinations

5.2. Acquisition Method

Impact of the Acquisition Method on Financial Statements, Post-Acquisition



Illustration: Acquisition Method Post-Combination Balance Sheet

Additional question: How would the excess purchase price be allocated?

Excess purchase price = €13,100,000

Goodwill = €9,600,000

FV – BV of net assets = €4,500,000

Comes from
differences in

Inventory = €1,300,000

PP&E (net) = €2,000,000

Long-term debt = €200,000

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

Acquisition



Company A



**50% - 100% of
Company B**



Company A



Company B

Parent–subsidiary relationship

Financial Statements Preparation

**Financial
Statements A
(The Parent)**



**Financial
Statements B
(The Subsidiary)**



**Consolidated
Financial Statements
(Made by the Parent)**

Both the parent and the subsidiary typically prepare their own financial records, but the parent also prepares consolidate financial statements.

**Items that need to
be combined:**

- Liabilities
- Assets
- Revenues
- Expenses

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

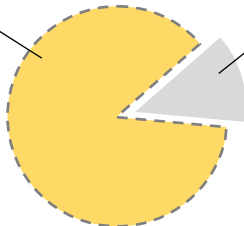
5.3. Consolidation Process

a. Non-controlling (Minority) Interest: Balance Sheet

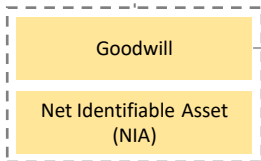
The pie chart represents the subsidiary.

**Parent's Controlling
Interest**

**Non-controlling (Minority)
Interest**



The parent
make a claim
on 2 essential
parts of the
target:



Full Goodwill
method

Partial Goodwill
method

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

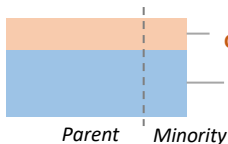
5. Business Combination

5.3. Consolidation Process

a. Non-controlling (Minority) Interest: Balance Sheet

Rule of thumb: “Return the parts unowned by the parent”

Full Goodwill method



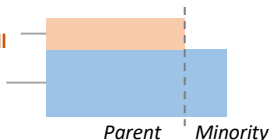
The parent company consolidate full goodwill and full net identifiable asset.

The parent returns minority investors' percent of goodwill and NIA.

NCI - Full Goodwill

$$\text{NCI} = \% \text{ minority} \times \text{FV}_{\text{subsidiary}}$$

Partial Goodwill method



The parent company consolidate their part of goodwill and full net identifiable asset.

The parent returns minority investors' percent of NIA only.

NCI - Partial Goodwill

$$\text{NCI} = \% \text{ minority} \times \text{FV}_{\text{NIA}}$$

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

a. Non-controlling (Minority) Interest: Balance Sheet



Illustration: Non-controlling Asset Valuation

On 1 January 2018, the hypothetical Parent Co. acquired 90% of the outstanding shares of the hypothetical Subsidiary Co. in exchange for shares of Parent Co.'s no par common stock with a fair value of €180,000. The fair market value of the subsidiary's shares on the date of the exchange was €200,000. Below is selected financial information from the two companies immediately prior to the exchange of shares (before the parent recorded the acquisition):

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process



Illustration: Non-controlling Asset Valuation

	Subsidiary		
	Parent BV	Book Value	Fair Value
Cash and receivables	€40,000	€15,000	€15,000
Inventory	125,000	80,000	80,000
PP&E (net)	235,000	95,000	155,000
	€400,000	€190,000	€250,000
Payables	55,000	20,000	20,000
Long-term debt	120,000	70,000	70,000
	175,000	90,000	90,000
Net assets	€225,000	€100,000	€160,000
Shareholders' equity:			
Capital stock (no par)	€87,000	€34,000	
Retained earnings	€138,000	€66,000	

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process



Illustration: Non-controlling Asset Valuation

1. Calculate the value of PP&E (net) on the consolidated balance sheet under both IFRS and US GAAP.

Answer:

- Under the acquisition method (IFRS and US GAAP), the parent company include 100% of the subsidiary's assets and liabilities at **fair value** on the consolidate balance sheet.

→ PP&E on consolidated balance sheet = ...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process



Illustration: Non-controlling Asset Valuation

1. Calculate the value of PP&E (net) on the consolidated balance sheet under both IFRS and US GAAP.

Answer:

- Under the acquisition method (IFRS and US GAAP), the parent company include 100% of the subsidiary's assets and liabilities at **fair value** on the consolidate balance sheet.

→ PP&E on consolidated balance sheet = 235,000 + 155,000 = €390,000.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process



Illustration: Non-controlling Asset Valuation

2. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the full goodwill method.

Answer:

Recall:

Full Goodwill = FV of the entity as a whole – FV of all identifiable assets

Fair value of the subsidiary

Fair value of subsidiary's identifiable net assets

Goodwill

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

 Illustration: Non-controlling Asset Valuation

2. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the full goodwill method.

Answer:

Recall:

Full Goodwill = FV of the entity as a whole – FV of all identifiable assets

Fair value of the subsidiary	€200,000
Fair value of subsidiary's identifiable net assets	- 160,000
Goodwill	€40,000

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

 Illustration: Non-controlling Asset Valuation

2. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the full goodwill method.

Answer:

Recall:

NCI - Full Goodwill

$$\text{NCI} = \% \text{ minority} \times \text{FV}_{\text{subsidiary}}$$

With:

- % minority = ...
- $\text{FV}_{\text{subsidiary}}$ = ...

$$\rightarrow \text{NCI} = \dots$$

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

Illustration: Non-controlling Asset Valuation

2. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the full goodwill method.

Answer:

Recall:

NCI - Full Goodwill

$$\text{NCI} = \% \text{ minority} \times \text{FV}_{\text{subsidiary}}$$

With:

- $\% \text{ minority} = 10\%$
- $\text{FV}_{\text{sub}} = \text{€}200,000$

$$\rightarrow \text{NCI} = 10\% \times \text{€}200,000 = \text{€}20,000$$

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process



Illustration: Non-controlling Asset Valuation

3. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the partial goodwill method.

Answer:

Recall: the full goodwill method (mandatory under US GAAP and optional under IFRS).

Partial Goodwill = FV of the acquisition – % acquire × FV of all identifiable assets

Acquisition price

90% of fair value

Goodwill

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process



Illustration: Non-controlling Asset Valuation

3. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the partial goodwill method.

Answer

Recall: the full goodwill method (mandatory under US GAAP and optional under IFRS).

Partial Goodwill = FV of the acquisition – % acquire × FV of all identifiable assets

Acquisition price	€180,000
90% of fair value	- 144,000
Goodwill	€36,000

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

 Illustration: Non-controlling Asset Valuation

3. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the partial goodwill method.

Answer:

Recall:

NCI - Partial Goodwill

$$\text{NCI} = \% \text{ minority} \times \text{FV}_{\text{NIA}}$$

With:

- $\% \text{ minority} = 10\%$
- $\text{FV}_{\text{NIA}} = \text{€}160,000$

→ NCI = ...

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

Illustration: Non-controlling Asset Valuation

3. Calculate the value of goodwill and the value of the non-controlling interest at the acquisition date under the partial goodwill method.

Answer:

Recall:

$$\text{NCI} = \% \text{ minority} \times \text{FV}_{\text{NIA}}$$

With:

- % minority = 10%
- $\text{FV}_{\text{NIA}} = \text{€}160,000$

$$\rightarrow \text{NCI} = 10\% \times \text{€}160,000 = \text{€}16,000$$

5.

Business Combination

5.3. Consolidation Process

	Full Goodwill	Partial Goodwill
Cash and receivables	€55,000	€55,000
Inventory	205,000	205,000
PP&E (net)	390,000	390,000
Goodwill	40,000	36,000
Total asset	€690,000	€686,000
Payables	€75,000	€75,000
Long-term debt	190,000	190,000
Total liability	€265,000	€265,000
Shareholders' equity		
Noncontrolling interests	€20,000	€16,000
Capital stock (no par)	€267,000	€267,000
Retained earnings	138,000	138,000
Total equity	€425,000	€425,000
Total liabilities and shareholders' equity	€690,000	€686,000

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5. Business Combination

5.3. Consolidation Process

b. Non-controlling (Minority) Interests: Income Statement

- On the income statement, non-controlling (minority) interests are presented as a line item reflecting the allocation of profit or loss for the period.
- All intercompany transactions must be eliminated.

Illustration: Non-controlling Interests: Income Statement

	Full Goodwill	Partial Goodwill
Sales	€250,000	€250,000
Cost of goods sold	137,500	137,500
Interest expense	10,000	10,000
Depreciation expense	39,000	39,000
Income from continuing operations	€63,500	€63,500
Non-controlling interest (10%)	(6,350)	(6,350)
Consolidated net income to parent's shareholders	€57,150	€57,150

	Full Goodwill	Partial Goodwill
Net Income	€57,150	€57,150
Total assets	€690,000	€686,000
Total equity	€425,000	€421,000
Return on assets	€57,150/€690,000 = 8.28	€57,150/€686,000 = 8.33
Return on equity	€57,150/€425,000 = 13.45	€57,150/€421,000 = 13.57

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment

- Goodwill is not amortized or depreciated; it is **tested for impairment** (at least annually).
- The impairment of goodwill is a one-way path. Once written down, goodwill cannot be later restored.

Goodwill Impairment

Under IFRS

Goodwill allocated to **Cash-generating Unit**

Impairment testing process: 1-step approach

Under US GAAP

Goodwill allocated to **Reporting Unit**

Impairment testing process: 2-step approach

Cash-generating unit and Reporting Unit have some minor differences under the 2 reporting standards. However, we do not focus on these differences. Just remember that goodwill is allocated among them.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

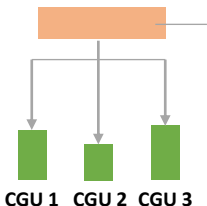
5.

Business Combination

5.3. Consolidation Process

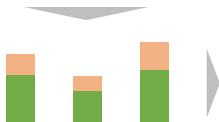
c. Goodwill impairment

IFRS - Cash generating unit



Goodwill from the new acquisition

On acquisition, the total amount of goodwill is allocated to the cash-generating units that benefit from the synergy with new asset



Goodwill impairment is conducted with the cash-generating unit.

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

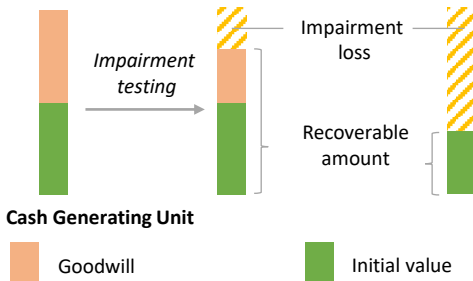
5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment

Goodwill Impairment - IFRS



- The Impairment loss is first absorbed by goodwill.
- Further loss is allocated to other **non-cash** assets on a pro rata basis.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment



Illustration: Goodwill Impairment: IFRS

The cash-generating unit of a French company has a carrying value of €1,400,000, which includes €300,000 of allocated goodwill. The recoverable amount of the cash-generating unit is determined to be €1,300,000, and the estimated fair value of its identifiable net assets is €1,200,000. Calculate the impairment loss.

Answer:

Recoverable amount of unit

Carrying amount of unit

Impairment loss

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment



Illustration: Goodwill Impairment: IFRS

The cash-generating unit of a French company has a carrying value of €1,400,000, which includes €300,000 of allocated goodwill. The recoverable amount of the cash-generating unit is determined to be €1,300,000, and the estimated fair value of its identifiable net assets is €1,200,000. Calculate the impairment loss.

Answer:

Recoverable amount of unit	€1,300,000
Carrying amount of unit	- 1,400,000
Impairment loss	- €100,000

The impairment loss of €100,000 is reported on the income statement, and the goodwill allocated to the cash-generating unit would be reduced by €100,000 to €200,000.

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment

Goodwill Impairment – US GAAP

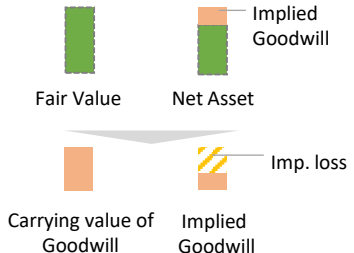
Step 1: Determination of an Impairment Loss

The carrying amount of the reporting unit (including goodwill) is compared to its fair value.



Step 2: Measurement of the Impairment Loss

The loss is the difference between the carrying value of the goodwill and the implied fair value of the goodwill.



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment



Illustration: Goodwill Impairment: US GAAP

A reporting unit of a US corporation (e.g., a division) has a fair value of \$1,300,000 and a carrying value of \$1,400,000 that includes recorded goodwill of \$300,000. The estimated fair value of the identifiable net assets of the reporting unit at the impairment test date is \$1,200,000. Calculate the impairment loss.

Answer:

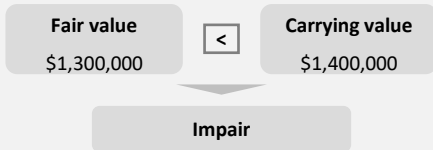
Step 1 – Determination of an Impairment Loss



[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...**5.****Business Combination****5.3. Consolidation Process****c. Goodwill impairment****Illustration: Goodwill Impairment: US GAAP**

A reporting unit of a US corporation (e.g., a division) has a fair value of \$1,300,000 and a carrying value of \$1,400,000 that includes recorded goodwill of \$300,000. The estimated fair value of the identifiable net assets of the reporting unit at the impairment test date is \$1,200,000. Calculate the impairment loss.

Answer:

Step 1 – Determination of an Impairment Loss

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment



Illustration: Goodwill Impairment: US GAAP

Answer:

Step 2 – Measurement of the Impairment Loss

Fair value of reporting unit

Less: net assets

Implied goodwill

Current carrying value of goodwill

Less: implied goodwill

Impairment loss

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

5.

Business Combination

5.3. Consolidation Process

c. Goodwill impairment



Illustration: Goodwill Impairment: US GAAP

Answer:

Step 2 – Measurement of the Impairment Loss

Fair value of reporting unit	€1,300,000
Less: net assets	1,200,000
Implied goodwill	€100,000
Current carrying value of goodwill	€300,000
Less: implied goodwill	100,000
Impairment loss	€200,000

The impairment loss of \$200,000 is reported on the income statement, and the goodwill allocated to the reporting unit would be reduced by \$200,000 to \$100,000.

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

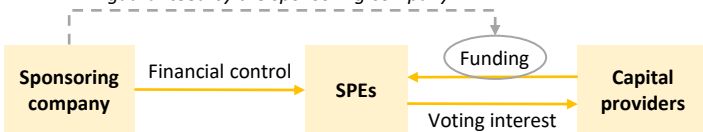
Variable interest and special purpose entities

Special purpose entities (SPEs)

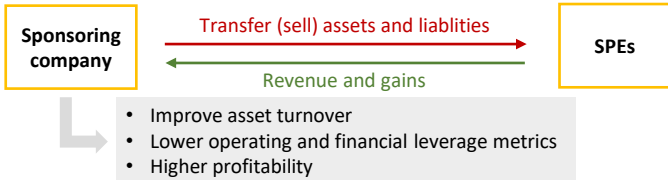
- Enterprises that are created to accommodate specific needs of the sponsoring entity.
- May take the form of a limited liability company, trust, partnership, or unincorporated entity.

Relationships between entities

The funding is often debt directly or indirectly guaranteed by the sponsoring company



Motivation and mechanism of creating SPEs



[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

Variable interest and special purpose entities

In the past, SPEs were often maintained off-balance-sheet

Enhancing the sponsor's financial statements and ratios

To address the accounting issues arising from the misuse of SPEs, the FASB uses the term variable interest entity (VIE) to more broadly define SPEs.

Under US GAAP, a VIE is an entity that

Total equity at risk is insufficient to finance activities without financial support from other parties

Equity investors lack any one of the following:

- Ability to make decisions
- Obligation to absorb losses
- The right to receive returns

The **primary beneficiary** of a VIE must consolidate it as a subsidiary

****Primary beneficiary:** the entity absorbs the majority of the risks or receives the majority of the rewards.*

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

Variable interest and special purpose entities



Illustration: Use of SPEs

Company P, a textile manufacturer, wants to borrow \$100 million. It has two options:

Option A

Borrow \$100 million from bank B

Option B

Sell \$100 million worth of accounts receivable to company S, an SPE created for this purpose. The SPE will fund the purchase by borrowing the money from bank B.

Using the financial statement information provided below, describe the effect of each Option on Company P balance sheet and income statement, assuming that Company P meets the definition of control and will consolidate the SPE.

Continue next slide →

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

Variable interest and special purpose entities



Illustration: Use of SPEs

Company P's balance sheet before the borrowing (\$mn)

Assets

Cash	\$50
Accounts receivable	\$200
Fixed assets	\$2,000
Total assets	\$2,250

Liabilities and Equity

Current liabilities	\$500
Debt	\$1,200
Equity	\$550
Total	\$2,250

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

Variable interest and special purpose entities



Illustration: Use of SPEs

Answer:

Option A: Company P's **cash and debt** will both increase by the new borrowing of \$100 million.

Company P's balance sheet after the borrowing

Assets

Cash	\$150
Accounts receivable	\$200
Fixed assets	\$2,000
Total assets	\$2,350

Liabilities and Equity

Current liabilities	\$500
Debt	\$1,300
Equity	\$550
Total	\$2,350

Its sales and net income will not change.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

Variable interest and special purpose entities



Illustration: Use of SPEs

Answer:

Option B: Company P's (nonconsolidated) balance sheet will reflect a reduction in accounts receivable of \$100 million and an increase in cash by the same amount.

Company P's balance sheet after the sale of accounts receivable to the SPE

Assets

Cash	\$150
Accounts receivable	\$100
Fixed assets	\$2,000
Total assets	\$2,250

Liabilities and Equity

Current liabilities	\$500
Debt	\$1,200
Equity	\$550
Total	\$2,250

Its revenue on income statement will increase by \$100 million as the sale of account receivable.

SPE's balance sheet after purchase of accounts receivable and bank loan

Assets

Accounts receivable	\$100
Total	\$100

Liabilities and Equity

Debt	\$100
Total	\$100

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

6.

Variable interest and special purpose entities



Illustration: Use of SPEs

Answer:

Option B:

After consolidation, the SPE's debt gets included with Company P's debt, and accounts receivable for Company P increase by the same amount.

Company P's balance sheet after consolidation

Assets

Cash	\$150
Accounts receivable	\$200
Fixed assets	\$2,000
Total assets	\$2,350

Liabilities and Equity

Current liabilities	\$500
Debt	\$1,300
Equity	\$550
Total	\$2,350

The balance sheet of Company P under either option is the same. Company P *cannot hide* the borrowing "off the books."

If Company P is able to sell the receivables to the SPE for more than their carrying value (for example, \$105 million), it would also report a gain on the sale in its profit and loss.

[LOS 1.a] *Describe* the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

7.

Additional issues in business combinations

2.1. Contingent Assets and Liabilities

Under IFRS

Only contingent liabilities are recognized (contingent *assets* are never recognized)

Recorded at **fair value** at the time of acquisition

Subsequently, contingent liabilities are measured at the higher of:

(1) *the value initially recognized,*
(2) *the best estimate of the amount needed to settle the liabilities.*

Under US GAAP

Divides contingent assets and liabilities into contractual and noncontractual

Contractual contingent assets and liabilities are recorded at their **fair values** on the acquisition date.

Noncontractual contingent assets and liabilities are also recorded if they meet the definition of an asset or liability.

Subsequently, measurement of **contingent liabilities** is similar under IFRS.

Contingent assets are recognized at the lower of (1) *the initial value* and (2) *the best estimate of the future settlement amount.*

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

7.

Additional issues in business combinations

2.2. Contingent Consideration

Contingent consideration (*e.g., a specific extra amount is payable to the former shareholders of the subsidiary if certain earnings or revenue targets are met*) may be negotiated as part of the acquisition price.

Is **recognized at fair value** under both IFRS and U.S. GAAP as an asset, liability, or equity.

- Subsequent ***changes in value are recognized in the income statement***
- Both IFRS and US GAAP do not remeasure equity classified contingent consideration; instead, settlement is accounted for within equity.

[LOS 1.a] Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS)...

7.

Additional issues in business combinations

2.3. In-Process R&D

**Accounting
treatment under
both U.S. GAAP
and IFRS**

- Is capitalized as an intangible asset and included as an asset.
- Subsequently ***amortized*** (if successful) ***or impaired*** (if unsuccessful).

2.4. Restructuring Costs

**Accounting
treatment under
both U.S. GAAP
and IFRS**

Restructuring costs are expensed when incurred and not capitalized as part of the acquisition cost

**MODULE 2:
EMPLOYEE COMPENSATION:
POST-EMPLOYMENT AND
SHARE-BASED**

LEARNING OUTCOMES

- 2.a.** Contrast types of employee compensation
- 2.b.** Explain how share-based compensation affects the financial statements
- 2.c.** Explain how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation
- 2.d.** Explain how post-employment benefits affect the financial statements
- 2.e.** Explain financial modeling and valuation considerations for post-employment benefits

[LOS 2.a] *Contrast* types of employee compensation

1.

Types of employee compensation

Accounting standards divide compensation into five general types, with distinctions based on:

- Time between employee service and payment.
- Form of payment

[LOS 2.a] *Contrast* types of employee compensation

1.

Types of employee compensation

Category	Definition	Common examples
Short-term benefits	Compensation expected to be paid within 12 months	- Salary and wages - Annual bonuses - Non-monetary benefits - Paid leaves
Long-term benefits	Compensation expected to be paid after 12 months	- Long-term paid leave - Long-term disability benefits
Termination benefits	Compensation paid in the event of employee termination	- Severance - Continued access to medical and other non-monetary benefits - Career counseling and outplacement services
Share-based compensation	Compensation in the form of, or in reference to, shares of the employer's stock	- Restricted stock - Stock options
Post-employment benefits	Compensation expected to be paid after employee retirement	- Pension and lump sum payments to retirees - Retiree life insurance and medical care

[LOS 2.a] *Contrast* types of employee compensation

2.

Accounting for types of compensation

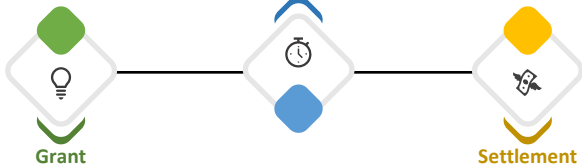
Employers' financial reporting principle across types of compensation

- Recognize compensation costs at *fair value* in the period that the employee provides services (**vesting period**).
- Most companies aggregate and report compensation expense on IS based on the employee's function.

Compensation timeline

Vesting

Employee earns compensation, thereby creating an obligation for the employer to pay that compensation



Employer communicates terms of compensation and employee accepts them.

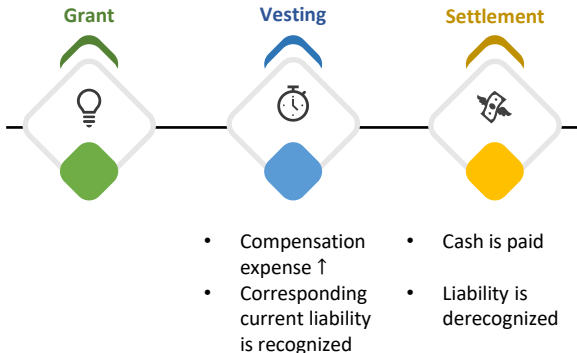
Employer pays compensation to employee

[LOS 2.a] *Contrast* types of employee compensation

2.

Accounting for types of compensation

Accounting for short-term benefits



Notes:

- Cash compensation is an outflow in operating activities on the statement of cash flows.
- Some compensation costs are capitalized as an asset → Compensation expense on the income statement deferred to when the employee service is consumed (*Illustrated on the next slide*).

[LOS 2.a] *Contrast* types of employee compensation

2.

Accounting for types of compensation



Illustration: Short-term benefits on the Financial Statements

Case 1: Company hires an employee in the legal department on 1 January, compensating them with an annual salary of SGD 82,200 paid every two weeks. The first payment date is 14 January.

	Grant 1 January	Vesting 1 January – 14 January	Settlement 14 January
<i>Income statement</i>	No impact	General and administrative expense +3,162*	No impact
<i>Balance sheet</i>	No impact	Accrued compensation +3,162	Accrued compensation (3,162)
<i>Statement of cash flow</i>	No impact	No impact	Cash flows from operations (3,162)

*SGD 82,200 / (52 weeks per year / 2 week pay period) = SGD 3,162.

[LOS 2.a] Contrast types of employee compensation

2.

Accounting for types of compensation



Illustration: Short-term benefits on the Financial Statements

Case 2: Company hires an employee in the manufacturing division on 1 January, compensating them with an annual salary of SGD 102,200 paid every two weeks. The first payment date is 14 January. The goods that the employee helped make are sold to customers on 3 April.

	Grant 1 January	Vesting 1 January – 14 January	Settlement 14 January	Sale 3 April
Income statement	No impact	No impact	No impact	Cost of sales +3,931
Balance sheet	No impact	Inventories +3,931 Accrued compensation +3,931	Accrued compensation (3,931)	Inventories (3,931)
Statement of cash flow	No impact	No impact	Cash flows from operations (3,931)	No impact

[LOS 2.a] *Contrast* types of employee compensation

2.

Accounting for types of compensation

Accounting for Share-based compensation and Post-employment benefits

- Compensation expense: Recognized on IS as the compensation earned by the employee, *similar to short-term benefits*.
- Measurement of the expense and the effect on the balance sheet and statement of cash flows *differ from short-term benefits*.

	Short-term benefits	Share-based compensation	Post- employment benefits
Typical vesting period	Days or weeks	Years	Years, decades
Form of payment	Cash	Shares	Cash
Amount recognized over the vesting period	Undiscounted salary, wage, etc.	Fair value, measured on the grant date	Present value of estimated future benefits

We will be looking into Share-based Compensation and Post-employment Benefits in the next sections.

[LOS 2.a] *Contrast* types of employee compensation

3.

Share-Based Compensation

Definition

Share-based compensation is typically awarded as a bonus to highly compensated employees, such as managers and those in technical roles.

3.1. Advantages over cash compensation



Aligns employees' financial interests with those of shareholders

- *Reducing principal-agency conflicts of interest.*
- *Allow employees to participate in firm value creation.*



Multi-year vesting periods improves employee retention



Requires no cash outlay, thereby preserving liquidity

- *Especially beneficial for younger companies that might otherwise struggle to attract top talent.*

[LOS 2.a] *Contrast* types of employee compensation

3. Share-Based Compensation

3.2. Disadvantages of share-based compensation



May not necessarily reward individual performance or influence their actions

- *Share-based compensation may have limited influence over the company's share price.*



May lead to suboptimal risk-taking by managers

- *Fearing large share price declines and loss of personal wealth → Seek less risky and less profitable investments.*
- *Attempting to maximize short-term gain over long-term viability → Excessive risk-taking.*



Severe price declines and prolonged underperformance can make shares less valuable to the employee

- *Share-based compensation means that employees also lose wealth from share price declines and underperformance.*

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

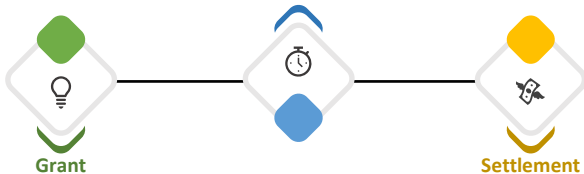
3.

Share-Based Compensation

3.3. *Financial reporting for Share-based compensation*

Vesting

- Recognize fair value as share-based compensation expense over the vesting period
- Adjust or reverse entries if needed for changes in estimates



Measure fair value of the award, adjusted for estimated number of awards expected not to vest

Share issued to employee

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

Illustration: Share-based Compensation Accounting

Company grants 25,000 shares to an employee in the R&D division on 1 January 20X1. The award vests three years from the grant date. The fair value of the award on 1 January 20X1 is BRL 273,000.

	Year ended 31 Dec 20X1	Year ended 31 Dec 20X2	Year ended 31 Dec 20X3
Income statement	R&D expense 91,000	R&D expense 91,000	R&D expense 91,000
Balance sheet	Share-based compensation reserve (equity) +91,000	Share-based compensation reserve (equity) +91,000	Share-based compensation reserve (equity) +91,000 Transfer 273,000 from <i>Share-based compensation reserve</i> to <i>Common stock</i> and <i>Paid-in capital</i> accounts upon settlement
Statement of CF	No impact	No impact	No impact

How is the accounting of Share-based compensation different from short-term benefits? (Compare with Illustration on slide 154)

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

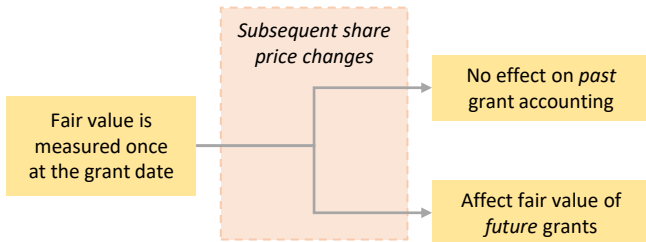
3.

Share-Based Compensation

3.3. *Financial reporting for Share-based compensation*

The accounting is similar to that for salaries but with 3 key differences.

- The offsetting entry to compensation expense is **made to an equity account**, not liability, because compensation is settled in shares not cash.
- Vesting is over several years, rather than weeks, so a single grant affects the financial statements **over multiple years**.
- Fair value is used as the **measurement basis**, rather than the undiscounted amount to be repaid at settlement.



[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. Share-based compensation plans

What must the employees do for the award to vest?

Conditions
Service

Compensation vests on a future date, requiring the employee to remain employed until that time.

Performance

Meeting or exceeding a target for EPS, ROIC, segment profit, for share prices, etc.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. *Share-based compensation plans*

What are the instruments used in Share-based compensation plans?

Instrument	Alternative names	Description
<i>Restricted stock</i>	- Restricted stock awards - RSUs - Performance shares or performance share units	Awards of shares or share-like units with sale and other restrictions that are lifted upon vesting.
<i>Stock options</i>	Share options	Awards of non-tradeable call options, typically at the money, on the employer's stock.
<i>Stock appreciation-based</i>	- Stock appreciation rights - Phantom shares	Award of cash or shares based on the performance of shares over a period.
<i>Stock purchase-based</i>	- Employee stock purchase plan - Employee stock ownership plan	Permits employees to purchase a limited number of newly issued shares at a discount.

**We will focus on the first two instruments.*

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. *Share-based compensation plans*

a. *Restricted stock*

Definition

Restricted stock involves common shares granted to employees but subject to selling and other restrictions.

*Restricted stock is also referred to as **performance shares** if vesting is based not only on service but also on performance conditions.*

Characteristics

- Has voting rights and dividend participation.
- Not tradeable.
- Upon settlement, restrictions are lifted → Recipient is free to sell their shares.

Restricted stock units (RSUs)

- Similar to restricted stocks, but RSUs are instruments which represent the right to receive shares upon settlement.
- Have neither voting rights nor dividend participation and are also not tradeable.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. Share-based compensation plans

a. Restricted stock

Accounting

- Grant-date fair value for restricted stock and RSU awards is the market price of the underlying shares.
- When RSU vests, settlement occurs automatically, converting to common stock.
- Accounting entry: Transferring amounts from the share-based compensation reserve account to *Common stock* and *Paid-in Capital accounts* (Equity)

Notes: If the RSU does not participate in dividends, share price is typically *adjusted downward* for dividends expected to be paid over the vesting period.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: Effect of RSU awards on the Financial statements

Workflow Corporation (“Workflow”) is a Japan-based company that designs, makes, and sells project management software for businesses. To motivate and retain employees, as well as preserve cash, Workflow pays bonuses to employees in management and technical roles in shares.

Under its Equity Compensation Plan approved by shareholders, Workflow grants RSUs representing one share of its no-par value common stock. The RSUs vest in three years, contingent on service. The company accounts for forfeitures as they occur and does not pay or expect to pay dividends. Threequarters of the grants were made to employees in the R&D division with the balance granted to executive management. RSU grants and share prices on the grant dates were as follows:

Date	Number of RSUs granted	Share price (JPY)
1 January 20X1	4,542,000	4,360
1 January 20X2	3,521,000	3,270
1 January 20X3	5,198,000	3,333

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

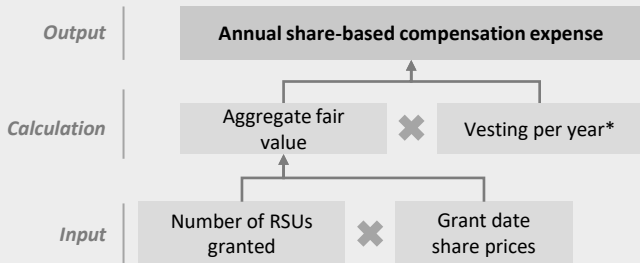


Illustration: Effect of RSU awards on the Financial statements

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 1: Calculate the annual share-based compensation expense

Recall: Grant-date fair value for restricted stock and RSU awards is the market price of the underlying shares.



**will be discussed further in Step 2.*

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 1: Calculate the annual share-based compensation expense

Date	RSUs granted	Share price (JPY)	Aggregate fair value (mil JPY)
1 January 20X1	4,542,000	4,360	$4.542 \times 4,360 = 19,803$
1 January 20X2	3,521,000	3,270	...
1 January 20X3	5,198,000	3,333	...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 1: Calculate the annual share-based compensation expense

Date	RSUs granted	Share price (JPY)	Aggregate fair value (mil JPY)
1 January 20X1	4,542,000	4,360	$4.542 \times 4,360 = 19,803$
1 January 20X2	3,521,000	3,270	$3.521 \times 3,270 = 11,514$
1 January 20X3	5,198,000	3,333	$5.198 \times 3,333 = 17,325$

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: Effect of RSU awards on the Financial statements

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 1: Calculate the annual share-based compensation expense

Date	Aggregate fair value (mil JPY)	Vesting per year	Annual compensation expense (mil JPY)
1 January 20X1	19,803	1/3	$19,803 \times 1/3 = 6,601$
1 January 20X2	11,514	1/3	...
1 January 20X3	17,325	1/3	...

[LOS 2.b] Explain how share-based compensation affects the financial statements



Illustration: Effect of RSU awards on the Financial statements

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 1: Calculate the annual share-based compensation expense

Date	Aggregate fair value (mil JPY)	Vesting per year	Annual compensation expense (mil JPY)
1 January 20X1	19,803	1/3	$19,803 \times 1/3 = 6,601$
1 January 20X2	11,514	1/3	3,838
1 January 20X3	17,325	1/3	5,775

[LOS 2.b] Explain how share-based compensation affects the financial statements



Illustration: Effect of RSU awards on the Financial statements

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

Three quarters of the grants were made to employees in the R&D division. Thus, annual compensation expense for each year is allocated as:

Date	Annual compensation expense (mil JPY)	Allocation to R&D expense	Allocation SG&A expense
1 January 20X1	6,601	$3/4 \times 6,601 = 4,951$	$6,601 \times 1/4 = 1,650$
1 January 20X2	3,838	...	...
1 January 20X3	5,775	...	...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

Three quarters of the grants were made to employees in the R&D division. Thus, annual compensation expense for each year is allocated as:

Date	Annual compensation expense (mil JPY)	Allocation to R&D expense	Allocation SG&A expense
1 January 20X1	6,601	$3/4 \times 6,601 = 4,951$	$6,601 \times 1/4 = 1,650$
1 January 20X2	3,838	2,878	960
1 January 20X3	5,775	4,331	1,444

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

The impact on items in the *Income statement* is as follows:

Date	Allocation to R&D expense	R&D expense	Allocation to SG&A expense	SG&A expense
Year ended 31 Dec 20X1	4,951	4,951	1,650	...
Year ended 31 Dec 20X2	2,878	$4,951 + 2,878 = 7,829$	960	...
Year ended 31 Dec 20X3	4,331	...	1,444	...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

The impact on items in the *Income statement* is as follows:

Date	Allocation to R&D expense	R&D expense	Allocation to SG&A expense	SG&A expense
Year ended 31 Dec 20X1	4,951	4,951	1,650	1,650
Year ended 31 Dec 20X2	2,878	$4,951 + 2,878 = 7,829$	960	$1,650 + 960 = 2,610$
Year ended 31 Dec 20X3	4,331	$7,829 + 4,331 = 12,161$	1,444	$2,610 + 1,444 = 4,054$

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

What is “Vesting per year”?

Recall:

- Recognize compensation costs at fair value in the period that the employee provides services (**vesting period**).
- The offsetting entry to compensation expense is **made to an equity account**.

“The RSUs vest in three years, contingent on service” → Illustrated on the next slide.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

What is “Vesting per year”?

Compensation expense and Share-based compensation reserve (equity) can be reported as:

Granted on	Annual compensation expense (mil JPY)	Vesting		
		20X1	20X2	20X3
1 January 20X1	6,601	6,601	6,601	6,601
1 January 20X2	3,838		3,838	3,838
1 January 20X3	5,775			5,775
Compensation expense or Increase in Share-based compensation reserve		+6,601	+10,439	+16,214
Total of Share-based compensation reserve (Equity)		6,601	17,040	33,254

Upon settlement, transfer **33,254** from **Share-based compensation** reserve to **Paid-in capital** account.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

1. Calculate the effect on the financial statements for the years ended 31 December 20X1, 20X2, 20X3.

Step 2: Determine the impacts on the financial statements

The impact on items in the *Balance sheet* and *Statement of cash flows* is as follows:

	Year ended 31 Dec 20X1	Year ended 31 Dec 20X2	Year ended 31 Dec 20X3
<i>Income statement</i>	R&D expense 4,951 SG&A expense 1,650	R&D expense 7,829 SG&A expense 2,610	R&D expense 12,161 SG&A expense 4,054
<i>Balance sheet</i>	Share-based compensation reserve (equity) +6,601	Share-based compensation reserve (equity) +10,439	Share-based compensation reserve (equity) +16,214 Transfer 33,254 from <i>Share-based compensation reserve</i> to <i>Paid-in capital accounts</i> upon settlement
<i>Statement of CF</i>	No impact	No impact	No impact

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 1: Calculate the annual share-based compensation expense

Assume the share prices on 1 January 20X2 and 20X3 were 25% higher:

- For 20X2: $3,270 \times \dots = \dots$
- For 20X3: $3,333 \times \dots = \dots$

Date	RSUs granted	Share price (JPY)	Aggregate fair value (mil JPY)
1 January 20X1	4,542,000	4,360	$4.542 \times 4,360 = 19,803$
1 January 20X2	3,521,000	...	...
1 January 20X3	5,198,000	...	...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 1: Calculate the annual share-based compensation expense

Assume the share prices on 1 January 20X2 and 20X3 were 25% higher:

- For 20X2: $3,270 \times 1.25 = 4,088$ JPY
- For 20X3: $3,333 \times 1.25 = 4,166$ JPY

Date	RSUs granted	Share price (JPY)	Aggregate fair value (mil JPY)
1 January 20X1	4,542,000	4,360	$4,542 \times 4,360 = 19,803$
1 January 20X2	3,521,000	4,088	14,394
1 January 20X3	5,198,000	4,166	21,655

[LOS 2.b] Explain how share-based compensation affects the financial statements



Illustration: Effect of RSU awards on the Financial statements

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 1: Calculate the annual share-based compensation expense

Date	Aggregate fair value (mil JPY)	Vesting per year	Annual compensation expense (mil JPY)
1 January 20X1	19,803	1/3	$19,803 \times 1/3 = 6,601$
1 January 20X2	11,514	1/3	...
1 January 20X3	17,325	1/3	...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 1: Calculate the annual share-based compensation expense

Date	Aggregate fair value (mil JPY)	Vesting per year	Annual compensation expense (mil JPY)
1 January 20X1	19,803	1/3	$19,803 \times 1/3 = 6,601$
1 January 20X2	14,394	1/3	4,798
1 January 20X3	21,655	1/3	7,218

[LOS 2.b] Explain how share-based compensation affects the financial statements



Illustration: Effect of RSU awards on the Financial statements

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 2: Determine the impacts on the income statement

Compensation expense can be reported as:

Granted on	Annual compensation expense (mil JPY)	Vesting		
		20X1	20X2	20X3
1 January 20X1	6,601	...	...	...
1 January 20X2	4,798		...	...
1 January 20X3	7,218		...	...
Compensation expense or Increase in Share-based compensation reserve		...	...	...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 2: Determine the impacts on the income statement

Compensation expense can be reported as:

Granted on	Annual compensation expense (mil JPY)	Vesting		
		20X1	20X2	20X3
1 January 20X1	6,601	6,601	6,601	6,601
1 January 20X2	4,798		4,798	4,798
1 January 20X3	7,218			7,218
Compensation expense or <i>Increase in</i> Share-based compensation reserve		+6,601	+11,399	+18,617

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of RSU awards on the Financial statements*

2. Assume the share prices on 1 January 20X2 and 20X3 were 25% higher. Calculate and explain the effect on the income statement.

Step 2: Determine the impacts on the income statement

Comparing to Compensation expense in Question 1:

	Vesting		
	20X1	20X2	20X3
Compensation expense (question 1)	+6,601	+10,439	+16,214
Compensation expense (question 2)	+6,601	+11,399	+18,617
% difference	0%	9%	15%

- Share price were 25% higher in 20X2 and 20X3 in question 2.
- Increase in share-based compensation expense in 20X2 and 20X3 is significantly lower (9% and 15%).

Reason: 20X1 share-based compensation expense does not change, regardless of how shares performed subsequently.

→ **Effect of Grant-date fair value and Vesting period in accounting.**

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. *Share-based compensation plans*

b. Stock options

Definition

Stock options are non-tradeable call options on the employer's stock typically issued at the money (i.e., strike price equal to the share price on the grant date).

Share price = Strike price

Grant

Vesting

Expiration



If Share price > Strike price
→ Recipient can exercise the option and earn the spread

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. *Share-based compensation plans*

b. Stock options

Accounting

- Stock option is also reported as fair value, however, its fair value must be estimated.
- When options vest, settlement does not occur until the options are exercised.
- At settlement, a cash inflow is recorded in financing activities on the statement of cash flows.

Option's fair value

Intrinsic value

The value that an option buyer will get from the option on exercising it

Time value

Premium in excess of intrinsic value, reflects hope of a favorable change in the underlying security's price

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.4. Share-based compensation plans

b. Stock options

Option valuation models is further discussed in the topic Derivatives, Module 2: Valuation of Contingent Claims.

Requirement of valuation method

Consistent with fair value measurement requirements

Based on established principles of financial economic theory

Reflect all substantive characteristics of the awards

Notes to financial statements must disclose material assumptions used to value the options: volatility, estimated life, risk-free interest rate, dividend yield, etc.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

Under its Equity Compensation Plan, Workflow Corporation grants 25 million stock options to executives on 1 January 20X1 that vest on 31 December 20X3. The options are granted at the money. The share price and fair value per option on the grant date are JPY 4,360 and JPY 1,288, respectively. The options expire seven years after the grant date.

- 1. Calculate the share-based compensation expense Workflow will recognize and its effect on the financial statements for the years ended 31 December 20X1, 20X2, and 20X3.*
- 2. Calculate the effect on the financial statements for the year ended 31 December 20X4 if the share price remains below JPY 4,360 that year.*
- 3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.*

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

1. Calculate the share-based compensation expense Workflow will recognize and its effect on the financial statements for the years ended 31 December 20X1, 20X2, and 20X3.

Step 1: Calculate the aggregated fair value of the option grants

**Aggregate fair
value of option
grants**

=

Number of
options granted

×

Grant-date fair
value

=

...

×

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

1. Calculate the share-based compensation expense Workflow will recognize and its effect on the financial statements for the years ended 31 December 20X1, 20X2, and 20X3.

Step 1: Calculate the aggregated fair value of the option grants

**Aggregate fair
value of option
grants**

=

Number of
options granted

×

Grant-date fair
value

=

25 million

×

JPY 1,288

=

JPY 32,200
million

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

1. Calculate the share-based compensation expense Workflow will recognize and its effect on the financial statements for the years ended 31 December 20X1, 20X2, and 20X3.

Step 2: Determine the impacts on the financial statements

- *The impact on items in the Income statement:* Vesting period of 3 years → Workflow will recognize the 1/3 of the aggregate fair value that vests.
- *The impact on item in the Balance sheet:* The offsetting entry is made to Share-based compensation reserve in equity on the balance sheet.

Financial statement	Impacts for each year ended 31 Dec 20X1, 20X2, 20X3	
<i>Income statement</i>	SG&A expense:	...
<i>Balance sheet</i>	Increase in equity:	...
<i>Statement of Cash flows</i>		...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

1. Calculate the share-based compensation expense Workflow will recognize and its effect on the financial statements for the years ended 31 December 20X1, 20X2, and 20X3.

Step 2: Determine the impacts on the financial statements

- *The impact on items in the Income statement:* Vesting period of 3 years → Workflow will recognize the 1/3 of the aggregate fair value that vests.
- *The impact on item in the Balance sheet:* The offsetting entry is made to Share-based compensation reserve in equity on the balance sheet.

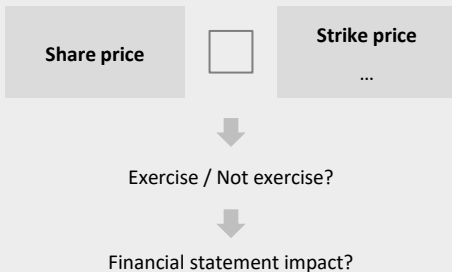
Financial statement	Impacts for each year ended 31 Dec 20X1, 20X2, 20X3
<i>Income statement</i>	SG&A expense: JPY 10,733 million
<i>Balance sheet</i>	Increase in equity: + JPY 10,733 million
<i>Statement of Cash flows</i>	Share-based compensation expense does not impact cash flows

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

2. Calculate the effect on the financial statements for the year ended 31 December 20X4 if the share price remains below JPY 4,360 that year.



[LOS 2.b] Explain how share-based compensation affects the financial statements**Illustration: Effect of Option awards on the Financial statements**

2. Calculate the effect on the financial statements for the year ended 31 December 20X4 if the share price remains below JPY 4,360 that year.

Share price

<

Strike price

JPY 4,360



Not exercise options



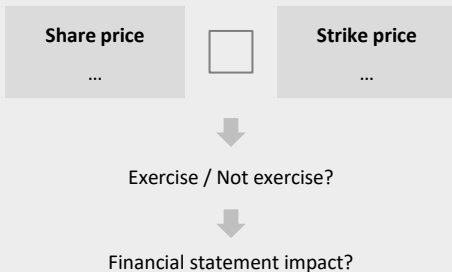
No financial statement impacts

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.

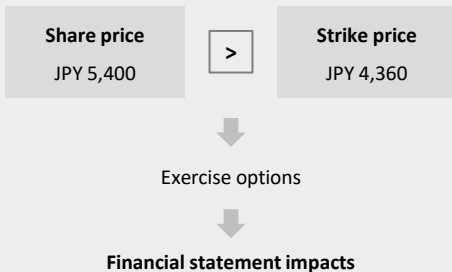


[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.



[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.

Financial statement impacts

- The impact on items in the Cash flow statement: Recognize cash inflow in financing activities.

Cash inflow

=

Strike price

×

Number of
options
exercised

=

...

×

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: Effect of Option awards on the Financial statements

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.

Financial statement impacts

- The impact on items in the Cash flow statement: Recognize cash inflow in financing activities.

Cash inflow

=

Strike price

×

Number of
options
exercised

=

JPY 4,360

×

6 million

=

JPY 26,160
million

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.

Financial statement impacts

- The impact on item in the Balance sheet:

Value of exercised
options

=

Option value

×

Number of options
exercised

=

...

×

...

=

...

→ Share-based compensation reserve (equity) is reduced by ...

The entry made to Share-based compensation reserve in equity on the balance sheet is transferred to Paid-in capital:

Cash inflow from financing = Share-based compensation reserve + Paid-in capital

⇒ Paid-in capital = ...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.

Financial statement impacts

- The impact on item in the Balance sheet:

Value of exercised
options

=

Option value

×

Number of options
exercised

=

JPY 1,288

×

6 million

=

JPY 7,728
million

→ Share-based compensation reserve (equity) is reduced by JPY 7,728 million. The entry made to Share-based compensation reserve in equity on the balance sheet is transferred to Paid-in capital:

Cash inflow from financing = Share-based compensation reserve + Paid-in capital

⇒ Paid-in capital = Cash inflow from financing - Share-based compensation reserve = 26,160 - (-7,728) = JPY 33,888 million

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Illustration: *Effect of Option awards on the Financial statements*

3. In 20X5, the share price increases to JPY 5,400 and 6 million options are exercised. Calculate the effect on the financial statements that year.

Financial statement impacts

- *The impact on items in the Cash flow statement:* Recognize cash inflow in financing activities.
- *The impact on item in the Balance sheet:* The entry made to Share-based compensation reserve in equity on the balance sheet is transferred to Paid-in capital.

Financial statement	Impacts for year ended 31 Dec 20X5
<i>Income statement</i>	No impact
<i>Balance sheet</i>	Share-based compensation reserve (equity) -7,728 Paid-in capital (equity) +33,888
<i>Statement of Cash flows</i>	Cash inflow from financing activities of JPY 26,160 million

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. *Tax and share count effects, note disclosure*

a. Tax effect

Share-based compensation is **deductible** for issuers' taxable income in most jurisdictions.

However, share-based compensation tax deduction differs from share-based compensation expense.

	Financial reporting: Share-based compensation expense	Tax return: Deduction for stock-based compensation
Timing	Over the vesting period	At settlement
Amount	Grant-date fair value	- Share price on the settlement date (RSUs) - Intrinsic value at exercise (options)

What is the effect of the differences if an issuer's share price at the grant date differs from its share price at settlement?

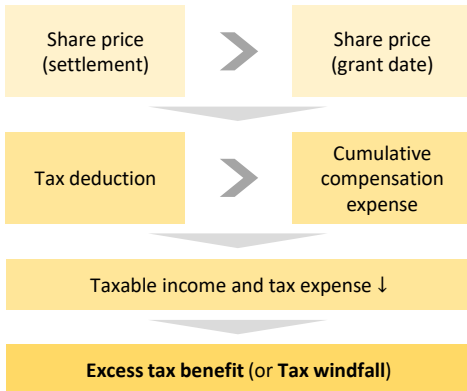
[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

a. Tax effect



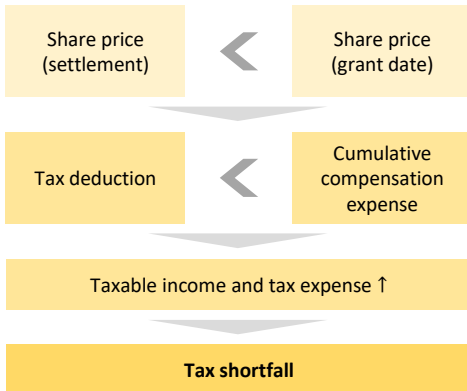
[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

a. Tax effect



[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

a. Tax effect

IFRS and US GAAP Treatment of Tax Windfalls and Shortfalls from Share-based compensation

	IFRS	US GAAP
<i>Tax windfall</i>	Gain recognized directly in stockholders' equity.	Decrease in income tax expense on the income statement.
<i>Tax shortfall</i>	Loss recognized directly in stockholders' equity.	Increase in income tax expense on the income statement.
	Tax windfalls and shortfalls are caused by changes in the stock price → Transactions between shareholders.	Introduces volatility in the effective tax rate and differences in effective and statutory tax rates.
	Not reported in earnings	

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

a. Tax effect

Implication for analysts

Analyzing US GAAP reporters:

- Closely examine the reconciliation of the statutory to effective tax rate for US GAAP reporters.
- Not assume the historical effective tax rate will persist if the company reported a tax windfall/shortfall from share-based compensation.

Analyzing IFRS reporters:

- Tax windfalls and shortfalls have comparatively more stable effective tax rates with less deviation from statutory tax rates.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

b. Share-based compensation and Shares outstanding

Basic shares outstanding

*Increases when share-based awards **settle***

Income statement

As weighted average for the period

Balance sheet

As of period end

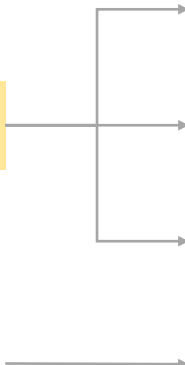
Statement of stockholders' equity

As of period end

*Share-based awards that **have not settled?***

Potentially dilutive securities

Treasury stock method



[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3. Share-Based Compensation

3.5. *Tax and share count effects, note disclosure*

b. Share-based compensation and Shares outstanding

*If shares are dilutive, they should be included in **diluted shares outstanding** using the treasury stock method.*

Treasury stock method

- 1 Proceeds from the exercise/conversion are assumed to be used to repurchase shares at the average share price for the period
→ Number of treasury shares
- 2 Nets the number of hypothetically repurchased shares against the total number of potentially dilutive securities → Dilutive securities
- 3 Adds the “net” amount of potentially dilutive securities to basic shares outstanding → **Diluted shares outstanding**

Continue next slide for calculation →

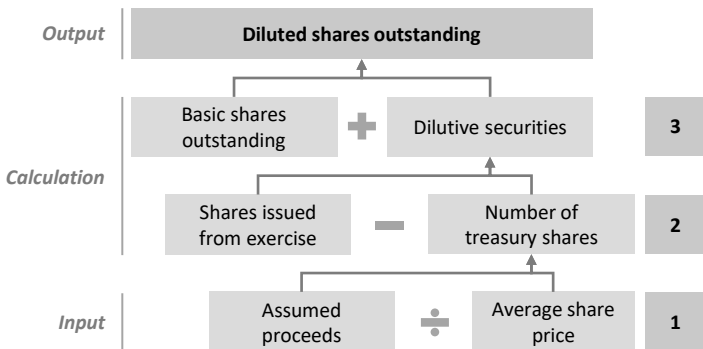
[LOS 2.b] Explain how share-based compensation affects the financial statements

3. Share-Based Compensation

3.5. Tax and share count effects, note disclosure

b. Share-based compensation and Shares outstanding

Treasury stock method



**The steps are arranged from bottom to top to demonstrate backward thinking, starting with output.*

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

b. Share-based compensation and Shares outstanding

Treasury stock method

What awards are included in the calculation?

- Awards with *service vesting* conditions are *usually included*.
- Awards with *performance* conditions that *have not been met* as of the end of the reporting period *are excluded*.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

b. Share-based compensation and Shares outstanding

Treasury stock method

What is Assumed proceeds from conversion or exercise of awards?

**Assumed
proceeds**

=

Cash proceeds
from exercise

+

Average unrecognized
share-based compensation
expense

Strike price × Number
of options

Assumes the vesting and settlement of
share-based awards today → Add back
the avoided compensation expense

- Unrecognized share-based compensation expense (end of period) = Unvested awards × Grant date fair value
- Average unrecognized share-based compensation expense = Average of the last two period-end values.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Zephire, Inc., has an employee stock option and a RSU plan for its senior management team. On January 1, 20X1, the company made a grant of 2 million at-the-money options (maturing in 5 years) and 1 million shares. On that date, the fair value of the options was \$2.85, and the stock price was \$23. Both vest after 4 years. The average stock price was \$24.50 during 20X1 and \$25 at the end of that year.

The unrecognized value of the compensation expense was \$4,275,000 for the options and \$17,250,000 for the stock grant.

Calculate number of dilutive shares.

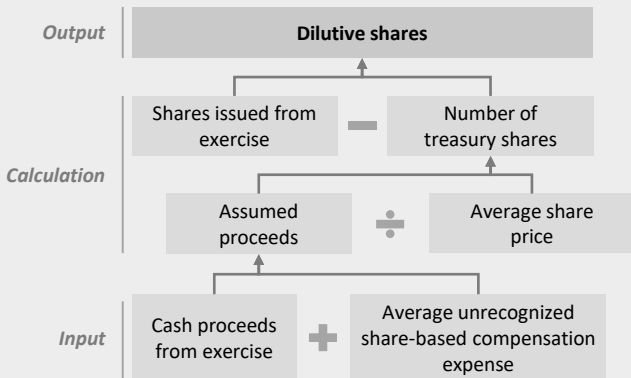
[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:



We have two types of potential dilutive securities:

1. *Stock options:* Because share price > strike price → Options are exercised.
2. *Stock grant*

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

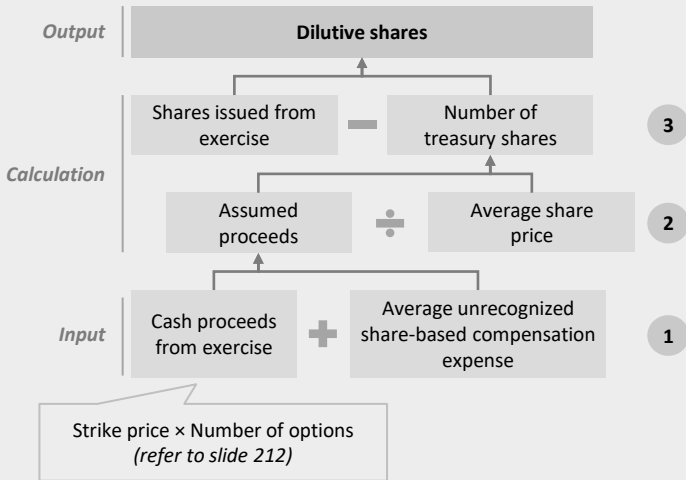


Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options



[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options

Step 1: Determine Assumed proceeds

Assumed
proceeds

=

Cash proceeds
from exercise

+

Average unrecognized
share-based compensation
expense

=

... × ...

+

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options

Step 1: Determine Assumed proceeds

Assumed
proceeds

=

Cash proceeds
from exercise

+

Average unrecognized
share-based compensation
expense

=

2 million × \$23
= \$46 million

+

(0 + \$4.275 million) / 2
= \$2.1375 million

=

\$48.1375 million

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options

Step 2: Determine Number of treasury shares

Number of
treasury shares

=

Assumed
proceeds

÷

Average share
price

=

...

÷

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options

Step 2: Determine Number of treasury shares

Number of
treasury shares

=

Assumed
proceeds

÷

Average share
price

=

\$48.1375 million

÷

\$24.50

=

1.965 million
shares

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options

Step 3: Determine Dilutive shares

Dilutive shares

=

Shares issued
from exercise

—

Number of
treasury shares

=

...

—

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

1. Stock options

Step 3: Determine Dilutive shares

Dilutive shares

=

Shares issued
from exercise

—

Number of
treasury shares

=

2 million

—

1.965 million

=

0.035 million or
35,000 shares

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

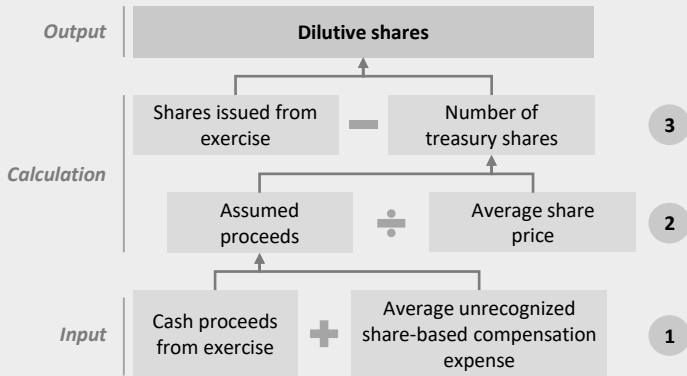


Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant



Steps for calculation are similar to stock options.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant

Step 1: Determine Assumed proceeds

Assumed
proceeds

=

Cash proceeds
from exercise

+

Average unrecognized
share-based compensation
expense

=

...

+

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant

Step 1: Determine Assumed proceeds

Assumed
proceeds

=

Cash proceeds
from exercise

+

Average unrecognized
share-based compensation
expense

=

0

+

$(0 + \$17.25 \text{ million}) / 2$
= \$8.625 million

=

\$8.625 million

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant

Step 2: Determine Number of treasury shares

Number of
treasury shares

=

Assumed
proceeds

÷

Average share
price

=

...

÷

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant

Step 2: Determine Number of treasury shares

Number of
treasury shares

=

Assumed
proceeds

÷

Average share
price

=

\$8.625 million

÷

\$24.50

=

0.352 million
shares

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant

Step 3: Determine Dilutive shares

Dilutive shares

=

Shares issued
from exercise

—

Number of
treasury shares

=

...

—

...

=

...

[LOS 2.b] *Explain* how share-based compensation affects the financial statements



Knowledge check

Example: Stock and Option grant

Answer:

2. Stock grant

Step 3: Determine Dilutive shares

Dilutive shares

=

Shares issued
from exercise

—

Number of
treasury shares

=

1 million

—

0.352 million
shares

=

0.648 million or
648,000 shares

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

b. Share-based compensation and Shares outstanding

Treasury stock method

- In-the-money options are dilutive and included in diluted shares outstanding.
- Out-of-the-money and at-the-money options are anti-dilutive and left out of diluted shares outstanding.
- RSUs and restricted stock grants are considered antidilutive only if the current stock price is significantly less than the price on the grant date
→ The unrecognized compensation expense per share is higher than the current market price.
- Rapid increases in the share price can result in more dilution (and vice versa), because the assumed number of shares that can be repurchased falls with a higher average share price.
- The analyst should add anti-dilutive securities to a company's diluted share count for valuation purposes.

[LOS 2.b] *Explain* how share-based compensation affects the financial statements

3.

Share-Based Compensation

3.5. Tax and share count effects, note disclosure

c. Disclosures for Share-based Compensation

In the Notes to the financial statements, disclose information that enables users of the financial statements to understand:

- The nature and extend of share-based payment arrangements that existed during the period.
- How the fair value of the equity instruments granted during the period was determined.
- The effect of share-based payment transactions on the company's net income (loss) during the period and on its financial position.

The proxy statement or other governance reports will contain disclosures on executive management and directors' compensation, which typically have significant share-based component.

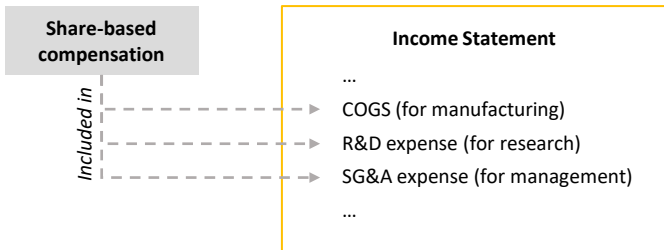
[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

a. Forecasting Share-based compensation



Relationship between expense categories and revenues

Remain static

Forecast share-based compensation implicitly while making operating expense or margin forecasts

Proportions do not hold

Separate share-based compensation expense forecasts from other items
(Demonstrated next slide)

[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

a. Forecasting Share-based compensation

Separating share-based compensation forecasts

Steps

1. Subtract the amount attributable to share-based compensation from each categories.
2. Forecast individual expenses as percentage of revenue (based on historical trends).
3. Forecast share-based compensation separately.

Sources of information that helps to predict changes in compensation:

- Historical data
- Management guidance
- Assumptions about reversion to industry mean

[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

a. Forecasting Share-based compensation

Separating share-based compensation forecasts

Benefits

Necessary for the statement of cash flows

Arrive at more accurate free cash flow forecasts

Not understate cash on the balance sheet

Compute non-GAAP metrics for comparisons and valuation

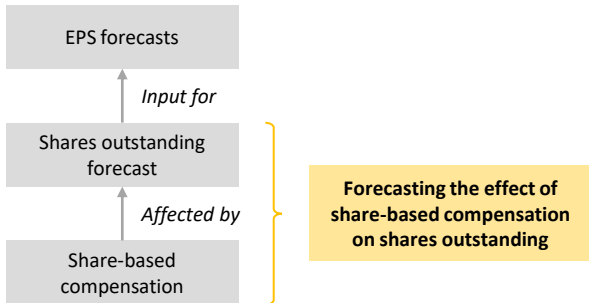
[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

b. Forecasting Shares Outstanding with Share-based awards



[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

b. Forecasting Shares Outstanding with Share-based awards

Forecasting the effect of share-based compensation on shares outstanding

1 *Forecast grants of share-based awards, net of forfeitures*

- Typically modeled using growth rates of historical values presented in note disclosures.
- Should be compatible with the forecast of share-based compensation expense.

2 *Forecast settlement of awards*

- Can be modeled the same way as (1) or by assuming that a percentage of outstanding awards settles each period.

3 *Model basic shares outstanding*

Continue next slide →

[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

b. Forecasting Shares Outstanding with Share-based awards

Forecasting the effect of share-based compensation on shares outstanding

3 *Model basic shares outstanding*

Basic shares outstanding, beg. of period

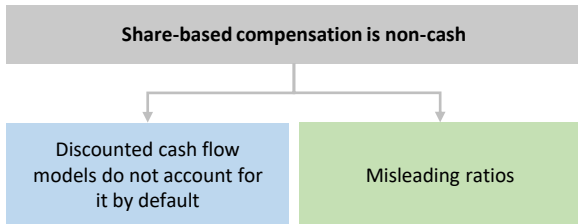
- + RSUs vested and/or shares options exercised
- + Share issuances from secondaries, acquisitions, etc.
- Share repurchases

Basic shares outstanding, end of period

Diluted shares outstanding is forecasted by adding a number of dilutive securities to the forecast of basic shares outstanding (*refer to Treasury stock method in Section 3.5 b*) → Challenging to forecast dilutive securities.

Common approach: Assume a percentage of outstanding awards are dilutive.

[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.
Share-Based Compensation
3.6. Financial Statement Modeling
c. Valuation Considerations with Share-based Compensation


[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.

Share-Based Compensation

3.6. Financial Statement Modeling

c. Valuation Considerations with Share-based Compensation

We need to modify the discounted cash flow models to account for the effect of:

Dilution from outstanding but unvested share-based awards

- Unvested grants can be accounted for by using the diluted number of shares outstanding as reported
- *More conservative approach:* Add the gross number of antidilutive shares

Dilution from future share-based awards

Can be estimated by:

- Discounting the estimated value of equity by a dilution factor
- Estimating an increase in the number of shares outstanding

[LOS 2.c] *Explain* how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

3.
Share-Based Compensation

3.6. Financial Statement Modeling

c. Valuation Considerations with Share-based Compensation

Share-based compensation is non-cash

Higher non-cash compensation component will
report higher free cash flow

Relative valuation using FCF-related ratios would
be misleading

[LOS 2.d] *Explain* how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.1. *Types of Post-employment Benefits*

Classification

Defined contribution
plan (DC)

Defined benefit plan
(DB)

Other post-
employment benefits
(OPEB)

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4. Post-employment Benefits

4.1. Types of Post-employment Benefits

Defined contribution plan

Based on:

- Years of service
- Age
- Compensation



Employer

Agreed-upon
contribution

**Defined
contribution
(DC) plan**

Based on:

- Percentage of salary



Employee

Contribution

No promise
regarding the
future value of
the plan assets



- Select
- Bear the risk



Investment

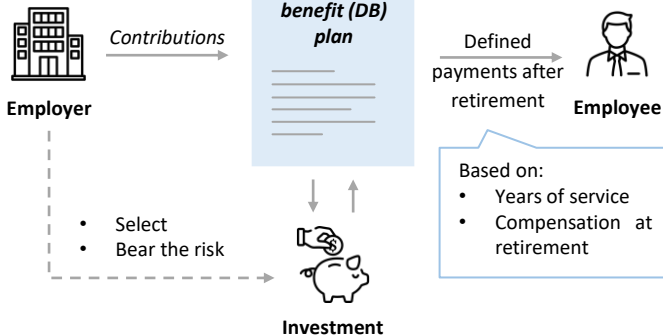
[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.1. Types of Post-employment Benefits

Defined benefit plan



[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.1. Types of Post-employment Benefits

	Defined contribution plan	Defined benefit plan
Contributing party	Employer and Employee	Employer
Amount of future benefit	Not guaranteed, based on plan asset investment performance	Fixed formula
Investment decision-maker	Employee	Employer
Risk-bearer	Employee	Employer
Obligation of sponsoring company	Defined in each period, no additional future obligation	Amount of future obligation, based on formula, must be estimated in current period
Prefunding	Not applicable	Contributing funds to a pension trust

[LOS 2.d] *Explain* how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.1. *Types of Post-employment Benefits*

Other post-employment benefits

- OPEB refer to *Defined benefit plans* that pay non-monetary benefits, such as life insurance and medical care for retirees.
- Companies are often not required by regulations to prefund OPEB plans
→ Make benefit payments as they arise (i.e., pay-as-you-go).
- OPEB obligates the employer to pay benefits in the future and thus exposes them to investment and actuarial risks.

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.2. Financial reporting for DC Plans

Substantially the same as for short-term benefits.

Employer's plan contribution

Income statement

*Functional costs in relevant
Operating expense category*

Balance sheet

*Accrued compensation
liability*

Statement of cash flows

*Cash outflow in operating
activities*

- The DC plan is a separate legal entity with its own financial statements.
- Plan assets, liabilities, and transactions such as withdrawals to employees are not recognized on the employer's financial statements.

[LOS 2.d] *Explain* how post-employment benefits affect the financial statements

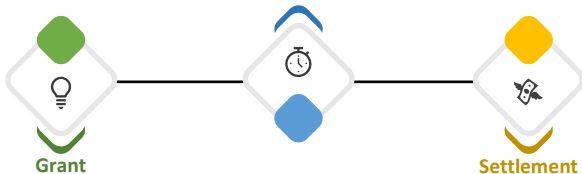
4.

Post-employment Benefits

4.2. Financial reporting for DC Plans

Vesting

- Recognize plan contribution as compensation expense and accrued compensation liability
- Adjust or reverse entries if needed for changes in estimates



Estimate undiscounted value of plan contribution for the period

- Employer makes contribution to the plan
- Accrued compensation liability is derecognized

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.2. Financial reporting for DC Plans



Illustration: Employer's Accounting for DC Post-Employment Benefits

Company makes contributions equal to 5% of an employee's salary to a DC plan every two weeks. The annual salary for an employee in the legal department is SGD 82,200. The first payment date is in two weeks, on 14 January.

	Grant 1 January	Vesting 1 - 14 January	Settlement 14 January
Income statement	No financial statement impact.	SG&A + 158	No impact
Balance sheet	Plan contribution is estimated at SGD 158	Accrued compensation +158	Accrued compensation (158)
Statement of CF		No impact	Cash flows from operations (158)

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

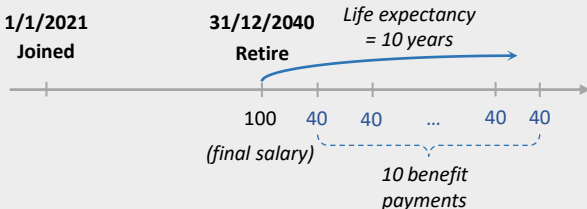
Post-employment Benefits

4.3. Financial reporting for DB Plans

- Employees earn retirement benefits through service.
- Size of benefit payments depends on years of service, final year salary. The aggregate benefit depends on their lifespan after retirement.

Example: An employee might earn a retirement benefit of 2% of a final salary for each year of service. This employee has had 20 years of service, her final salary is \$100,000.

$$\begin{aligned} \text{Benefit payment} &= \text{Final salary} \times 2\% \times \text{Years of service} \\ &= \$100,000 \times 2\% \times 20 = \$40,000 \text{ per year} \end{aligned}$$



[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

Same underlying principle as the other forms: Recognize the fair value of compensation in the period that employees perform services.

DB plans accounting requires some modifications.

a. Balance sheet

Both IFRS and US GAAP require a DB plan's **funded status** to be reported on the balance sheet.

Funded status

Funded status = Fair value of plan assets – Pension obligation

Fair value of plan assets

Assets held by the plan exclusively for paying benefits, measured at the price that would be received in an orderly sale, or quoted market price if available.

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

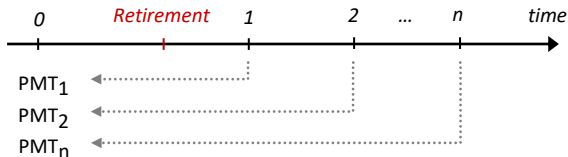
a. Balance sheet

Funded status

Funded status = Fair value of plan assets – Pension obligation

Pension obligation

The present value of expected future payments required to settle the obligation resulting from employee service in the current and prior periods.



Pension obligation

- The discount rate is the yield on investment grade corporate bonds.
- Actuarial assumptions required: Salary growth rates, Retirement dates, Mortality.

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

a. Balance sheet

Funded status

Funded status = Fair value of plan assets – Pension obligation

Funded status

Plan

Treatment

Negative

Underfunded plan

Reported on the balance sheet as a net pension liability

Positive

Overfunded plan

Reported on the balance sheet as a net pension asset

Notes: Different plans' funded statuses cannot be netted (e.g., an overfunded plan cannot be netted against an underfunded plan).

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4. Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement

The company recognizes pension expense each period for the cost of providing benefits.

The Pension expense differs between US GAAP and IFRS.

Under IFRS

Service cost

Net interest
expense/income

Remeasurement

Under US GAAP

Current service cost

Interest cost

Expected return on
plan asset

Amortization of past
service cost

Amortization of net
gains or losses

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement - IFRS

Service cost

Current service cost

Amount by which a company's pension obligation increases as a result of employees' service in the current period.

Past service cost

Incurred if plan amendments are made that change the pension obligation relating to employees' service in prior periods.

Service cost

Grouped
with

Income Statement

COGS (for manufacturing)

R&D expense (for research)

SG&A expense (for management)

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement - IFRS

Net interest expense/income

Definition

Represents the accretion of the pension obligation from the passage of time.

Net interest expense (income)

Net interest expense (income) = Beg. Net pension liability (asset) × Discount rate

Net interest
expense/income

Income Statement

...

Operating income

→ Net interest expense/income

...

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement - IFRS

Remeasurement

Differences between: Actual return on plan assets & Amount assumed in the net interest expense/income calculation

Actuarial gains losses: Changes in the pension obligation from changes in actuarial assumptions

Obligation ↑

Actuarial losses

Obligation ↓

Actuarial gains

Income Statement

Net income

Statement of comprehensive income

Net income

Other comprehensive income (OCI)

Remeasurement



[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement – US GAAP

Current service costs

Similar to IFRS

Interest cost

This represents the passage of time or unwinding the discount, and is a *gross* amount rather than *net*.

Interest cost

$$\text{Interest cost} = \text{Discount rate} \times \text{Beg. Pension obligation}$$

Interest cost

Income Statement

...
Operating income
Interest expense
...

[LOS 2.d] *Explain* how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. *Financial reporting for DB Plans*

b. Income statement – US GAAP

Expected return on plan assets

Expected return on plan assets

$$\text{Expected return on PA} = \text{Expected rate of return} \times \text{Beg. Fair value of PA}$$

- Expected return on plan assets is used as *an offset* for the computation of reported pension expense.
- Expected rate of return should be consistent with historical data and the plan's asset allocation.

[LOS 2.d] *Explain* how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement – US GAAP

Amortization of past service cost

Amendments
in pension
plans

Benefits ↑

Pension
obligation ↑

Past service
cost

Past service
cost

Amendment occurs

**Statement of
comprehensive income**

Net income

Other comprehensive
income (OCI)

Remaining service life

Income statement

...
Amortization of past
service cost

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement – US GAAP

Net gains or losses

Differences
between
Expected and
Actual return on
pension assets

Actuarial gains
losses

Income statement

or

Other comprehensive
income

Amortization of net gains or losses

Income statement
Corridor approach

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement – US GAAP

Corridor approach

At the beginning of the period

Fair value
of PA



Pension
obligation

Fair value of
PA



Pension
obligation

Net cumulative
unrecognized
G/L



10%

Fair value
of PA

Net cumulative
unrecognized
G/L



10%

Pension
obligation

Excess amount is amortized over the expected average remaining working life

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

b. Income statement - Comparison

Component	US GAAP	IFRS
Current service cost	Income statement	Income statement
Past service cost	OCI, amortized over service life in subsequent years	Income statement
Interest cost	Income statement	Income statement*
Expected return	Income statement	Income statement*
Actuarial gains/losses	Amortized portion in income Unamortized in OCI	All in OCI – not amortized

**Under IFRS, the expected rate of return on plan assets equals the discount rate, and net interest expense/income is reported.*

[LOS 2.d] Explain how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.3. Financial reporting for DB Plans

c. Cash flow statement

Plan contributions are recognized on the statement of cash flows, typically in operating activities.

[LOS 2.d] *Explain* how post-employment benefits affect the financial statements

4.

Post-employment Benefits

4.4. Disclosure for Post-Employment Benefit Plans

Under IFRS, plans are required to:

- Disclose the main characteristic of the plan and the risks involved
- Identify and explain the figures in the financial statements arising from them
- Describe the amount, timing, and uncertainty of future cash flows

Analysts should check the disclosed assumptions over time and against other companies for reasonableness.

For example:

Aggressive accounting:

- High discount rate, low salary growth rate, low life expectancy, low inflation rate, ... → Lower pension obligation and lower pension expense.
- Higher expected rate of return on PA → Lower reported pension expense but does not affect pension obligation and fair value of PA.

[LOS 2.e] *Explain* financial modeling and valuation considerations for post employment benefits

4. Post-employment Benefits

4.5. *Financial modeling and Valuation considerations*

a. Financial modeling

DC plan

Typically done implicitly by making operating expense forecasts.

DB plan

Model service cost, net interest expense/income, remeasurements, and the employer's plan contributions in the future periods.

b. Valuation considerations for DB plans

Valuation
must
account for

1

The plan's funded status, either a net liability of the company to plan beneficiaries or net asset if overfunded.

2

Future service costs, which are future increases in the pension obligation from employee service

[LOS 2.e] *Explain* financial modeling and valuation considerations for post employment benefits

4. Post-employment Benefits

4.5. *Financial modeling and Valuation considerations*

b. Valuation considerations for DB plans

1

The plan's funded status, either a net liability of the company to plan beneficiaries or net asset if overfunded.

Analysts must apply an asymmetrical treatment of underfunded and overfunded plans:

- Underfunded plan: Considered debt in an enterprise value calculation and/or bridge from enterprise value to equity value.
- Overfunded plan: Ignored in valuation.

[LOS 2.e] *Explain* financial modeling and valuation considerations for post employment benefits

4.

Post-employment Benefits

4.5. Financial modeling and Valuation considerations

b. Valuation considerations for DB plans

2

Future service costs, which are future increases in the pension obligation from employee service

Future service costs are not included in the plan's funded status. Analyst should:

- Deduct service cost from free cash flow in a discounted cash flow model.
- Net interest expense/income should not be included in the discounted cash flow model.
- Valuation is done on a present value basis, and the present value of an underfunded pension is already considered by deducting the net pension liability from enterprise value.



Publication 505
For 2018

Tax Withholding and Estimated

Capital Gains and Losses

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Exemptions
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MODULE 3: MULTINATIONAL OPERATIONS

LEARNING OUTCOMES

- 3.a.** Compare and contrast presentation in (reporting) currency, functional currency, and local currency
- 3.b.** Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction gains and losses
- 3.c.** Analyze how changes in exchange rates affect the translated sales of the subsidiary and parent company
- 3.d.** Compare the current rate method and the temporal method, evaluate how each affects the parent company's balance sheet and income statement, and determine which method is appropriate in various scenarios
- 3.e.** Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and income statement into the parent company's presentation currency
- 3.f.** Analyze how the current rate method and the temporal method affect financial statements and ratios

LEARNING OUTCOMES

- 3.g.** Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect financial statements and ratios
- 3.h.** Describe how multinational operations affect a company's effective tax rate
- 3.i.** Explain how changes in the components of sales affect the sustainability of sales growth
- 3.j.** Analyze how currency fluctuations potentially affect financial results, given a company's countries of operation

[LOS 3.a] *Compare and contrast* presentation in (reporting) currency, functional currency, and local currency

1.

Introduction to multinational operations

Currently, many companies engage in transactions that cross national borders.

→ The parties to these transactions must agree on the *currency* in which to settle the transaction.

Foreign currency can affect a multinational firm's financial statements in two ways, in both cases, special accounting treatment is required:

1

The multinational firm may engage in business transactions that are denominated in a foreign currency.

2

The multinational firm may invest in subsidiaries that maintain their books and records in a foreign currency.

To prepare consolidated financial statements, a multinational company must translate the foreign currency amounts into the currency in which the company presents its financial statements.

What do the analysts need to understand?

1

The effects of foreign exchange rate fluctuations on the financial statements of a multinational company.

2

How a company's financial statements reflect foreign currency gains and losses, whether realized or not.

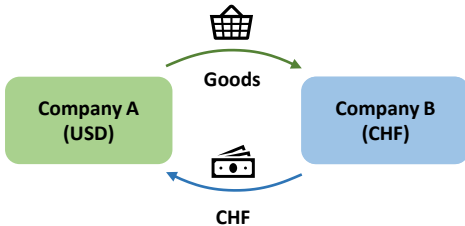
[LOS 3.a] *Compare and contrast* presentation in (reporting) currency, functional currency, and local currency

1.

Introduction to multinational operations

In the illustration below, the exporter (Company A) receives payment in foreign currency (CHF) from the importer (Company B)
→ **The transaction is denominated in CHF.**

Foreign currency transactions



To account for the sales revenue and account receivable in CHF, Company A must translate the CHF amounts into USD using appropriate exchange rates.

Company B does not have a foreign currency transaction and no translation is necessary.

[LOS 3.a] *Compare and contrast presentation in (reporting) currency, functional currency, and local currency*

2.

The currencies involved in multinational accounting

*We need to define **three** type of currencies that are involved in multinational accounting:*

1

The **presentation (reporting) currency**: the currency in which the parent company prepares its financial statements.

In most cases, a company's presentation currency will be the currency of the country where the company is located.

2

The **functional currency**: the currency of the primary economic environment in which an entity operates.

- Normally, the functional currency is the currency in which an entity primarily generates and expends cash.
- In most cases, an organization's functional currency will be the same as its presentation currency.

3

The **local currency**: the currency of the country being referred to.

Because the local currency generally is an entity's functional currency, a multinational corporation with subsidiaries in a variety of different countries is likely to have a variety of different functional currencies.

[LOS 3.a] *Compare and contrast presentation in (reporting) currency, functional currency, and local currency*

2.

The currencies involved in multinational accounting

By definition, for accounting purposes:

- *A foreign currency is any currency other than a company's functional currency.*
- *Foreign currency transactions* are those denominated in a currency other than the company's functional currency.

Foreign currency transactions occur when a company:

1

Makes an import purchase or an export sale that is denominated in a foreign currency.

2

Borrows or lends funds where the amount to be repaid or received is denominated in a foreign currency.

→ *In each of these cases, the company has an asset or a liability denominated in a foreign currency.*

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

1.

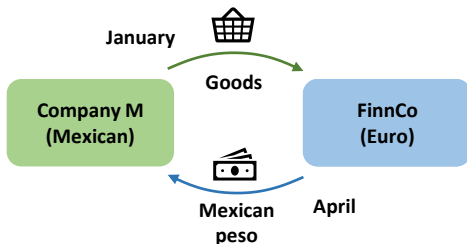
Exposure to foreign exchange risk



Illustration

Assume that FinnCo, a Finland-based company, imports goods from Mexico in January under 45-day credit terms, and the purchase is denominated in Mexican pesos.

By deferring payment until April, FinnCo runs the risk that from the date the purchase is made until the date of payment, the value of the Mexican peso might increase relative to the euro. FinnCo would then need to spend more euro to settle its Mexican peso account payable.



→ In this case, FinnCo is said to have an **exposure to foreign exchange risk**. Specifically, FinnCo has a foreign currency **transaction exposure**.

[LOS 3.b] *Describe* foreign currency transaction exposure, including accounting for and *disclosures* about foreign currency transaction...

1.

Exposure to foreign exchange risk

Transaction exposure related to imports and exports can be summarized as follows:

1 Import purchase

The importer is exposed to the risk that from the purchase date until the payment date the foreign currency might increase in value.

→ Increasing the amount of functional currency to settle the account payable.

2 Export sale

The exporter is exposed to the risk that from the purchase date until the payment date, the foreign currency might decrease in value.

→ Decreasing the amount of functional currency when it is received.

The major issue in accounting for *foreign currency transactions* is how to account for the *foreign currency risk*.

Both IFRS and US GAAP require the change in the value of the foreign currency asset or liability resulting from a foreign currency transaction to be treated as a **gain or loss reported on the income statement**.

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and *disclosures* about foreign currency transaction...

2.

Accounting with settlement before balance sheet date

The basic principle:

All transactions are recorded at the spot rate on the date of the transaction.

→ The *foreign currency risk* on transactions, therefore, arises ***only when the transaction date and the payment date are different.***



Illustration

FinnCo purchases goods from its Mexican supplier on 1 November 20X1; the purchase price is 100,000 Mexican pesos. Credit terms allow payment in 45 days, and FinnCo makes payment of 100,000 pesos on 15 December 20X1. FinnCo's functional and presentation currency is the euro.

Spot exchange rates between the euro (EUR) and Mexican peso (MXN) are as follows:

1 November 20X1

MXN1 = EUR0.0684

15 December 20X1

MXN1 = EUR0.0703

FinnCo's fiscal year end is 31 December. How will FinnCo account for this foreign currency transaction, and what effect will it have on the 20X1 financial statements?

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and *disclosures* about foreign currency transaction...

2.

Accounting with settlement before balance sheet date



Illustration

1 November 20X1

MXN1 = EURO.0684

15 December 20X1

MXN1 = EURO.0703

We can see that from the purchase date (1 November 20X1) until the payment date (15 December 20X1), the foreign currency (Mexican peso) increases in value.

→ The importer (FinnCo) is exposed to the risk (**loss**) by increasing the amount of functional currency to settle the account payable.

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

2.
Accounting with settlement before balance sheet date

Illustration
Income statement
Revenues and gains
Expenses and losses

Foreign
exchange
loss =

Total

Total

Balance Sheet
Assets
Liabilities + Stockholders' Equity

Cash =

Inventory =

Retained
earnings

Total

Total

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

2.

Accounting with settlement before balance sheet date



Illustration

Income statement

Revenues and gains
Expenses and losses

 Foreign
exchange
loss

$$= 100,000 \times (0.0684 - 0.0703)$$

$$= - \text{EUR } 190$$
Total
0
Total
- EUR 190

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

2.
Accounting with settlement before balance sheet date

Illustration

Balance Sheet

Assets		Liabilities + Stockholders' Equity	
Cash	$= -100,000 \times 0.0703$ $= - \text{EUR } 7,030$	Retained earnings	- EUR 190
Inventory	$= 100,000 \times 0.0684$ $= \text{EUR } 6,840$		
Total	- EUR 190	Total	- EUR 190

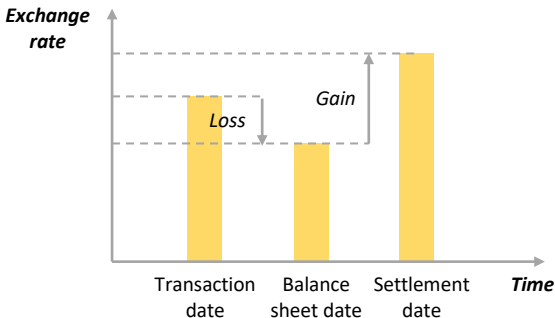
[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

3.

Accounting with intervening balance sheet dates

If the balance sheet date occurs before the transaction is settled, gains and losses on foreign currency transactions are recognized.

- The balance sheet amounts are adjusted based on the exchange rate on the balance sheet date, and an unrealized gain or loss is recognized in the income statement.
- Once the transaction is settled, additional gain or loss is recognized if the exchange rate changes after the balance sheet date.



[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

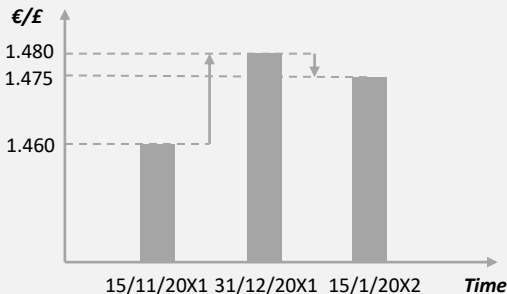
3.

Accounting with intervening balance sheet dates



Illustration

FinnCo sells goods to a customer in the United Kingdom for £10,000 on 15 November 20X1, with payment to be received in British pounds on 15 January 20X2. FinnCo's functional and presentation currency is the euro. Spot exchange rates between the euro (€) and British pound (£) are as follows:



FinnCo's fiscal year end is 31 December. How will FinnCo account for this foreign currency transaction, and what effect will it have on the 20X1 and 20X2 financial statements?

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

3.
Accounting with intervening balance sheet dates

Illustration

The euro value of the British pound account receivable at each of the three relevant dates is determined as follows:

Account Receivable (£10,000)

Date	€/£ Exchange Rate	Euro Value	Change in Euro Value
15 Nov 20X1	€1.460	=	=
31 Dec 20X1	€1.480	=	=
15 Jan 20X2	€1.475	=	=

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

3.

Accounting with intervening balance sheet dates



Illustration

The euro value of the British pound account receivable at each of the three relevant dates is determined as follows:

Account Receivable (£10,000)

Date	€/£ Exchange Rate	Euro Value	Change in Euro Value
15 Nov 20X1	€1.460	14,600	N/A
31 Dec 20X1	€1.480	14,800	+200
15 Jan 20X2	€1.475	14,750	-50

From the transaction date to the settlement date, the British pound has increased in value by €0.015 ($€1.475 - €1.460$), which generates a realized foreign currency transaction gain of €150. A gain of €200 was recognized in 20X1 and a loss of €50 is recognized in 20X2.

Over the two-month period, the net gain recognized in the financial statements is equal to the actual realized gain on the foreign currency transaction.

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction...

3.

Accounting with intervening balance sheet dates

Whether a change in exchange rate results in a foreign currency transaction gain or loss (measured in local currency) depends on:

- The nature of the exposure to foreign exchange risk (asset or liability)
- The direction of change in the value of the foreign currency (strengthens or weakens).

Foreign currency

Transaction	Type of Exposure	Strengthens	Weakens
Export sale	Asset (account receivable)	Gain	Loss
Import purchase	Liability (account payable)	Loss	Gain

[LOS 3.b] *Describe* foreign currency transaction exposure, including accounting for and *disclosures* about foreign currency transaction...

4.

Analytical issues

Issue 1: *Accounting standards do not provide guidance on the placement of foreign currency transaction gains and losses on the income statement.*

The two most common treatments are either

- As a component of other operating income/expense or
- As a component of non-operating income/ expense, in some cases as a part of net financing cost.

Consequence: Companies are free to choose among the alternatives, which would *distort* the direct comparison of operating profit and operating profit margins between those companies.

Issue 2: *Unrealized foreign currency transaction gains and losses are included in net income when the balance sheet date falls between the transaction and settlement dates.*

Consequence: It may not accurately reflect the company's ultimate net gain or loss, which may vary dramatically because of the possibility for changes in trend and volatility of currency prices.

[LOS 3.b] *Describe* foreign currency transaction exposure, including accounting for and *disclosures* about foreign currency transaction...

5.

Disclosures gains and losses

IFRS

US GAAP

Require disclosure of “the amount of exchange differences recognized in profit or loss”

Require disclosure of “the aggregate transaction gain or loss included in determining net income for the period”

Neither standard specifically requires disclosure of the line item in which these gains and losses are located.

Disclosures related to foreign currency are commonly found both in the Management Discussion & Analysis (MD&A) and the Notes to Financial Statements sections of an annual report.

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Scenario: Many companies have operations in foreign countries. Most operations located in foreign countries keep their accounting records and prepare financial statements in the local currency.

Under IFRS and US GAAP, to prepare worldwide consolidated statements, parent companies must *translate* the foreign currency financial statements of their foreign subsidiaries into the parent company's presentation currency.

*In translating foreign currency financial statements into the parent company's presentation currency, **two** questions must be addressed:*

- 1 What is the appropriate exchange rate to use in translating each financial statement item?
- 2 How should the translation adjustment that inherently arises from the translation process be reflected in the consolidated financial statements?

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1.

Translation conceptual issues



Illustration

Spanco is a hypothetical Spain-based company that uses the euro as its presentation currency. Spanco establishes a wholly owned subsidiary, Amerco, in the United States on 31 December 20X1 by investing €10,000 when the exchange rate between the euro and the US dollar is €1 = US\$1. The equity investment of €10,000 is physically converted into US\$10,000 to begin operations. In addition, Amerco borrows US\$5,000 from local banks on 31 December 20X1. Amerco purchases inventory that costs US\$12,000 on 31 December 20X1 and retains US\$3,000 in cash.

Amerco's balance sheet at 31 December 20X1 thus appears as follows:

Cash	\$3,000	Notes payable	\$5,000
Inventory	\$12,000	Common stock	\$10,000
Total	\$15,000	Total	\$15,000

To prepare a consolidated balance sheet in euro as of 31 December 20X1, Spanco must translate all of the US dollar balances on Amerco's balance sheet at the **€1 = US\$1** exchange rate.

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Illustration

The translation worksheet as of 31 December 20X1 is as follows:

	USD	Exchange Rate (€)	EUR
Cash	\$3,000	1.00	€3,000
Inventory	\$12,000	1.00	€12,000
Total	\$15,000		€15,000
Notes payable	\$5,000	1.00	€5,000
Common stock	\$10,000	1.00	€10,000
Total	\$15,000		€15,000

→ By translating each US dollar balance at the same exchange rate (€1.00), Amerco's translated balance sheet in euro reflects an equal amount of total assets and total liabilities plus equity and remains in balance.

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1.

Translation conceptual issues



Illustration

During the first quarter of 20X2, Amerco engages in no transactions. During that period, however, the US dollar weakens against the euro such that the exchange rate on 31 March 20X2 is €0.80 = US\$1.

→ Let's consider the first question: To prepare a consolidated balance sheet at the end of the first quarter of 20X2, ***what is the appropriate exchange rate to use in translating each financial statement item: €0.80 or €1.00?***

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

 Illustration

Answer:

- The original investment made by Spanco of €10,000 is a historical fact, so the company wants to translate Amerco's ***common stock*** in such a way that it continues to reflect this amount by using the ***historical exchange rate of €1 = US\$1***.
- There are two approaches for translating the foreign subsidiary's ***assets and liabilities*** are as follows:

1 All assets and liabilities are translated at the ***current exchange rate*** (the spot exchange rate on the balance sheet date).

2 Only ***monetary assets and liabilities*** are translated at the ***current exchange rate***; ***non-monetary assets and liabilities*** are translated at ***historical exchange rates*** (the exchange rates that existed when the assets and liabilities were acquired).

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Illustration

- All assets and liabilities are translated at the **current exchange rate**

The translation worksheet on 31 March 20X2, in which all assets and liabilities are translated at the current exchange rate (€0.80), is as follows:

	US Dollar	Exchange Rate (€)	Euro	Change in Euro Value since 31 Dec 20X1
Cash	\$3,000	0.80	= ...	=
Inventory	\$12,000	0.80	= ...	=
Total	\$15,000		= ...	=
Notes payable	\$5,000	0.80	= ...	=
Common stock	\$10,000	1.00	= ...	=
Subtotal	\$15,000		= ...	=
Translation adjustment			= ...	=
Total			= ...	=

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Illustration

- All assets and liabilities are translated at the **current exchange rate**

The translation worksheet on 31 March 20X2, in which all assets and liabilities are translated at the current exchange rate (€0.80), is as follows:

	US Dollar	Exchange Rate (€)	Euro	Change in Euro Value since 31 Dec 20X1
Cash	\$3,000	0.80	€2,400	- €600
Inventory	\$12,000	0.80	€9,600	- €2,400
Total	\$15,000		€12,000	- €3,000
Notes payable	\$5,000	0.80	€4,000	- €1,000
Common stock	\$10,000	1.00	€10,000	0
Subtotal	\$15,000		€14,000	- €1,000
Translation adjustment			(€2,000)	- €2,000
Total			€12,000	- €3,000

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Illustration

1 All assets and liabilities are translated at the **current exchange rate**

We can see that, to keep the euro translated balance sheet in balance, a *negative translation adjustment* of €2,000 is created and included in stockholders' equity on the consolidated balance sheet.

The negative translation adjustment (net translation loss) does not result in a cash outflow of €2,000 for Spanco and thus is **unrealized**.

It could be **realized**, however, if Spanco were to sell Amerco at its book value of US\$10,000 = €8,000.

€8,000 cash
inflow
(proceeds
from the sale)

—

€10,000 cash
originally
invested

=

€2,000
realized loss

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Illustration

- 2 Only monetary assets and monetary liabilities are translated at the ***current exchange rate***

The worksheet at 31 March 20X2, in which only monetary assets and liabilities are translated at the current exchange rate (€0.80), is as follows:

	US Dollar	Exchange Rate (€)	Euro	Change in Euro Value since 31 Dec 20X1
Cash	\$3,000	0.80	= ...	=
Inventory	\$12,000	1.00	= ...	=
Total	\$15,000		= ...	=
Notes payable	\$5,000	0.80	= ...	=
Common stock	\$10,000	1.00	= ...	=
Subtotal	\$15,000		= ...	=
Translation adjustment			= ...	=
Total			= ...	=

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1. Translation conceptual issues

Illustration

- 2 Only monetary assets and monetary liabilities are translated at the ***current exchange rate***

The worksheet at 31 March 20X2, in which only monetary assets and liabilities are translated at the current exchange rate (€0.80), is as follows:

	US Dollar	Exchange Rate (€)	Euro	Change in Euro Value since 31 Dec 20X1
Cash	\$3,000	0.80	€2,400	- €600
Inventory	\$12,000	1.00	€12,000	0
Total	\$15,000		€14,400	- €600
Notes payable	\$5,000	0.80	€4,000	- €1,000
Common stock	\$10,000	1.00	€10,000	0
Subtotal	\$15,000		€14,000	- €1,000
Translation adjustment			€400	€400
Total			€14,400	- €600

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

1.

Translation conceptual issues



Illustration

2

Only monetary assets and monetary liabilities are translated at the ***current exchange rate***

We can see that, to keep the balance sheet in balance, a *positive translation adjustment* of €400 must be included in stockholders' equity. The translation adjustment reflects the net translation gain or loss related to monetary items only.

The positive translation adjustment (net translation gain) also is ***unrealized***. The gain could be ***realized***, however, if:

- The subsidiary uses its cash (US\$3,000) to pay as much of its liabilities as possible, and
- The parent sends enough euro to the subsidiary to pay its remaining liabilities (US\$5,000 – US\$3,000 = US\$2,000).

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

2. Balance Sheet Exposure

The assets and liabilities translated in terms of the parent company's presentation currency at:

- **Current exchange rate** → **be exposed** to translation adjustment.
 - Exposed assets > Exposed liabilities → **Net asset balance sheet exposure**
 - Exposed assets < Exposed liabilities → **Net liability balance sheet exposure**
- **Historical exchange rate** → **be not exposed** to translation adjustment.

*The sign (positive or negative) of the current period's translation adjustment is a function of **two** factors:*

- 1** The nature of the balance sheet exposure (asset or liability)
- 2** The direction of change in the exchange rate (strengthens or weakens)

[LOS 3.c] *Analyze* how changes in exchange rates affect the translated sales of the subsidiary and parent company

2. Balance Sheet Exposure

Balance Sheet Exposure	Foreign Currency (FC)	
	Strengthens	Weakens
Net asset	Positive translation adjustment	Negative translation adjustment
Net liability	Negative translation adjustment	Positive translation adjustment

After the initial period of operations, a *cumulative translation adjustment* is required to keep the translated balance sheet in balance.

The *cumulative translation adjustment* will be the *sum of the translation adjustments* that arise over successive accounting periods.

[LOS 3.d] Compare the current rate method and the temporal method, evaluate how each affects the parent company's...

1.

Translation methods

There are **two** methods used to remeasure or translate the financial statements of a foreign subsidiary to the parent's presentation (reporting) currency:

Translation

Converting the functional currency into the parent's presentation (reporting) currency using the **current rate method (all-current method)**

Remeasurement

Converting the local currency into functional currency using the **temporal method**



Which method is appropriate for an individual foreign entity

It depends on that entity's **functional currency**. Since the functional currency is chosen by management, it may not be completely objective.



Which factors should be considered in determining the functional currency

Continue next slide →

[LOS 3.d] Compare the current rate method and the temporal method, evaluate how each affects the parent company's...

1. Translation methods

*In accordance with IASB, the following factors should be considered in determining an entity's **functional currency**:*

- 1 The currency that mainly influences sales prices for goods and services
- 2 The currency of the country whose competitive forces and regulations mainly determine the sales price of its goods and services
- 3 The currency that mainly influences labour, material, and other costs of providing goods and services
- 4 The currency in which funds from financing activities are generated
- 5 The currency in which receipts from operating activities are usually retained

The FASB provides similar guidance.

[LOS 3.d] *Compare* the current rate method and the temporal method, *evaluate* how each affects the parent company's...

1.

Translation methods

Additional factors to consider in determining whether the foreign entity's functional currency is the same as the parent's functional currency are:

6

Whether the activities of the foreign operation are an extension of the parent's or are carried out with a significant amount of autonomy

7

Whether transactions with the parent are a large or a small proportion of the foreign entity's activities

8

Whether cash flows generated by the foreign operation directly affect the cash flow of the parent and are available to be remitted to the parent

9

Whether operating cash flows generated by the foreign operation are sufficient to service existing and normally expected debt or whether the foreign entity will need funds from the parent to service its debt

[LOS 3.d] Compare the current rate method and the temporal method, evaluate how each affects the parent company's...

1.

Translation methods

In summary, we can use the following to determine the appropriate translation method:

Company's characteristics

A subsidiary is self-contained, independent whose operating, investing, and financing activities are decentralized from the parent

A subsidiary is well integrated with the parent (i.e., the parent makes the operating, investing, and financing decisions)

Functional currency

Differ from the parent's presentation currency

Same as the parent's presentation currency

Method

The **current rate method** is used to translate the foreign currency financial statements

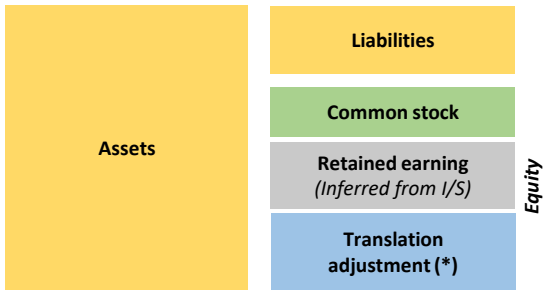
The **temporal method** is used to remeasure the foreign currency financial statements

[LOS 3.d] Compare the current rate method and the temporal method, evaluate how each affects the parent company's...

2.

Current rate method

*The foreign entity's foreign currency financial statements are translated into the parent's presentation currency under the **current rate method** as follows:*



Are translated at the **current exchange rate** at the balance sheet date

(*) The **plug figure** so that the balance sheet is balanced

Are translated at **historical exchange rates**

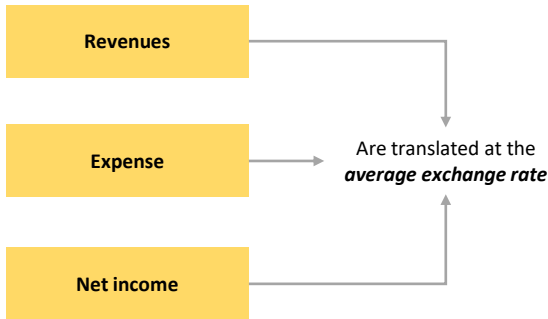
Translation treatments

[LOS 3.d] Compare the current rate method and the temporal method, *evaluate* how each affects the parent company's...

2.

Current rate method

*The foreign entity's foreign currency financial statements are translated into the parent's presentation currency under the **current rate method** as follows:*



[LOS 3.d] Compare the current rate method and the temporal method, evaluate how each affects the parent company's...

3.
Temporal method

Assets	Monetary assets	Monetary liabilities	Liabilities
	Non-monetary assets measured at historical cost	Non-monetary liabilities measured at historical cost	
	Non-monetary assets measured at current value	Non-monetary liabilities measured at current value	Equity
		Common stock	
		Retained earning (*)	

*Are translated at the **current exchange rate** at the balance sheet date*

*Are translated at **historical exchange rates***

() The plug figure so that the balance sheet is balanced*

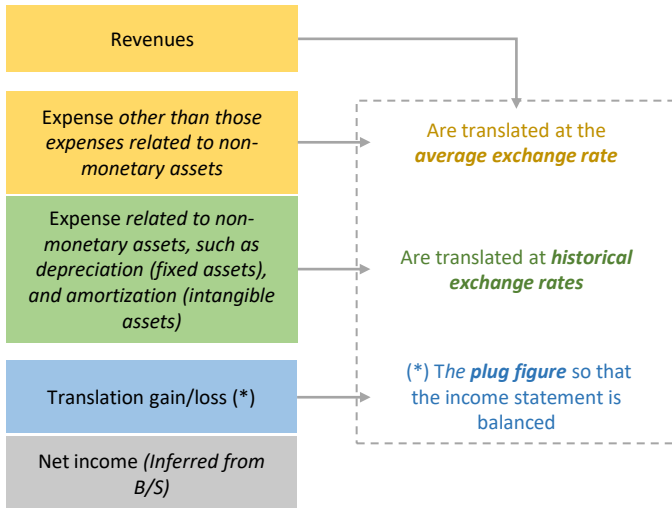
Remeasurement treatments

- **Monetary items** are cash and receivables (payables) that are to be received (paid) in a fixed number of currency units.
- **Non-monetary assets** include inventory, fixed assets, and intangibles, and **non-monetary liabilities** include deferred revenue.

[LOS 3.d] Compare the current rate method and the temporal method, *evaluate* how each affects the parent company's...

3.

Temporal method



[LOS 3.d] Compare the current rate method and the temporal method, evaluate how each affects the parent company's...

4.

Summary of temporal method and current rate method

	Temporal method	Current rate method
Monetary assets and liabilities	Current rate	Current rate
Non-monetary assets and liabilities	Historical rates	Current rate
Common stock	Historical rates	Historical rates
Equity (taken as a whole)	Mixed	Current rate
Revenues and SG&A	Average rate	Average rate
COGS	Average rate	Average rate
Depreciation and amortization	Historical rates	Average rate
Net income	Mixed rate	Average rate
Exposure	Net monetary assets	Net assets
Exchange rate gain and loss	Income statement	Equity

[LOS 3.e] *Calculate* the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

Assume that Interco is a Europe-based company that has the euro as its presentation currency.

On 1 January 20X1, Interco establishes a wholly owned subsidiary in Canada, Canadaco. In addition to Interco making an equity investment in Canadaco, a long-term note payable to a Canadian bank was negotiated to purchase property and equipment.

The subsidiary begins operations with the following balance sheet in Canadian dollars (C\$):

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

Assets

Cash	C\$1,500,000
------	--------------

Property and equipment	C\$3,000,000
------------------------	--------------

Total	C\$4,500,000
--------------	---------------------

Liabilities and Equity

Long-term note payable	C\$3,000,000
------------------------	--------------

Capital stock	C\$1,500,000
---------------	--------------

Total	C\$4,500,000
--------------	---------------------

Canadaco purchases and sells inventory in 20X1, generating net income of C\$1,180,000, out of which C\$350,000 in dividends are paid.

[LOS 3.e] *Calculate* the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.
Illustration of translation methods

Illustration

The company's income statement and statement of retained earnings for 20X1 is as follows:

Sales	C\$12,000,000
COGS	(9,000,000)
Selling expenses	(750,000)
Depreciation expense	(300,000)
Interest expense	(270,000)
Income tax	(500,000)
Net income	C\$1,180,000
Less: Dividends, 1 Dec 20X1	(350,000)
Retained earnings, 31 Dec 20X1	C\$830,000

[LOS 3.e] *Calculate* the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

The company's balance sheet at 31 December 20X1 is as follows:

Assets	
Cash	C\$980,000
Accounts receivable	900,000
Inventory	1,200,000
Total current assets	C\$3,080,000
Property and equipment	3,000,000
Less: accumulated depreciation	(300,000)
Total	C\$5,780,000

[LOS 3.e] *Calculate* the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

The company's balance sheet at 31 December 20X1 is as follows:

Liabilities and Equity

Accounts payable	C\$450,000
------------------	------------

Total current liabilities	450,000
---------------------------	---------

Long-term notes payable	3,000,000
-------------------------	-----------

Total liabilities	C\$3,450,000
-------------------	--------------

Capital stock	1,500,000
---------------	-----------

Retained earnings	830,000
-------------------	---------

Total	C\$5,780,000
--------------	---------------------

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

Note that inventory here is measured at historical cost on a FIFO basis.

To translate Canadaco's Canadian dollar financial statements into euro for consolidation purposes, the following exchange rate information was gathered:

Date	€ per C\$
1 January 20X1	0.70
Average, 20X1	0.75
Weighted-average rate when inventory was acquired	0.74
1 December 20X1 when dividends were declared	0.78
31 December 20X1	0.80

→ During 20X1, the Canadian dollar strengthened steadily against the euro from an exchange rate of €0.70 at the beginning of the year to €0.80 at year-end.

[LOS 3.e] *Calculate* the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

We will consider Canadaco's translated financial statements under each of the two translation methods:

1 Using the ***current rate method***

→ In this case, we assume first that Canadaco's functional currency is the Canadian dollar.

The Canadaco's translated *Income Statement and Statement of Retained Earnings, 20X1* is as follows:

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods

	C\$	Exch. Rate		€
Sales	12,000,000		=	
COGS	(9,000,000)		=	
Selling expenses	(750,000)		=	
Dep expense	(300,000)		=	
Interest expense	(270,000)		=	
Income tax	(500,000)		=	
Income before trans. gain (loss)	1,180,000		=	
Translation gain (loss)	N/A			N/A
Net income	1,180,000		=	
Less: Dividends, 1 Dec 20X1	(350,000)		=	
Retained earnings, 31 Dec 20X1	830,000		=	

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

The Canadaco's translated *balance sheet* at 31 December 20X1 is as follows:

	C\$	Exch. Rate		€
Assets				
Cash	980,000		=	
Accounts receivable	900,000		=	
Inventory	1,200,000		=	
Total current assets	3,080,000		=	
Property and equipment	3,000,000		=	
Less: accumulated depreciation	(300,000)		=	
Total assets	5,780,000		=	

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

The Canadaco's translated *balance sheet* at 31 December 20X1 is as follows:

	C\$	Exch. Rate	€
Assets			
Cash	980,000	0.80 C	784,000
Accounts receivable	900,000	0.80 C	720,000
Inventory	1,200,000	0.80 C	960,000
Total current assets	3,080,000		2,464,000
Property and equipment	3,000,000	0.80 C	2,400,000
Less: accumulated depreciation	(300,000)	0.80 C	(240,000)
Total assets	5,780,000		4,624,000

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

The Canadaco's translated *balance sheet* at 31 December 20X1 is as follows:

	C\$	Exch. Rate	€
Liabilities and Equity			
Accounts payable	450,000		=
Total current liabilities	450,000		=
Long-term notes payable	3,000,000		=
Total liabilities	3,450,000		=
Capital stock	1,500,000		=
Retained earnings	830,000	from I/S	=
Translation adjustment	N/A	to balance	=
Total	5,780,000		=

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

The Canadaco's translated *balance sheet at 31 December 20X1* is as follows:

	C\$	Exch. Rate	€
Liabilities and Equity			
Accounts payable	450,000	0.80 C	360,000
Total current liabilities	450,000		360,000
Long-term notes payable	3,000,000	0.80 C	2,400,000
Total liabilities	3,450,000		2,760,000
Capital stock	1,500,000	0.70 H	1,050,000
Retained earnings	830,000	from I/S	612,000
Translation adjustment	N/A	to balance	202,000
Total	5,780,000		4,624,000

[LOS 3.e] Calculate the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

We will consider Canadaco's translated financial statements under each of the two translation methods:

2

Using the ***temporal method***

→ In this case, we assume that Canadaco's functional currency is the euro (the parent's presentation currency).

Note that with the temporal method, the inventory and COGS is translated by the weighted-average rate when inventory was acquired.

The Canadaco's translated *balance sheet at 31 December 20X1 and Income Statement and Statement of Retained Earnings, 20X1* are as follows:

[LOS 3.e] Calculate the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.
Illustration of translation methods

Illustration

	C\$	Exch. Rate		€
Assets				
Cash	980,000		=	
Accounts receivable	900,000		=	
Inventory	1,200,000		=	
Total current assets	3,080,000		=	
Property and equipment	3,000,000		=	
Less: accumulated depreciation	(300,000)		=	
Total assets	5,780,000		=	

[LOS 3.e] Calculate the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.
Illustration of translation methods

Illustration

	C\$	Exch. Rate	€
Assets			
Cash	980,000	0.80 C	784,000
Accounts receivable	900,000	0.80 C	720,000
Inventory	1,200,000	0.74 H	888,000
Total current assets	3,080,000		2,392,000
Property and equipment	3,000,000	0.70 H	2,100,000
Less: accumulated depreciation	(300,000)	0.70 H	(210,000)
Total assets	5,780,000		4,282,000

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

	C\$	Exch. Rate	€
Liabilities and Equity			
Accounts payable	450,000		=
Total current liabilities	450,000		=
Long-term notes payable	3,000,000		=
Total liabilities	3,450,000		=
Capital stock	1,500,000	...	=
Retained earnings	830,000	to balance	=
Translation adjustment	N/A		N/A
Total	5,780,000		

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods



Illustration

	C\$	Exch. Rate	€
Liabilities and Equity			
Accounts payable	450,000	0.80 C	360,000
Total current liabilities	450,000		360,000
Long-term notes payable	3,000,000	0.80 C	2,400,000
Total liabilities	3,450,000		2,760,000
Capital stock	1,500,000	0.70 H	1,050,000
Retained earnings	830,000	to balance	472,000
Translation adjustment	N/A		N/A
Total	5,780,000		4,282,000

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods

	C\$	Exch. Rate		€
Sales	12,000,000		=	
COGS	(9,000,000)		=	
Selling expenses	(750,000)		=	
Dep expense	(300,000)		=	
Interest expense	(270,000)		=	
Income tax	(500,000)		=	
Income before trans. gain (loss)	1,180,000		=	
Translation gain (loss)	N/A	to balance	=	
Net income	1,180,000		=	
Less: Dividends, 1 Dec 20X1	(350,000)		=	
Retained earnings, 31 Dec 20X1	830,000	from B/S		

[LOS 3.e] Calculate the translation effects and *evaluate* the translation of a subsidiary's balance sheet and...

1.

Illustration of translation methods

	C\$	Exch. Rate	€
Sales	12,000,000	0.75 A	9,000,000
COGS	(9,000,000)	0.74 H	(6,660,000)
Selling expenses	(750,000)	0.75 A	(562,500)
Dep expense	(300,000)	0.70 H	(210,000)
Interest expense	(270,000)	0.75 A	(202,500)
Income tax	(500,000)	0.75 A	(375,000)
Income before trans. gain (loss)	1,180,000		990,000
Translation gain (loss)	N/A	to balance	(245,000)
Net income	1,180,000		745,000
Less: Dividends, 1 Dec 20X1	(350,000)	0.78 H	(273,000)
Retained earnings, 31 Dec 20X1	830,000	from B/S	472,000

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

1.

Translation analytical issues

Firstly, we will consider the effects of the *current rate method* to *pure balance sheet and pure income statement ratios*.

	Pure balance sheet ratios	Pure income statement ratios
Definition	All of the components of the ratio are from the balance sheet	All of the components of the ratio are from the income statement
Example	Current ratio = Current assets / Current liabilities	Interest coverage = EBIT / Interest expense
Effects of the current rate method	Recall: All assets, liabilities and equity taken as a whole are translated at the <i>current exchange rate</i> at the <i>balance sheet date</i> → The ratios are <i>unaffected</i>	Recall: All revenues and expenses are translated at the <i>average exchange rate</i> → The ratios are <i>unaffected</i>

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

1.

Translation analytical issues

Secondly, we will consider the effects of the **current rate method** to **mixed balance sheet/income statement ratios**

Mixed balance sheet/income statement ratios

Definition

The components of the ratio are from both the income statement and balance sheet

Example

Fixed asset turnover ratio = Revenue / Fixed assets

Effects of the current rate method

- We can't make any definitive statements about specific ratios after translation unless we make the assumption that all mixed ratios are calculated using **end-of-period balance sheet figures**.
- With an income statement item in the numerator and an end-of-period balance sheet item in the denominator, the translated mixed ratios will be **larger** than the original ratio if the foreign currency is **depreciating**, and vice versa.

[LOS 3.f] *Analyze* how the current rate method and the temporal method affect financial statements and ratios

1.

Translation analytical issues

*Thirdly, we will compare the results using the **temporal method** and **current rate method**.*

The basic procedure of analyzing the effect on the financial ratios of the choice of accounting method is as follows:

- 1 Determine whether the foreign currency is appreciating or depreciating
- 2 Determine which rate (historical rate, average rate, or current rate) is used to convert the *numerator* under both methods and then compare them
- 3 Determine which rate (historical rate, average rate, or current rate) is used to convert the *denominator* under both methods and then compare them
- 4 Determine whether the ratio will increase, decrease, or stay the same based on the direction of change in the numerator and the denominator

[LOS 3.f] *Analyze* how the current rate method and the temporal method affect financial statements and ratios

1.

Translation analytical issues

Thirdly, we will compare the results using the **temporal method** and **current rate method**.

Example: Let's analyze the fixed asset turnover ratio. Assume the foreign currency is depreciating.

Answer:

Recall that Fixed asset turnover ratio = Revenue / Fixed assets

Temporal method

Current rate method

1 As given, foreign currency is depreciating

2 The numerator (revenue) is converted at the average rate

The numerator (revenue) is converted at the average rate

3 The denominator (fixed assets) is converted at the historical rate (> the current rate)

The denominator (fixed assets) is converted at the current rate (< the historical rate)

4 Turnover will be lower (higher denominator)

Turnover will be higher (lower denominator)

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

1.

Translation analytical issues

This exhibit below summarizes the typical effect that a strengthening or weakening of the foreign currency has on financial statement:

	Temporal Method, Net Monetary Liability Exposure	Temporal Method, Net Monetary Asset Exposure	Current Rate Method
Foreign currency strengthens relative to parent's presentation currency	↑ Revenues ↑ Assets ↑ Liabilities ↓ Net income ↓ Shareholders' equity Translation loss	↑ Revenues ↑ Assets ↑ Liabilities ↑ Net income ↑ Shareholders' equity Translation gain	↑ Revenues ↑ Assets ↑ Liabilities ↑ Net income ↑ Shareholders' equity Positive translation adjustment
Foreign currency weakens relative to parent's presentation currency	↓ Revenues ↓ Assets ↓ Liabilities ↑ Net income ↑ Shareholders' equity Translation gain	↓ Revenues ↓ Assets ↓ Liabilities ↓ Net income ↓ Shareholders' equity Translation loss	↓ Revenues ↓ Assets ↓ Liabilities ↓ Net income ↓ Shareholders' equity Negative translation adjustment

[LOS 3.g] *Analyze* how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

What is a hyperinflationary environment?

US. GAAP

One where cumulative inflation exceeds 100% over a 3-year period.

IASB

Does not specifically define hyperinflation but cumulative inflation of over 100% in a 3-year period is one indication.

In a **hyperinflationary environment**, the local currency will rapidly depreciate relative to the parent's presentation currency because of a *deterioration of purchasing power*.

IFRS and US GAAP differ substantially in their approach to translating the foreign currency financial statements of foreign entities operating in the currency of a hyperinflationary economy:

US. GAAP

Functional currency is considered to be the parent's presentation currency

→ **Temporal method** is used.

IFRS

Foreign currency financial statements are restated for inflation and then translated using the **current exchange rate**. (see next slide)

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1. Hyperinflation

Under IFRS, the procedures of restating for inflation will be illustrated by the following example:

Example : Adjusting financial statements for inflation

Imagine that a foreign subsidiary was created on December 31, 2021. The LC is the currency of the country where the foreign subsidiary is located. The subsidiary's balance sheets for 2021 and 2022, and income statement for the year-ended 2022, are shown in the following:

in LCs	2021	2022
Cash	5,000	8,000
Supplies	25,000	25,000
Total assets	30,000	33,000
Accounts payable	20,000	20,000
Common stock	10,000	10,000
Retained earnings	0	3,000
Liabilities and equity	30,000	33,000

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

 **Example : *Adjusting financial statements for inflation***

in LCs	2022
Revenue	15,000
Expenses	(12,000)
Net income	3,000

Also, use the following price indices:

December 31, 2021: 100

December 31, 2022: 150

Average for 2022: 125

Prepare an inflation adjusted balance sheet and income statement for 2022.

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

Example : Adjusting financial statements for inflation

Answer:

Firstly, we prepare an inflation adjusted balance sheet for 2022.

The procedures for adjusting the assets are as follows:

- 1 **Nonmonetary assets** are restated for inflation using a price index by multiplying the original cost by the **change in the price index** for the period between the *acquisition date* and the *balance sheet date*
- 2 **Monetary assets** are not necessary to restate

in LCs	2022	Adjustment factor (<i>change in the price index</i>)	Inflation Adjusted
Cash	8,000		
Supplies	25,000		
Total assets	33,000		

[LOS 3.g] *Analyze* how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

Example : *Adjusting financial statements for inflation*

Answer:

Firstly, we prepare an inflation adjusted balance sheet for 2022.

The procedures for adjusting the assets are as follows:

3 Nonmonetary liabilities are restated for inflation using a price index by multiplying the original cost by the ***change in the price index*** for the period between the acquisition date and the balance sheet date

4 **Monetary liabilities** are not necessary to restate

5 The **components of shareholders' equity** (other than retained earnings) are restated by applying the change in the price index from the *beginning of the period* or the *date of contribution* if later

6 **Retained earnings** is the ***plug figure*** that balances the balance sheet

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

 **Example : Adjusting financial statements for inflation**

Answer:

in LCs	2022	Adjustment factor (<i>change in the price index</i>)	Inflation Adjusted
Accounts payable (1)	20,000		...
Common stock (2)	10,000	...	...
Retained earnings	3,000	Plug figure	$= (3) - (1) - (2)$ $= \dots$
Total liabilities and equity (3)	33,000		...

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

 **Example : *Adjusting financial statements for inflation***

Answer:

in LCs	2022	Adjustment factor	Inflation Adjusted
Cash	8,000		8,000
Supplies	25,000	150/100	37,500
Total assets	33,000		45,500
Accounts payable	20,000		20,000
Common stock	10,000	150/100	15,000
Retained earnings	3,000		10,500
Total liabilities and equity	33,000		45,500

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

Example : Adjusting financial statements for inflation

Answer:

Secondly, we prepare an inflation adjusted income statement for 2022.

The procedures for adjusting the income statement items are as follows:

- 1** In the statement of retained earnings, **net income** is the **plug figure**
- 2** The **income statement items** are restated by multiplying by the change in the price index from the date the transactions occur
- 3** The **net purchasing power gain or loss** is recognized in the income statement based on the net monetary asset or liability exposure

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

 **Example : *Adjusting financial statements for inflation***

Answer:

in LCs	2022	Adjustment factor (<i>change in the price index</i>)	Inflation Adjusted
Revenue (1)	15,000		
Expenses (2)	(12,000)		
Net purchasing power gain			$= (3) - (1) - (2)$ $= \dots$
Net income (3)	3,000	Plug figure (from balance sheet)	

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

 **Example : *Adjusting financial statements for inflation***

Answer:

in LCs	2022	Adjustment factor	Inflation Adjusted
Revenue	15,000	150/125	18,000
Expenses	(12,000)	150/125	(14,400)
Net purchasing power gain			6,900
Net income	3,000		10,500

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect...

1.

Hyperinflation

Notice the similarities of adjusting the financial statements for inflation and remeasuring the financial statements using the temporal method.

Temporal method

The monetary assets and liabilities are exposed to changing exchange rates

An exchange rate gain is recognized if a subsidiary has net monetary liability exposure in a depreciating environment

The exchange rate gain or loss from remeasurement is recognized in the income statement

Adjusting for inflation

The monetary assets and liabilities are exposed to risk of inflation

A purchasing power gain is recognized when a net monetary liability exposure is adjusted for the effects of inflation

The net purchasing power gain or loss from inflation is recognized in the income statement

[LOS 3.h] *Describe* how multinational operations affect a company's effective tax rate

1.

Tax implications of multinational operations

Multinational corporations are subject to *multiple tax jurisdictions* with differing laws and tax rates. In the United States, for example, the tax code allows a ***credit for foreign taxes*** paid by U.S. companies.

The effect of the ***tax credit***:

Only when *US corporate tax rate > the foreign tax rate*

→ Multinational company owes taxes on the foreign income.

Accounting standards require companies to provide a *reconciliation* between:

- ***Effective tax rate***: Tax expense / Pretax profit
- ***Statutory tax rate***: provided by the tax code of the home country

This reconciliation disclosure can be used by the analyst to project future tax expense.

Changes in effective tax rate on account of foreign operations can be due to:

- Changes in the *mix of profits* from different countries (with varying tax rates)
- Changes in the *applicable tax rates*

[LOS 3.h] *Describe* how multinational operations affect a company's effective tax rate

1.

Tax implications of multinational operations

Example : *Analysis of reconciliation of effective tax rate*

The reconciliation between the statutory tax rate and effective tax rate for two companies (Amco and Bianco) for 2024 is provided.

Item	Amco	Bianco
Statutory tax rate	25.0%	30.0%
Effect of disallowed expenses	3.0%	1.0%
Effect of exempt income	(2.0%)	(0.5%)
Effect of taxes in foreign jurisdictions	3.4%	(1.2%)
Effect of recognition of prior losses	(0.8%)	(3.0%)
Effective tax rate	28.6%	26.3%

[LOS 3.h] *Describe* how multinational operations affect a company's effective tax rate

1.

Tax implications of multinational operations

Example : *Analysis of reconciliation of effective tax rate*

1. *Which company benefited from the lowering of tax expense on account of its foreign operations?*
2. *If the mix of foreign operations for both companies is expected to increase over time, which company is most likely to report lower effective tax rate in the future? Assume that the statutory tax rates do not change.*

Answer:

1. The effect of foreign operations resulted in an increase in effective tax rate for Amco by 3.4% and a decrease for Bianco by 1.2%. Hence, Bianco benefited from its foreign operations in reducing its effective tax rate and tax expense.
2. If the mix of foreign operations increases, assuming no change to the tax rate, Bianco would see its effective tax rate decrease further and Amco would see its effective tax rate increase.

[LOS 3.i] *Explain* how changes in the components of sales affect the sustainability of sales growth

1.

Analysis of sales components

Financial statements for multinational companies report sales denominated in the *reporting currency* even though the actual sales may have occurred in many different currencies.

→ *Changes in currency values affect the value of translated sales.*

There are two components affect sales growth:

- Increase in volumes or prices → *More sustainable*
- Appreciation of the foreign currencies in which sales were made

Companies often include disclosures about the effect of exchange rates on sales growth in the MD&A.

This information can be used by analysts to improve the accuracy of their forecasts of future sales.

[LOS 3.i] *Explain* how changes in the components of sales affect the sustainability of sales growth

1.

Analysis of sales components

Example : *Analysis of reconciliation of effective tax rate*

BCN, Inc., an American multinational, reported the following in its MD&A for 2024:

Region	% Change in sales	Impact of currency	% Change in sales excluding currency effect
North America ex-U.S.	3%	-2%	5%
Asia (incl. Japan)	2%	0%	2%
Europe	2%	3%	-1%

Comment on the growth rates observed in different markets for BCN.

[LOS 3.i] *Explain* how changes in the components of sales affect the sustainability of sales growth

1.

Analysis of sales components

Example : *Analysis of reconciliation of effective tax rate*

Comment on the growth rates observed in different markets for BCN.

Answer:

BCN experienced *highest* growth in North America ex-U.S (3%). Excluding the loss in translation (due to depreciation of the foreign currency in which sales are denominated) for North America ex-U.S., the revenue growth would have been 5% due to an increase in volumes and/or price.

In Europe, the revenue growth is 2%, but that was due to appreciation of the currencies in which sales occurred. Excluding the currency effect, sales were down in Europe by 1%.

In Asia, the revenue growth of 2% is solely due to price/volume changes.

[LOS 3.j] *Analyze* how currency fluctuations potentially affect financial results, given a company's countries of operation

1.

Foreign exchange risk

Definition

Foreign exchange risks reflect the effect of changes in currency values on the assets and liabilities of a business as well as on future sales.

Managing foreign exchange risk

- Disclosures as part of MD&A may include the impact of currency value changes on profits that help an analyst estimate the potential impact of currency value changes on a company's earnings going forward.
- If such information is not provided, analysts can conduct their own sensitivity analysis to further improve their forecasts and understand risks faced by the company.
- Analysts also should inquire about any hedging tools employed by the company to manage its currency exposures.



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Tax Withholding and Estimated

Capital Gains and Losses

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MODULE 4: ANALYSIS OF FINANCIAL INSTITUTIONS

LEARNING OUTCOMES

- 4.a.** Describe how financial institutions differ from other companies
- 4.b.** Describe key aspects of financial regulations of financial institutions
- 4.c.** Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations
- 4.d.** Analyze a bank based on financial statements and other factors
- 4.e.** Describe other factors to consider in analyzing a bank
- 4.f.** Describe key ratios and other factors to consider in analyzing an insurance company

[LOS 4.a] *Describe* how financial institutions differ from other companies

1.

Introduction to Financial institutions

Financial institutions provide a wide range of financial products and services.



Capital providers
Provide funding



Financial institutions
Serve as intermediaries



Borrowers
Receive funding



- Given the diversity of financial services, there are numerous types of financial institutions exist.
- This reading focuses primarily on *two types* of financial institutions: **banks** and **insurance companies**.

[LOS 4.a] *Describe* how financial institutions differ from other companies

2.

What Makes Financial Institutions Different?

2.1. Systematic importance

The smooth functioning of financial institutions is essential to the overall health of an economy.

*Financial institutions' role as intermediaries of capital creates **financials inter-linkages** across all types of entities.*

Financial inter-linkages



Failure of one bank



Affect other
financial and non-
financial entities



Negative affect the
entire economy

Systemic risk

A risk of
disruption
to financial
services
that is

Caused by an impairment of all or parts of
the financial system

Has the potential to have serious negative
consequences for the economy as a whole.

[LOS 4.a] *Describe* how financial institutions differ from other companies

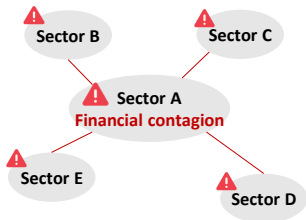
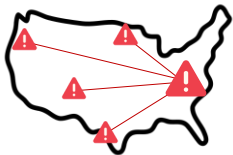
2.

What Makes Financial Institutions Different?

2.1. Systematic importance

Financial contagion

Financial shocks spread from their place or sector of origin to other locales or sectors.



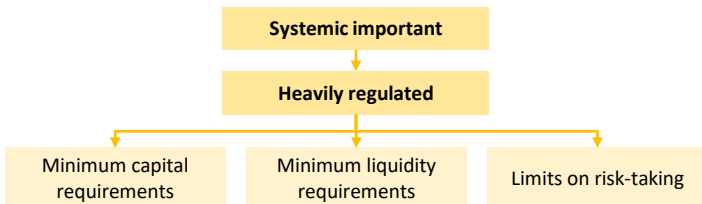
Globally, a faltering economy may infect other, healthier economies.

[LOS 4.a] Describe how financial institutions differ from other companies

2.

What Makes Financial Institutions Different?

2.1. Systematic importance



2.2. The nature of the liabilities

The liabilities of most banks are made up primarily of deposits

The failure of a bank to honor its deposits could have negative consequences

Could cause a **bank run***

Deposits are often *insured by the government*

***Bank run:** a large number of customers rapidly withdraw their deposits from a bank due to concerns about its solvency, potentially leading to the bank's financial collapse.

[LOS 4.a] *Describe* how financial institutions differ from other companies

2.
What Makes Financial Institutions Different?

2.3. The nature of the assets


Financial Institution

 Mostly **Financial** assets

 Record at **Fair value**

Other companies

 Mostly **Tangible** assets

 Record at **Historical costs**
Direct exposure to a variety of risks

Credit risks

Liquidity risks

Market risks

Interest rate risks

[LOS 4.b] Describe key aspects of financial regulations of financial institutions

2.

What Makes Financial Institutions Different?

Goal of global/regional regulatory bodies

Minimizing systemic risk

Harmonization and globalization
of regulatory rules, standards,
and oversight

Consistency of standards and regulations helps minimize regulatory arbitrage

3.

Basel III

One of the most important global organizations focus on financial stability is the Basel Committee on Banking Supervision.

→ Developed the **international regulatory framework for banks** known as Basel III.

Purposes of the measures contained in Basel III

Absorb shocks arising
from financial and
economic stress

Improve risk
management and
governance

Strengthen banks'
transparency and
disclosures

[LOS 4.b] *Describe* key aspects of financial regulations of financial institutions

3.

Basel III

3 pillars of Basel III

Minimum capital
requirement

Minimum liquidity

Stable funding

Pillar	Measurement	Requirement
Minimum capital requirement	Minimum % of its risk-weighted assets (RWA) that a bank must fund with equity capital	The riskier a bank's assets are, the higher its required capital
Minimum liquidity	High-quality liquid assets to cover liquidity needs in a 30-day liquidity stress scenario	Should have enough cash to cover a partial loss of funding sources or a cash outflow from off-balance-sheet funding commitments
Stable funding	Minimum amount of stable funding relative to the bank's liquidity needs over a 1-year horizon	Based on: • Tenor of deposits • Type of depositors

[LOS 4.b] *Describe* key aspects of financial regulations of financial institutions

3.

Basel III

Basel III has forced banks to

Focus on asset
quality

Hold capital against
other types of risk

Develop improved
risk assessment
processes

4.

Other organizations involved financial stability

**Other international organizations
to coordinate regulations**

Functions

Financial Stability Board

Coordinate actions of participating jurisdictions in identifying and managing systemic risks.

**International Association of
Deposit Insurers**

Improve the effectiveness of deposit insurance systems.

**International Organization of
Securities Commissions (IOSCO)**

Promote fair and efficient security markets.

**International Association of
Insurance Supervisors (IAIS)**

Improve supervision of the insurance industry.

[LOS 4.c] *Explain* the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

What does CAMELS stand for?

C
Capital adequacy (1.)
A
Asset quality (2.)
M
Management capabilities (3.)
E
Earnings sufficiency (4.)
L
Liquidity position (5.)
S
Sensitivity to market risk (6.)

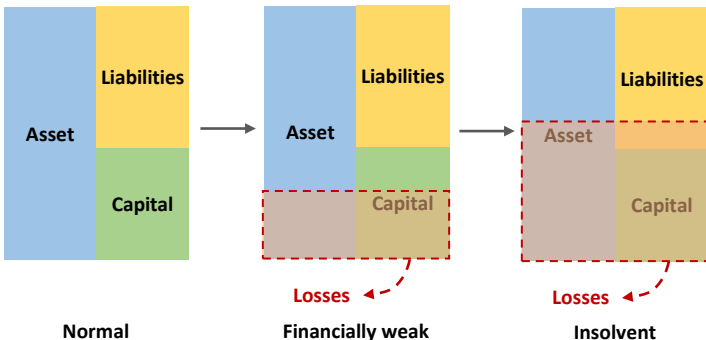
[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy

Definition

Potential losses is first absorbed by capital → Adequate capital helps bank to avoid become financially weak or insolvent when it incur losses.



A strong capital position lowers the probability of insolvency and bolsters public confidence in the bank.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy

Capital adequacy

- Described in terms of the proportion of the bank's assets funded with capital.
- Assets are adjusted based on their risk, with riskier assets requiring a higher weighting.

Asset

Weighting

Cash

0%

Corporate loans

100%

Lower quality assets

> 100%

Off-balance-sheet exposures

Also assigned risk weights

Example:

Consider a hypothetical bank with three assets: \$10 in cash, \$1,000 in performing loans, and \$10 in non-performing loans.

The bank's risk-weighted assets (RWAs) would equal

$$(\$10 \times 0\%) + (\$1,000 \times 100\%) + (\$10 \times 150\%) = \$1,015.$$

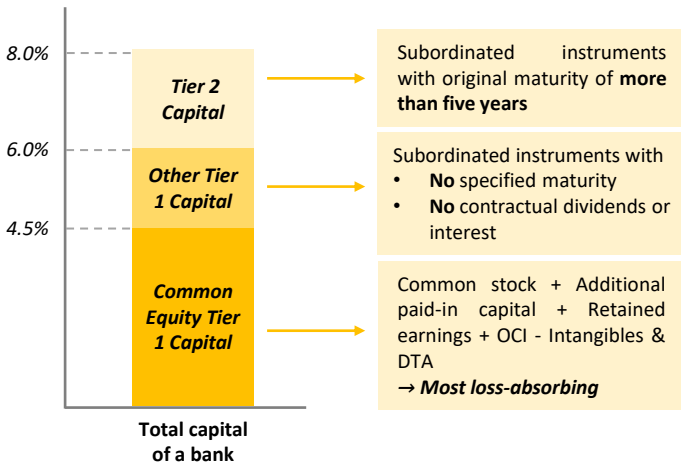
[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy

Basel III capital requirements

Minimum % of RWA



[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy


Illustration: Capital Position

The example presents an excerpt from an annual report disclosure by HSBC Holdings plc about its capital position. The excerpt shows the group's capital ratios, amount of capital by tier, and risk-weighted assets by type.

Table 1: Capital ratios

At 31 Dec.		
	2016 (%)	2015 (%)
Common equity tier 1 ratio	13.6	11.9
Tier 1 ratio	16.1	13.9
Total capital ratio	20.1	17.2

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy


Illustration: *Capital Position*

Table 2: Total Regulatory Capital and Risk-Weighted Assets

	At 31 Dec.	
	2016 (\$m)	2015 (\$m)
Regulatory Capital		
Common equity tier 1 capital	116,552	130,863
Additional tier 1 capital	21,470	22,440
Tier 2 capital	34,336	36,530
Total regulatory capital	172,358	189,833
 Risk-weighted assets	 857,181	 1,102,995

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1. Capital adequacy



Illustration: *Capital Position*

Table 3: Risk-weighted assets (RWAs) by risk types

	RWA (\$bn)	Capital required (\$bn)
Credit risk	655.7	52.5
Counterparty credit risk	62.0	5.0
Market risk	41.5	3.3
Operational risk	98.0	7.8
At 31 Dec 2016	857.2	68.6

1. *Did HSBC's capital ratios strengthen or weaken in 2016?*
2. *What was the primary reason for the change in HSBC's capital ratios in 2016?*

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy


Illustration: Capital Position
Answer:

1. Did HSBC's capital ratios strengthen or weaken in 2016?

We need to compare the capital ratios of 2016 to 2015.

Table 1: Capital ratios

At 31 Dec.		
	2016 (%)	2015 (%)
Common equity tier 1 ratio 	13.6	11.9
Tier 1 ratio 	16.1	13.9
Total capital ratio 	20.1	17.2

→ HSBC's capital ratios strengthened in 2016.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy



Illustration: *Capital Position*

Answer:

2. What was the primary reason for the change in HSBC's capital ratios in 2016?

- Possible reasons:
- ☐ More capital
 - ☐ Fewer risk-weighted assets

We need to examine Table 2 to see the changes in capital.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.

Capital adequacy


Illustration: Capital Position

Answer:

2. What was the primary reason for the change in HSBC's capital ratios in 2016?

Table 2: Total Regulatory Capital and Risk-Weighted Assets

	At 31 Dec.	
	2016 (\$m)	2015 (\$m)
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Common equity tier 1 capital ↓↓	116,552	130,863
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Tier 2 capital ↓↓	34,336	36,530
Total regulatory capital	172,358	189,833
Risk-weighted assets ↓↓	857,181	1,102,995

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

1.
Capital adequacy

Illustration: *Capital Position*
Answer:

2. What was the primary reason for the change in HSBC's capital ratios in 2016?

The amount of capital in each tier decreased, which leads to an increase in capital ratios in Table 1.

Possible reasons:



More capital



Fewer risk-weighted assets

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality

Definition

Asset quality pertains to the amount of existing and potential credit risk associated with a bank's assets, focusing primarily on financial assets.

Bank assets include *loans* and *investments in securities*.

2.1. Loan

Loans typically constitute the largest portion of a bank's assets.

Asset quality

Creditworthiness of the borrowers

Adequacy of adjustments for
expected loan losses

Measurement & Reporting

- Measured at amortized cost.
- Shown on the balance sheet net of allowances for loan losses.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality

2.2. Investment in securities issued by other entities

Measurement depends on how the security is categorized.



Recall: Classification of investment in financial assets under IFRS

Debt Securities

Held to maturity → Amortized cost

Held for trading → FVPL

Available for sale → FVOCI

Equity Securities

Held for trading → FVPL

Other equity securities → FVPL or FVOCI

Once classified, the choice is **irrevocable**

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality

2.2. Investment in securities issued by other entities

Measurement depends on how the security is categorized.

Classification of investment in financial assets under US GAAP

Debt Securities

Held to maturity → Amortized cost

Held for trading → FVPL

Available for sale → FVOCI

Equity Securities

Equity investments → FVPL

[LOS 4.c] *Explain* the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality

Asset quality can be addressed from the perspective of **overall asset composition** or **credit quality**.

Overall asset composition

Invest in
highly liquid
assets



Improve
liquidity



Asset quality
↑

Consideration

How much of total asset is allocated to loan and to investments in securities?

Credit quality

Consideration

How much of the total asset is allocated to loans at different credit quality?

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality


Example: Credit analysis

The following table provides information for loans held by Small Bank at the end of 20X8 and 20X7.

Loans	20X8	20X7
Very strong credit quality	661	596
Strong credit quality	882	894
Good credit quality	1,488	1,292
Satisfactory credit quality	606	696
Sub-standard credit quality	992	894
Past due but not yet impaired	276	199
Impaired	441	298
Seriously impaired	165	99
Total	5,511	4,968

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality



Example: *Credit analysis*

Question 1: It appears that from 20X7 to 20X8, the credit quality of Small Bank's loan assets has *most likely*:

- A. worsened.
- B. improved.
- C. remained essentially the same.

Question 2: The proportion of loans at the end of 20X8 that are either past due or impaired is *closest to*:

- A. 12%.
- B. 16%.
- C. 19%.

We need to calculate the percentage of loans in each category for both years.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality



Example: Credit analysis

The percentage of loans in each category for both years:

Loans	20X8	20X7
Very strong credit quality	...	...
Strong credit quality	...	...
Good credit quality	...	...
Satisfactory credit quality	...	...
Sub-standard credit quality	...	...
Past due but not yet impaired	...	...
Impaired	...	...
Seriously impaired	...	...
Total	...	...

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality


Example: Credit analysis

The percentage of loans in each category for both years:

Loans		20X8	20X7
Very strong credit quality	—	12%	12%
Strong credit quality	↓	16%	18%
Good credit quality	↑	27%	26%
Satisfactory credit quality	↓	11%	14%
Sub-standard credit quality	—	18%	18%
Past due but not yet impaired	↑	5%	4%
Impaired	↑	8%	6%
Seriously impaired	↑	3%	2%
Total		100%	100%

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

2.

Asset Quality



Example: *Credit analysis*

Question 1: It appears that from 20X7 to 20X8, the credit quality of Small Bank's loan assets has *most likely*:

- A. worsened.
- B. improved.
- C. remained essentially the same.

Answer: A

Proportion of better-quality loans (at least satisfactory credit quality) has decline while that of poor credit quality (past due, impaired and seriously impaired) has increased.

Question 2: The proportion of loans at the end of 20X8 that are either past due or impaired is *closest to*:

- A. 12%.
- B. 16%.
- C. 19%.

Answer: B

The total of loans that were either past due or impaired = $5\% + 8\% + 3\% = 16\%$.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

3. Management Capabilities

Indicators of management effectiveness in general

Sound internal
controls

Transparent
management
communication

Financial reporting
quality

Financial institutions

A particularly important aspect of management capability is the ability to **identify and control risk**.

Directors of
banks



Bank's senior
managers

- Set overall guidance on risk exposure levels
 - Set implementation policies
 - Provide oversight of bank management
-
- Develop and implement effective procedures for measuring and monitoring risks consistent with guidance.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

4.

Earnings

**High quality
earnings –
Adequate and
sustainable**

Trend in earnings should be upward.

Underlying **accounting estimates** should be unbiased.

Earnings should be derived from **recurring sources**.

Major sources of earnings

Net interest income

Service income

Trading income

Most volatile

- Banks with proportionally higher net interest income and service income would have more sustainable earnings.
- Lower volatility within net interest income is desirable.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

4.

Earnings

Accounting estimates

Valuation of investment in securities (4.1.)

Subjective estimates (4.2.)

4.1. Valuation of investment in securities

Fair value hierarchy

The three “levels” of the fair value hierarchy pertain to the observability of the inputs used to establish the fair value.

Inputs

Level 1

Inputs are quoted prices for identical financial instruments in active markets

Level 2

Inputs are observable but are not the quoted prices for identical financial instruments in active markets

Level 3

Inputs are non-observable → Subjective
FV of a financial instrument is based on a model and unobservable inputs

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

4.
Earnings
4.2. Subjective estimates
Estimates
Requirement

**Goodwill
impairment**

Estimating future cash flows of a business unit.

**Deferred tax
asset**

Making assumptions about the probability of future taxes.

**Liability
versus
contingencies**

Professional expert advice and management judgment.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

5.

Liquidity position

Banks' systemic importance increases the importance of adequate liquidity.
→ Liquidity is a key focus of regulators.

(Refer to Systemic importance in LOS 4.a, Section 2.1)

2 minimum liquidity standards from Basel III

*Liquidity coverage ratio – LCR
(5.1)*

*Net stable funding ratio - NSFR
(5.2)*

2 liquidity-monitoring metrics from Basel III

*Concentration of funding
(5.3)*

*Contractual maturity mismatch
(5.4)*

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

5.
Liquidity position
5.1. Liquidity coverage ratio
Definition

The minimum percentage of a bank's expected cash outflows that must be held in highly liquid assets.

Liquidity coverage ratio

$$\text{LCR} = \frac{\text{Highly liquid assets}}{\text{Expected cash outflows}}$$

- Highly liquid assets are those that are easily convertible into cash
- Expected cash flows are the estimated one-month liquidity needs in a stress scenario.
- The standards recommend a minimum LCR of 100%.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

5.

Liquidity position

5.2. Net stable funding ratio

Definition

The minimum percentage of a bank's required stable funding that must be sourced from available stable funding

Net stable funding ratio

$$\text{NSFR} = \frac{\text{Available stable funding}}{\text{Required stable funding}}$$

- Required stable funding is a function of the composition and maturity of a bank's *asset base*.
- Available stable funding is a function of the composition and maturity of a bank's *funding sources*.

NSFR relates the liquidity needs of the financial institution's assets to the liquidity provided by the funding sources.

[LOS 4.c] *Explain* the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

5.
Liquidity position

5.3. Concentration of funding

Definition

Refers to the proportion of funding that is obtained from a single source.

Excessive concentration of funding exposes a bank to the risk that a single funding source could be withdrawn.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

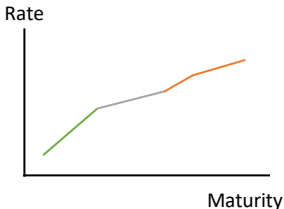
5.

Liquidity position

5.4. Contractual maturity mismatch

Definition

Refers to the maturity dates of a bank's assets compared to the maturity dates of a bank's fund sources.



Long-term rates > **Short-term rate**

Borrowing short-term + Lending long-term → Net interest income

⚠ Expose the bank to **liquidity risk**

Need to monitor maturity mismatch

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

6.

Sensitivity to Market Risk

Bank's operation

Loans and deposits

affected by

Maturity mismatch

Repricing frequency

Reference rates

Currency of loans/deposits

Off-balance-sheet activities

affected by

Interest rates

Exchange rates

Equities

Commodities

Sensitivity to market risk is an important consideration for analysts:

- How an adverse change in the market would affect a bank's earnings.
- Evaluate the strength of a bank's ability to manage market risks.

[LOS 4.c] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

6.

Sensitivity to Market Risk

Bank's risk management

1

Yield curve shape changes impact bank's earnings based on assets and liabilities compositions. Bank often disclose exposure to a variety of market and nonmarket risks.

2

Banks respond to opportunities presented in the market and alter their balance sheets.

Example:

Following the financial crisis of 2008, central banks around the world reduced short-term interest rates, allowing banks to borrow at lower rates.

→ Many banks increased their duration risk (borrowing short-term, lending long-term) to benefit from interest rate scenario.

The impact of market risk can be captured by value at risk (VaR), which will be discussed in Portfolio Management topic later on.

**[LOS 4.d,e] Describe other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

1.

Bank specific factors

1.1. Government support

Inter-linkages* in the banking sector

Government's interest to have a healthy banking system

Government support

Close banks that might fail

Assist banks to keep them afloat

Factors to consider in terms of government support:

- Size of the bank: “Too big to fail” problem
- Status of the country's banking system: Is the nation's banking system healthy enough to handle a particular bank's failure?

**Refer to Financial Inter-linkage in LOS 4.a, Section 2.1*

**[LOS 4.d,e] Describe other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

1.

Bank specific factors

1.2. Government ownership

Reasons for government ownership

**Development
view**

Government ownership aids financial development of the banks, leading to bread economic growth.

**Pessimistic
view**

A nation's banking system is not strong enough to stand on its own, because of a lack of public confidence.

Role of government ownership

Government is more likely to intervene on the bank's behalf in the event of economic distress

Adds another dimension of security for a bank investor.

**[LOS 4.d,e] Describe other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

1.

Bank specific factors

1.3. Mission of banking entity

The mission of the bank and the economics of its constituents will affect the way the bank manages its assets and liabilities.

Example:

**Community
banks**

- Primarily serve the needs of the immediate community in which they operate.
- If the community is dependent on a primary industry, it may lead to concentration of risk in the bank's asset portfolio.

Global banks

- Absorb deposits around the world while investing globally.
- The global bank have well-diversified asset bases, reducing their overall risk.

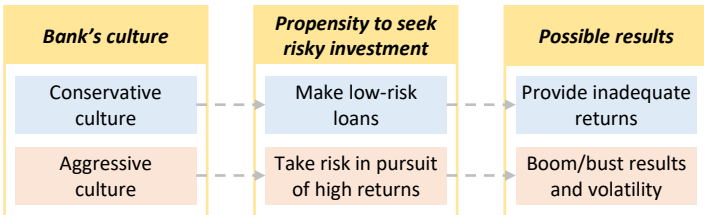
**[LOS 4.d,e] Describe other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

1.

Bank specific factors

1.4. Culture

A bank's culture influences its propensity to seek risky investments.



Culture evaluation can be conducted by a review of:

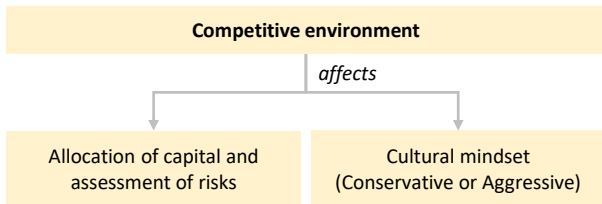
- **Diversity of a bank's assets:** Losses due to investment strategy with narrow focus → Aggressive.
- **Accounting restatements** due to failures of internal controls pertaining to financial reporting → Unethical culture.
- **Management compensation:** Excessive compensation tied with performance → Excessive risk-taking
- **Speed of adjusting loan loss provisions:** Slower response rate → Aggressive and risk-taking

**[LOS 4.d,e] Describe other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

2.

Non-bank specific factors

2.1. Competitive environment



Example:

Regional bank

- Satisfied with its current market position.
- May not be tempted to take excessive risk.

Global bank

- Management seeks to outdo their large rivals.
- End up taking excessive risk.

**[LOS 4.d,e] Describe other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

2.

Non-bank specific factors

2.2. Off-balance-sheet items

- Off-balance-sheet assets and liabilities pose a risk to entities and their investors if they should unexpectedly drain resources.
- Key information about a bank's off-balance-sheet exposures may be opaque or not readily available to analysts, making it very difficult to examine.

Example: Off-balance sheet items

- *Variable interest entities (VIE)*
- *Benefit plans*
- *Asset under management (AUM)*

**[LOS 4.d,e] *Describe* other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

2.

Non-bank specific factors

2.3. Segment information

- Segment information should illustrate the information used by the chief operating decision maker in the entity.
- Helps investors decide whether capital is being allocated well within the bank's internally competing operations.

2.4. Currency exposure

- Floating currency exchange rates can create problems for global banks.
- When a bank's home currency strengthens against the functional currencies of its foreign subsidiaries, the translation of balance sheet accounts may lead to currency translation adjustments, reducing capital.

**[LOS 4.d,e] *Describe* other factors to consider in analyzing a bank;
Analyze a bank based on financial statements and other factors**

2.

Non-bank specific factors

2.5. Risk factors

The risk factors section of a company's filing can fill gaps in an investor's knowledge about legal and regulatory issues that might not otherwise be uncovered.

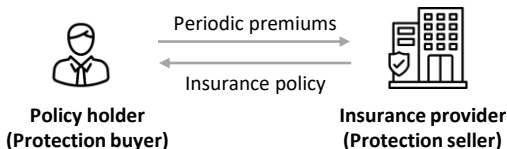
2.6. Basel III disclosures

Extensive disclosures that complement the minimum risk-based capital requirements and other quantitative requirements.

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

1.

Background of Insurance company



Categories

Property and Casualty (P&C) (2.)

- Short-term
- Variable and lumpier claims due to unpredictable events

Life and Health (L&H) (3.)

- Long-term
- More predictable due to relatively stable actuarially based mortality rates

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.

Property and Casualty Insurance Companies

2.1. Business profile

Property insurance policies protect against loss or damage to property – buildings, automobiles, environmental damage, and other tangible objects of value.

Methods of distributing insurance

Direct writing

- Direct writers have their own sales and marketing staff.
- Higher fixed costs.

Agency writing

- Agency writers use independent agents, exclusive agents, and insurance brokers to sell policies.
- Lower fixed costs, incur commissions, high variable costs..

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

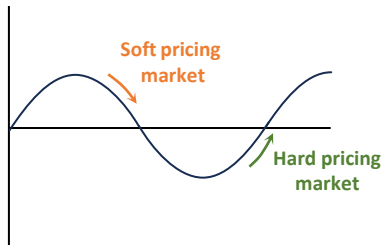
2.

Property and Casualty Insurance Companies

2.2. *Earnings characteristics*

The property and casualty insurance business is cyclical and price sensitive.

Profitability



High profit

Attracts new entrants → Cut prices to obtain market shares → Profitability decreases → Soft pricing market

Low profit

Competition lessens → Underwriting standards tighten → Hard pricing market → Premium and profitability rises

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.
Property and Casualty Insurance Companies

2.2. Earnings characteristics

Underwriting cycle is driven largely by the expense of the participants, indicated in the **combined ratio**.

Combined ratio

$$\text{Combined ratio} = \frac{\text{Total insurance expenses}}{\text{Net premiums earned}}$$

- Low combined ratio: Hard pricing market → Attracting new entrants
- High combined ratio: Soft pricing market → Competitors leave the market
- Combined ratio > 100%: Indicates an underwriting loss

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.
Property and Casualty Insurance Companies

2.2. Earnings characteristics

In the US, insurance companies must prepare financial reports according to statutory accounting rules, which define the combined ratio as the sum of two ratios:

Combined ratio = Underwriting loss ratio + Expense ratio

Underwriting loss ratio

$$\text{Underwriting loss ratio} = \frac{\text{Claims paid} + \Delta \text{loss reserves}}{\text{Net premiums earned}}$$

→ Indicator of the quality of the underwriting activities.

Expense ratio

$$\text{Expense ratio} = \frac{\text{Underwriting expenses including commissions}}{\text{Net premiums written}}$$

→ Indicator of the efficiency of a company's operations in acquiring and managing underwriting business

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.

Property and Casualty Insurance Companies

2.2. Earnings characteristics

Loss reserves

Estimated value of unpaid claims, or reserves put aside against policy losses.

- Underestimation may lead to undercharging for risks assumed.
- Loss reserves is subject to management discretion.

Net premiums written

Insurer's direct premiums written net of reinsurance premiums paid.

Net premiums earned

Net premiums written earned over a relevant accounting period.

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.

Property and Casualty Insurance Companies

2.2. *Earnings characteristics – Profitability ratios*

Loss and loss adjustment expense ratio

$$\frac{\text{Loss expense} + \text{Loss adjustment expense}}{\text{Net premiums earned}}$$

→ Degree of success an underwriter has achieved in estimating risk insured

Underwriting expense ratio

$$\text{Underwriting expense ratio} = \frac{\text{Underwriting expense}}{\text{Net premiums written}}$$

→ Efficiency of money spent in obtaining new premiums

Combined ratio

$$\text{Combined ratio} = \text{Loss and loss adj. expense ratio} + \text{Underwriting expense ratio}$$

→ Overall efficiency of an underwriting operation

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2. Property and Casualty Insurance Companies

2.2. Earnings characteristics – Profitability ratios

Dividends to policyholders/shareholders ratio

$$\frac{\text{Dividends to policyholders}}{\text{Net premiums earned}}$$

→ A measure of liquidity, relates cash outflow of dividends to the premiums earned in the same period.

Combined ratio after dividends

$$\text{Combined ratio after dividends} = \text{Combined ratio} + \text{Dividends to policyholders ratio}$$

→ Stricter measure of efficiency than the ordinary combined ratio, takes into account the cash satisfaction of policyholders

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.
Property and Casualty Insurance Companies

2.3. Investment returns

Risk profile

Face much uncertainty, price competitiveness.

Favored investment return

Steady return, low risk assets, avoid low-liquidity investments.

Investment performance

$$\text{Investment performance} = \frac{\text{Total investment income}}{\text{Invested assets}}$$

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

2.

Property and Casualty Insurance Companies

2.4. Liquidity

- Liquidity is an important consideration for P&C insurers as they stand ready to meet their claim obligations.
- One way to gauge the liquidity of the investment portfolio is to look at their fair value hierarchy reporting. (*Refer to “Fair value hierarchy” in LOS 4.c, Section 4.1*)
 - **Level 1:** Based on readily available prices for traded securities → Most liquid.
 - **Level 2:** Lower liquidity.
 - **Level 3:** Least liquid.

2.5. Capitalization

- There are no global risk-based capital requirement standards for insurers.
- Various jurisdictions have their own capital standards.

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

3.

Life and Health Insurance Companies

3.1. Business profile

Life insurance

Mainly provide a benefit upon the death of the insured and other types provide a savings vehicle.

Health-related insurance

Cover specific medical expenses and treatments and provide income payments if the policy holder is injured or becomes ill.

Methods of distributing insurance

Direct writing

- Direct writers have their own sales and marketing staff.
- Higher fixed costs.

Agency writing

- Agency writers use independent agents, exclusive agents, and insurance brokers to sell policies.
- Lower fixed costs, incur commissions, high variable costs..

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

3.

Life and Health Insurance Companies

3.2. Earnings characteristics

L&H insurers' earnings reflect a number of accounting items that require a significant amount of judgement and estimates.

Actuarial assumptions

Affect the value of the future liabilities due to policyholders.

Current period claim expense

Include actual claim payments and interest on the estimated liability to policyholders.

Cost of acquiring and renew policies

Capitalize and amortize based on actual and estimated future profits from that business.

Valuation of assets and liabilities

Mismatches between the valuation approaches can distort values when the interest rate environment changes.

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

3.
Life and Health Insurance Companies

3.2. Earnings characteristics

Apart from standard ratios such as ROE, ROA, and EBIT margins, specific profitability ratio is:

Combined ratio

$$\frac{\text{Total benefits paid}}{\text{Net premiums written + Deposits}} + \frac{\text{Commission and expenses}}{\text{Net premiums written + Deposits}}$$

Cost of *having*
the deal

Cost of *getting*
the deal

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

3. Life and Health Insurance Companies

3.3. Investment returns

Key aspects in evaluating L&H companies' investment activities include:

- **Diversification:** Assessment and evaluation of allocation across asset classes and its correspondence to liabilities to policyholders
- **Investment performance:** Similar to P&C insurance companies
- **Interest rate risk:** Comparison of the duration of the company's assets with the duration of its liabilities

Investment performance

$$\text{Investment performance} = \frac{\text{Total investment income}}{\text{Invested assets}}$$

[LOS 4.f] *Describe* key ratios and other factors to consider in analyzing an insurance company

3.

Life and Health Insurance Companies

3.4. Liquidity

- Requirements for liquidity are driven by its liability structure → Fairly predictable.
- Sources of liquidity: Operating cash flow and Investment assets.

3.5. Capitalization

- No global standards.
- Varies by jurisdictions.
- L&H insurers typically have lower capital requirements than P&C, do not need as high equity cushion.
- Due to exposure to interest rate risk, the calculation of risk-based capital for an L&H company incorporates interest rate risk.

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Tax Withholding and Estimated

Capital Gains and Losses

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MODULE 5: EVALUATING QUALITY OF FINANCIAL REPORTS

LEARNING OUTCOMES

- 5.a.** Demonstrate the use of a conceptual framework for assessing the quality of a company's financial reports
- 5.b.** Explain potential problems that affect the quality of financial reports
- 5.c.** Describe how to evaluate the quality of a company's financial reports
- 5.d.** Evaluate the quality of a company's financial reports
- 5.e.** describe indicators of earnings quality
- 5.f.** describe the concept of sustainable (persistent) earnings
- 5.g.** explain mean reversion in earnings and how the accruals component of earnings affects the speed of mean reversion
- 5.h.** evaluate the earnings quality of a company
- 5.i.** evaluate the cash flow quality of a company
- 5.j.** describe indicators of balance sheet quality
- 5.k.** evaluate the balance sheet quality of a company

LEARNING OUTCOMES

5.l. Describe indicators of cash flow quality

5.m. Describe sources of information about risk

[LOS 5.a] *Demonstrate* the use of a conceptual framework for assessing the quality of a company's financial reports

1.

Conceptual Framework

The quality of financial reports can be viewed along two highly related dimensions: ***reporting quality*** and ***earning quality***.

Reporting quality

Pertains to the information disclosed in financial reports.

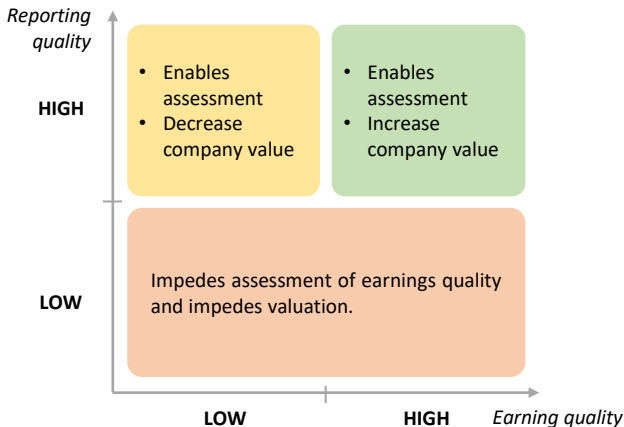
Earning quality

Pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition relative to expectations of current and future financial performance.

[LOS 5.a] *Demonstrate* the use of a conceptual framework for assessing the quality of a company's financial reports

1. Conceptual Framework

The quality of financial reports can be viewed along two highly related dimensions: **reporting quality** and **earning quality**.



[LOS 5.a] *Demonstrate* the use of a conceptual framework for assessing the quality of a company's financial reports

1.

Conceptual Framework

***Two basic questions
needs to be
considered***

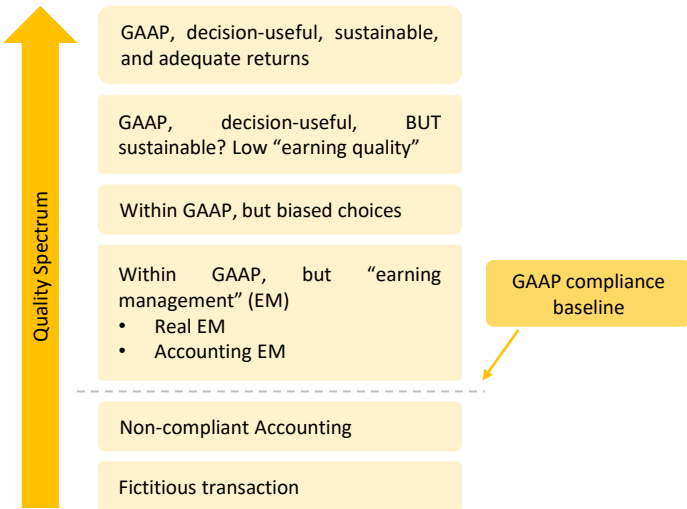
Are the results (earnings) of high quality? Do they provide an adequate level of return, and are they sustainable?

Are the financial reports GAAP-compliant and decision-useful?

[LOS 5.a] *Demonstrate* the use of a conceptual framework for assessing the quality of a company's financial reports

1. Conceptual Framework

The quality of financial reports can be viewed along two highly related dimensions: **reporting quality** and **earning quality**.



[LOS 5.b] Explain potential problems that affect the quality of financial reports

1.

Potential Problems

Problems in Financial Reporting can hinder analysts to get a clear view about the company's financial position, hence the information is not decision useful.

Potential problems of Financial Reports

***Reported
Amounts and
Timing of
Recognition***

Classification



Possible Consequences

*GAAP
compliance*

Financial statement
diverge from
economics reality

*Not GAAP
compliance*

Reporters may
present fraudulent
reports

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition

Though accounting choices may focus on a single element, it might have a statement-wide effect because the financial statement is interrelated.

Example of Choices	Result
Aggressive, premature, and fictitious revenue recognition	Overstated income, equity, receivable
Conservative revenue recognition	Understated income, equity, assets
Omission and delayed recognition of expenses	Understated: expenses, liabilities Overstated: Income, equity, assets
Understatement of contingent liabilities	Understated expenses Overstated equity, income, OCI
Overstatement of financial assets and understatement of financial liabilities	Overstated equity, unrealized gains Understated losses
Differing payments on payables, accelerating payments from customers	Overstated cashflows from operations

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Satyam Computer Services Limited, an Indian information technology company, was founded in 1987 and grew rapidly by providing business process outsourcing (BPO) on a global basis. In 2007, its CEO, Ramalinga Raju, was named “Entrepreneur of the Year” by Ernst & Young, and in 2008, the World Council for Corporate Governance recognized the company for “global excellence in corporate accountability.” In 2009, the CEO submitted a letter of resignation that outlined a massive financial fraud at the company. The company’s decline was so rapid and significant that it came to be referred to as “India’s Enron.”

In late 2008, the World Bank terminated its relationship with the company after finding that Satyam gave kickbacks to bank staff and billed for services that were not provided. These initial revelations of wrongdoing had the effect of putting the company under increased scrutiny. Among other misconduct, the CEO eventually admitted that he created fictitious bank statements to inflate cash and to show interest income. The CEO also created fake salary accounts and took the money paid to those “employees.”

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Fictitious Reports: Satyam Computer Services Limited

The company's head of internal auditing created fictitious customer accounts and invoices to inflate revenues.

The external auditors did not independently verify much of the information provided by the company. Even when bank confirmations, which were sent to them directly as opposed to indirectly through Satyam, contained significantly different balances than those reported by Satyam, they did not follow up.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Fictitious Reports: Satyam Computer Services Limited

1. *Based on the information provided, characterize Satyam's financial reports, with reference to the quality spectrum of financial reports.*

Solution:

Satyam:

- Gave kickbacks to bank staff;
- Billed for services that were not provided.

→ They clearly are at the bottom of the quality spectrum of financial reports: reports based on fictitious information.



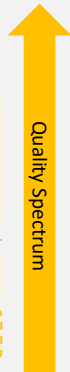
Within GAAP, but “earning management” (EM)

- Real EM
- Accounting EM

Non-compliant Accounting

Fictitious transaction

Quality Spectrum



[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

2. Explain each of the following misconducts with reference to the basic accounting equation:

- A. Transactions with World Bank
- B. Fictitious interest income
- C. CEO's embezzlement
- D. Fictitious revenue

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Solution:

The effects on the basic accounting equation of the different acts of misconduct are as follows:

Transactions with World Bank	Effects	Examples of account
Billing for fictitious services	BS - Increase asset	Accounts receivable
Kickbacks to the customer's staff	IS - Increase expense	Commissions paid
	BS - Increase liability	Commissions payable
	BS - Decrease asset	Cash
Net Effect	Overstated Income, net asset, equity	

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Solution:

The effects on the basic accounting equation of the different acts of misconduct are as follows:

Fictitious interest rate income	Effects	Examples of account
Fictitious interest income	IS - Overstate Income	
	BS - Overstate Asset	Cash Interest Receivable
	BS - Overstate Equity	
These overstatements were hidden by falsifying revenue and cash balances		

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Solution:

The effects on the basic accounting equation of the different acts of misconduct are as follows:

CEO's embezzlement	Effects	Examples of account
Creating fictitious employees	IS - Increase expense	Wages Salaries
	BS - Decrease Asset	Cash

→ The resulting understatement of income and equity was offset by a real but fraudulent decrease in cash, which was hidden by falsifying revenue and cash balances.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

2.

Reported Amounts and Timing of Recognition



Illustration: Fictitious Reports: Satyam Computer Services Limited

Solution:

The effects on the basic accounting equation of the different acts of misconduct are as follows:

Fictitious revenues	Effects	Examples of account
Fictitious revenues	IS - Overstate Revenues	
	IS - Overstate Income	
	BS - Overstate Asset	Cash Account Receivable
	BS - Overstate Equity	Cash

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

3. Classification

Classification issues refer to how an individual financial statement element is categorized within a particular financial statement.

Examples of Misclassification	
Misclassification	Effect
Removing account receivable by selling or transferring receivables to a related entity or treating them as long-term receivables.	Reduce day's sales outstanding and Enhances the receivables turnover ratio. Done to mask aggressive revenue recognition practices.
Reclassifying inventory as other (long-term) assets.	Increases inventory turnover ratio. Decrease current ratio.
Reclassifying non-core revenues as revenues from core continuing operations.	Misleads analysts about the sustainability of future revenues.
Reclassifying expenses as non-operating.	Analysts treat them as one-time costs.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

4.

Biased Accounting and Warning Signs

Biased accounting choices result in financial reports that do not faithfully represent economic phenomena. Biased choices can be made not only in the context of reported amounts but also in the context of how information is presented.

Examples: Biased Accounting and Warning Signs

Misstated elements	Mechanisms	Warnings signs
Profitability	• Aggressive revenue recognition • Lessor use of finance lease classification. • Classifying non-operating revenue/income as operating, and operating expenses as non-operating.	• Revenue growth higher than peers'. • Receivables growth higher than revenue growth. • High rate of customer returns.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

4.

Biased Accounting and Warning Signs

Examples: Biased Accounting and Warning Signs

Misstated elements	Mechanisms	Warnings signs
Assets/ Liabilities	• Choosing inappropriate models to estimate values of financial statement elements • Reclassification from current to non-current. • Over- or understating allowances and reserves. • Understating identifiable assets (and overstating goodwill) in acquisition method accounting for business combinations.	• Inconsistency in model inputs for valuation of assets versus valuation of liabilities. • Typical current assets (e.g., inventory, receivables) being classified as non-current. • High goodwill relative to total assets. • Use of special purpose entities.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

4.

Biased Accounting and Warning Signs

Examples: Biased Accounting and Warning Signs

Misstated elements	Mechanisms	Warnings signs
Cash flows	- Managing activities to affect cash flow from operations (e.g., stretching payables). - Misclassifying investing cash flow as cash flow from operations.	- Increase in payables combined with decreases in inventory and receivables. - Capitalized expenditures (which flow through investing activities). - Increases in bank overdraft.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.1. M&A Issues

1 *Companies with faltering cash-generating ability*

Companies with declining operating cash flow are motivated to acquire other cash-generating entities to increase cash flow from operations.



Company A want to increase the reporting cash flows

Company A acquires Company B. The acquisition have no negative effect on Company A's OCF

Company A now can consolidate company B's OCF, concealing its own cash flow problems

If paid in cash:

Recorded in the Investing Cash Flows

If paid with Equity:

Not recorded at all on the CF Statement

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.1. M&A Issues

2 *Acquisition create incentive to misreporting*

Both parties in a potential acquisition have the incentives by pursuing aggressive accounting or even misreport to maximize their own benefits.



For the acquirer, increase earnings can:

- Make the acquisition on more favorable terms
- Influence investors after the acquisition.
- Inflate share price if the acquisition is equity settled.

For the acquiree, increase earnings can:

- Help securing a more favorable price for their company

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.1. M&A Issues

3 *Misreporting create incentive to acquisition*

Company with reporting issues

Acquisition's effects on Financial Statement

Complicating financial statement

Hindering comparability

Company with reporting issues is motivated to make acquisitions to conceal their prior misreporting.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.1. M&A Issues

4

Acquisition give opportunities to make choices that affect the initial consolidated BS and consolidated IS in the future

Acquiring companies often underestimate the value of identifiable net assets, thereby overestimating goodwill on acquisition. This result in:

- Less depreciation, which increase future profits.
- Goodwill is not amortized, and the loss from writing down can be timed.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.2. Divergence from Economic Reality

Certain accounting standards may give rise to financial reporting that an analyst may find less useful because he or she does not view it as reflective of economic reality.



An analyst should:

- If possible: adjust the reported information to better reflect his or her view of economic reality.
- If not possible: make a qualitative assessment of the effect

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.2. Divergence from Economic Reality

When faced with a restructuring charge, an impairment charge, or a combination of the two, an analyst should consider whether:

- Similar events occur regularly enough such that they should be factored into estimates of permanent earnings, or;
- They should be regarded as one-off items that provide little information about the future earnings of the remaining activities of the company.

Regular occurrence

Analyst should attempt to “normalize” earnings by essentially spreading the current restructuring / impairment charge(s) over past periods as well as the current period.

One-off items

Analyst is justified in “normalizing” earnings by excluding the item from earnings.

This process will require a significant amount of judgment, best informed by knowledge of the underlying facts and circumstances.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.2. Divergence from Economic Reality

Items that disrupt economic reality

The item

What to consider

Revisions to ongoing estimates

An earlier change in estimate would have been more appropriate

Sudden increases to allowances and reserves

The prior estimates resulted in overstatement of prior periods' earnings instead of an unbiased picture of economic reality.

Large accruals for losses

Prior periods' earnings may have been overstated because of the failure to accrue losses earlier.

- Management may use items such as reserves and allowances to manage or smooth earnings.
- No basis of accounting can be expected to recognize all of the economic assets and liabilities for an entity.

[LOS 5.b] *Explain* potential problems that affect the quality of financial reports

5.

M&A Issues and Divergence From Economic Reality

5.2. Divergence from Economic Reality

OCI items: should it be added to net income?

It is the analyst's subjective choice to include an item from OCI in net income

The item is decided to be appropriately added to net income

The analyst can adjust reported and forecasted amount

The item is decided **not** to be appropriately added to net income

Bypass the item

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of a company's financial reports

1.

Prior to the Evaluation

**Questions to
clarify the
purpose and
context**

- 1 What is the ***purpose*** of the analysis?
- 2 What ***level of detail*** will be needed to accomplish this purpose?
- 3 What ***data*** are available for the analysis?
- 4 What are the ***factors or relationships*** that will influence the analysis?
- 5 What are the analytical ***limitations***, and will these limitations potentially impair the analysis?

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of a company's financial reports

2.

General Steps to Evaluate the Quality of Financial Reports

- 1** *Develop an understanding* of the company and its industry
- 2** *Learn about management*
- 3** *Identify material areas* of accounting that are vulnerable to subjectivity
- 4** *Make cross-sectional and time series comparisons* of financial statements and important ratios.
- 5** *Check for warning signs* as discussed previously.
- 6** *Consider specific items* – if the firm operates in multiple segments.
- 7** *Use appropriate quantitative tools* to assess the likelihood of misreporting (described in the following section)

[LOS 5.c,d] *Describe* how to evaluate the quality of a company's financial reports; *Evaluate* the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

3.1. Beneish Model

Beneish Model

$$M\text{-score} = -4.84 + 0.920 (\text{DSR}) + 0.528 (\text{GMI}) + 0.404 (\text{AQI}) + 0.892 (\text{SGI}) + 0.115 (\text{DEPI}) - 0.172 (\text{SGAI}) + 4.679 (\text{Accruals}) - 0.327 (\text{LEVI})$$

Variables in Beneish Model

M-score

A score indicating the probability of earnings manipulation
→ The higher the M-score, the higher probability of manipulation

+ 0.920

Days Sales Receivable Index

$$\text{DSR} = \frac{\text{Receivables}_t / \text{Sales}_t}{\text{Receivables}_{t-1} / \text{Sales}_{t-1}}$$

Changes in the relationship between receivables and sales could indicate inappropriate revenue recognition. *DSR is positively correlated to M-score.*

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; **Evaluate** the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

Explain the variables in Beneish Model

+ 0.528

Gross Margin Index

$$GMI = \frac{\text{Gross margin}_t - 1}{\text{Gross margin}_t}$$

Deterioration in margins could predispose companies to manipulate earnings. *GMI is positively correlated to M-score.*

+ 0.404

Asset Quality Index

$$AQI = \frac{1 - (\text{PPE}_t + \text{CA}_t)/\text{TA}_t}{1 - (\text{PPE}_{t-1} + \text{CA}_{t-1})/\text{TA}_{t-1}}$$

Where:

- PPE: Property, plant and equipment
- CA: Current assets
- TA: Total assets

Change in the percentage of assets other than in PPE and CA could indicate excessive expenditure capitalization. *AQI is positively correlated to M-score.*

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; **Evaluate** the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

Explain the variables in Beneish Model

+ 0.892

Sales Growth Index

$$SGI = \frac{Sales_t}{Sales_{t-1}}$$

Managing the perception of continuing growth and capital needs from actual growth could predispose companies to manipulate sales and earnings. *SGI is positively correlated to M-score.*

+ 0.115

Depreciation Index

$$DEPI = \frac{Depreciation\ rate_{t-1}}{Depreciation\ rate_t}$$

Where: Depreciation rate = Depreciation / (Depreciation + PPE)

Declining depreciation rates could indicate understated depreciation as a means of manipulating earnings. *DEPI is positively correlated to M-score.*

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; **Evaluate** the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

Explain the variables in Beneish Model

- 0.172

Sales, general, and administrative expenses index

$$SGAI = \frac{SGA_t / Sales_t}{SGA_{t-1} / Sales_{t-1}}$$

An increase in fixed SGA expenses suggests decreasing administrative and marketing efficiency, which could predispose companies to manipulate earnings. *SGAI is negatively correlated to M-score.*

+ 4.679

Accruals

$$\text{Accruals} = \frac{\text{Income before extraordinary items} - \text{Cash from operations}}{\text{Total assets}}$$

Higher accruals can indicate earnings manipulation. *Accruals is positively correlated to M-score.*

[LOS 5.c,d] *Describe* how to evaluate the quality of a company's financial reports; *Evaluate* the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

Explain the variables in Beneish Model

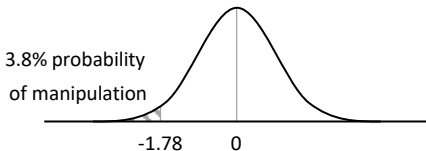
- 0.327

Leverage Index

$$LEVI = \frac{\text{Leverage}_t}{\text{Leverage}_{t-1}}$$

Increasing leverage could predispose companies to manipulate earnings.
LEVI is negatively correlated to M-score.

The M-score in Beneish model is a normally distributed random variable with a mean of 0 and standard deviation of 1



-1.78 (corresponding to 3.8%) is the cutoff value:

M-score > -1.78 → Company is likely a manipulator

[LOS 5.c,d] *Describe* how to evaluate the quality of a company's financial reports; *Evaluate* the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

3.2. Other Quantitative Models

Researchers have examined numerous factors that contribute to assessing the probability that a company is engaged in accounting manipulation.

Some of the useful variables are:

Accruals quality

Deferred taxes

Auditor change

Market-to-book value

Public or private
company?

Growth rate
differences between
financial and non-
financial variables

Aspects of corporate
governance

Incentive
compensation

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; **Evaluate** the quality ...

3.

Quantitative Tools to Assess the Likelihood of Misreporting

3.3. Limitations of Quantitative Models

1

Accounting is a partial representation of economic reality



Financial models based on accounting numbers aren't decision useful.

2

Manipulators are also aware of the power of quantitative models



Predictive power of the models is declining over time.



- It is necessary for analysts to use qualitative tools, not just quantitative tools, to assess quality.
- The search for more powerful analytical tools continues.

[LOS 5.e] *Describe* indicators of earnings quality.

1.

Elements Characterize High-quality Earnings

1.1. Recurring Earnings

Recurring Earnings: An Elements of High-quality Earnings

High-quality
earnings
assumed **high-
quality
reporting**

Sustainable: high-quality
earnings tend to persist
in the future.

Adequate: high-quality
earnings cover the
company's cost of capital.

Analysts
should focus
on earnings
that are
expected to
recur
**(Recurring
Earnings)**

[LOS 5.f] *Describe* the concept of sustainable (persistent) earnings

1. Elements Characterize High-quality Earnings

1.2. Earnings Persistence

Definition

Sustainability of earnings excluding items that are obviously non-recurring and persistence of growth in those earnings.

Using current and future earnings

Earnings Persistence Regression

$$\text{Earnings}_{(t+1)} = \alpha + \beta_1 \text{earnings}_{(t)} + \varepsilon$$

β_1 is the coefficient on current earnings – an expression of earnings persistence.

→ A higher coefficient (β_1) represents more persistent earnings.

[LOS 5.f] *Describe* the concept of sustainable (persistent) earnings

1. Elements Characterize High-quality Earnings

1.3. Accruals Components

Accruals

Accruals are revenues earned or expenses incurred, but cash related to the transaction has not yet changed hands.

This is contrary to the Cash flow component, which account for cash related transactions.

Earnings Persistence Regression

$$\text{Earnings}_{(t+1)} = \alpha + \beta_1 \text{Cash flow}_{(t)} + \beta_2 \text{Accruals}_{(t)} + \varepsilon$$

In the model, the coefficient on cash flow (β_1) has been shown to be higher than the coefficient on accruals (β_2), indicating that the cash flow component of earnings is more persistent. There are two types of accruals:

Non-discretionary (Normal) Accruals

- Accruals that arise from normal transaction in the period.
- Total accruals are regressed on the factors expected to give rise to normal accruals.

Discretionary (Abnormal) Accruals

- Accruals that arise from accounting choices, often with intention of distort.
- Regressed as residual (error).

[LOS 5.f] *Describe* the concept of sustainable (persistent) earnings

1. Elements Characterize High-quality Earnings

1.4. External indicators of poor-quality earnings

Other metrics used to identify potentially low-quality earnings:

- Companies repeatedly meet or barely beat consensus estimates;
- Companies face enforcement actions by regulatory authorities (external indicators)

[LOS 5.g] *Explain* mean reversion in earnings and how the accruals component of earnings affects the speed of mean reversion

1.

Mean Reversion

Definition

Extreme levels of earnings (both high and low) tend to revert to normal levels over time.

Extreme low earnings

The competitive marketplace corrects poor performance

Firms eliminate negative value projects
→ Losses are eliminated

Extreme high earnings

Successful projects attract capital

Increase competition
→ Lowering return

Analysts should not expect extreme earnings to continue indefinitely

- One cannot simply extrapolate either very high or very low earnings into the future and expect to construct useful forecasts.
- In order to be useful, analysts' forecasts need to take into account normalized earnings over the relevant valuation time frame.

[LOS 5.h] *Evaluate* the earnings quality of a company

1.

Revenue Recognition Issues

Revenue is the largest and most important element in the income statement.

However, revenue recognition is subjective, makes it highly vulnerable to manipulation.

Investors should always thoroughly and skeptically analyze revenues

Some aspects about revenues analysts should consider:

- Growth of revenues and whether growth came from acquisitions or organically;
- How revenues was generated.

→ Focus on the qualitative, not quantitative, aspects of revenues.

[LOS 5.h] *Evaluate* the earnings quality of a company

2. Looking for Quality in Revenues

1 *Understand the basics*

From the information disclosed, understand the **revenue recognition** practices followed by the company including relevant shipping terms, return policies, rebates, and the existence of multiple deliverables.

2 *Evaluate and question ageing receivables*

Compare receivables metrics with those from the past and with the industry median.

3 *Cash versus accruals*

Evaluate the proportion of earnings that are cash-based versus accruals-based.

4 *Compare financials with physical data provided by the company*

For example, correlate sales with capacity utilization data.

5 *Evaluate revenue trends and compare with peers*

Narrow down such analysis by segments.

6 *Check for related party transactions*

Check and take care of sales to affiliated entity.

[LOS 5.h] *Evaluate* the earnings quality of a company

3. Looking for Quality in Expense Recognition

1 *Understand the basics*

Understand the company's cost capitalization policies based on information disclosed in the annual report. Also gather information about depreciation policies and how they compare with those of the company's peers.

2 *Trend and comparative (peer) analysis*

Evaluate changes in non-current assets over time to see if there are any anomalies which could be explained by cost capitalization.

- Stable or improving profit margins coupled with a buildup of non-current assets would be a warning sign.
- Steady or rising revenue coupled with declining asset turnover ratios is another warning sign of cost capitalization.

3 *Check for related party transactions*

An example of a problematic transaction would be if the company is shifting resources to a privately held company that is owned by senior managers.

On the other hand, analysts should also watch for propping practices whereby profits from related entities temporarily prop up an ailing public company.

[LOS 5.I] *Describe* indicators of cash flow quality

1.

Indicators of Cash Flow Quality

**Operating Cash Flow
(OCF)**

**Investing Cash Flow
(ICF)**

**Financing Cash Flow
(FCF)**

Among the cash flow components, OCF is the most important. Discussions of cashflow quality, therefore, typically focus on OCF.

Similar to the term “earnings quality”, when **reported cash flows are described as being of high quality**:

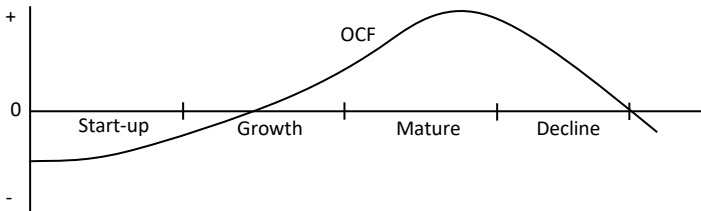
The company had
high reporting quality

The company’s underlying **economic performance was good**

[LOS 5.I] *Describe* indicators of cash flow quality

1. Indicators of Cash Flow Quality

The corporate life cycle and industry profile affect cash flow and must be considered when analyzing the statement of cash flows.



Characteristics of High-quality Cash Flow

Positive OCF

OCF derived from sustainable sources

OCF adequate to cover capital expenditures, dividends, and debt repayments

OCF with relatively low volatility (relative to industry participants)

[LOS 5.I] *Describe* indicators of cash flow quality

1.

Indicators of Cash Flow Quality

Subjectivity to manipulation

- Overall, cash flows are less amenable to manipulation.
- However, management can affect cash flows via strategic decisions (timing issues).
- Management may also try to shift positive cash flows from investing or financing activities into operating activities to boost OCF (classification issues).

Analysts can identify such practices by reviewing activity ratios (receivables and payables turnover ratios).

[LOS 5.i] *Evaluate* the cash flow quality of a company

1. Evaluate the cash flow quality of a company

1 *Checking for any unusual items or items that have not shown up in prior years*

2 *Checking revenue quality*

- Aggressive revenue recognition practices typically result in an increase in receivables - which reduces operating cash flow.
- Another common indicator of aggressive revenue recognition is an increase in inventories (and hence a cash outflow) when sham sales are reversed (i.e., treated as returns from customers).

3 *Checking for strategic provisioning*

Provisions for restructuring charges show up as an inflow (i.e., a non-cash expense) in the year of the provision and then as an outflow when ordinary operating expenses are channeled through such reserves.

[LOS 5.j,k] *Describe* indicators of balance sheet quality; *Evaluate* the balance sheet quality of a company

High-quality financial balance sheet reporting is evidenced by

Completeness

Unbiased
measurement

Clarity of
presentation

1.

Completeness



Definition: Completeness means that all financial information that is relevant and necessary to provide a fair view of the company should be included.

Off-balance-sheet item deteriorate completeness → Significant amounts of off-balance-sheet obligations could be a concern for an analyst.

[LOS 5.j,k] Describe indicators of balance sheet quality; Evaluate the balance sheet quality of a company

High-quality financial balance sheet reporting is evidenced by

Completeness

**Unbiased
measurement**

**Clarity of
presentation**

2. Unbiased Measurement

The balance sheet reflects subjectivity in the measurement of several assets and liabilities:

Items

Value of the pension liability

Value of investment

Goodwill value

Inventory valuation

PP&E and other assets

Affected by

Several actuarial assumptions

Unavailable market price

Impairment testing

Impairment testing

Impairment

[LOS 5.j,k] *Describe* indicators of balance sheet quality; *Evaluate* the balance sheet quality of a company

High-quality financial balance sheet reporting is evidenced by

Completeness

**Unbiased
measurement**

**Clarity of
presentation**

3.

Clear Presentation

Although accounting standards specify many aspects of what appears on the balance sheet, companies have discretion on how to report them.

For example:

- Companies have discretion regarding which items they present as a single-line item versus those that are grouped together.
- When these items are separated (i.e. more clearly presented,) analysts can gain more information.

[LOS 5.j,k] *Describe* sources of information about risk

1.

Sources of Information About Risk

Financial Statements

Financial statements contain information regarding the leverage used by the company and the variability of cash flows and earnings over time.

Auditor's report

Involuntary changes in auditors, a small-sized audit firm relative to the size of the company being audited, and a lack of auditor independence are red flags that an analyst should pay special attention to.

Foot notes

Companies are required to make certain risk related disclosures in the notes to financial statements.

MD&A

Companies should include principal risks that are unique to the business (as opposed to risks faced by most businesses) in their MD&A.

SEC Form 'NT'

In the United States, SEC form 'NT' is filed when a firm is unable to file required reports in a timely manner.

Financial press

Often the initial information about accounting irregularities at a company is obtained from the financial press.



Publication 505
2018-19



Tax Withholding and Estimated

Contents

Introduction
What's New for 2018
Exemptions
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MODULE 6: INTEGRATION OF FINANCIAL STATEMENT ANALYSIS TECHNIQUES

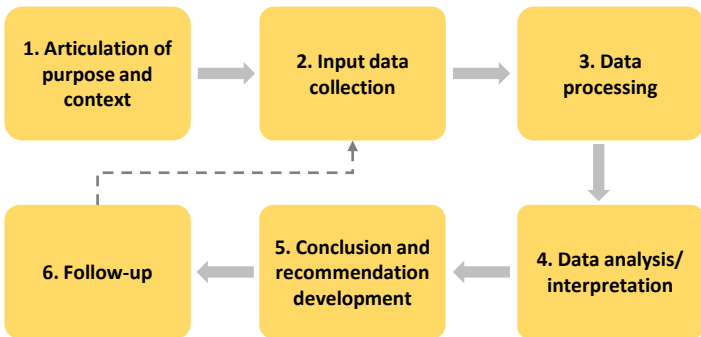
LEARNING OUTCOMES

- 6.a.** Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose (e.g., valuing equity based on comparables, critiquing a credit rating, obtaining a comprehensive picture of financial leverage, evaluating the perspectives given in management's discussion of financial results)
- 6.b.** Identify financial reporting choices and biases that affect the quality and comparability of companies' financial statements and explain how such biases may affect financial decisions
- 6.c.** Evaluate the quality of a company's financial data and recommend appropriate adjustments to improve quality and comparability with similar companies, including adjustments for differences in accounting standards, methods, and assumptions
- 6.d.** Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements and ratios
- 6.e.** Analyze and interpret how balance sheet modifications, earnings normalization, and cash flow statement related modifications affect a company's financial statements, financial ratios, and overall financial condition

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

Some equity analysts want to evaluate potential investments in a company's equity securities as a basis for deciding whether a prospective investment is attractive and what an appropriate purchase price might be. This section presents a generic framework for financial statement analysis that can be used in these various tasks.

A financial statement analysis framework



[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

Phase	Input	Output
1. Establish the objectives	- Perspective of the analyst - Needs or concerns communicated by the client or supervisor - Institutional guidelines	- Purpose statement - Specific questions to be answered - Nature and content of the final report - Timetable and resource budget
2. Collect data	- Financial statements - Communication with management, suppliers, customers and competitors	Organized financial information
3. Process data	Data from step 2	- Adjusted financial statements - Common-size statements - Ratios - Forecasts
4. Analyze data	Data from step 2 and 3	Results
5. Develop and communicate conclusions	- Results from analysis - Published report guidelines	- Report answering questions posed in step 1 - Recommendations
6. Follow up	Periodically updated information	Updated analysis and recommendations

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

Nestlé S.A. case study

The portfolio manager for the food sector of a large public employee pension fund wants to take a long-term equity position in a publicly traded food company and has become interested in Nestlé S.A., a global company.

In its 2014 annual report, Nestlé’s management outlined its long-term objectives for organic growth, margin and earnings per share improvement, and capital efficiency. The management report indicated the following general strategic direction: “Our ambition is not just to be the leader but the industry reference for Nutrition, Health and Wellness. In recent years we have built on the strong foundations of our unrivalled food and beverage portfolio, exploring the benefits of nutrition’s therapeutic role with Nestlé Health Science.” Nestlé’s stated objectives, including expansion of the company’s mission into “nutrition’s therapeutic role,” captured the portfolio manager’s attention: She became intrigued with Nestlé as an investment possibility.

She asks an analyst to evaluate Nestlé for consideration as a large core holding.

 *What does the portfolio manager want to know from the announcement of Nestlé?*

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

 *What does the portfolio manager want to know from the announcement of Nestlé?*

In its 2014 annual report, Nestlé's management outlined its long-term objectives for organic growth, margin and earnings per share improvement, and capital efficiency. The management report indicated the following general strategic direction: "Our ambition is not just to be the leader but the industry reference for Nutrition, Health and Wellness. In recent years we have built on the strong foundations of our unrivalled food and beverage portfolio, exploring the benefits of nutrition's therapeutic role with Nestlé Health Science." Nestlé's stated objectives, including expansion of the company's mission into "nutrition's therapeutic role," captured the portfolio manager's attention: She became intrigued with Nestlé as an investment possibility.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

 *What does the portfolio manager want to know from the announcement of Nestlé?*

Earnings

- What are Nestlé's sources of **earnings growth**?
- How **sustainable** is Nestlé's performance?
- Do the company's reported earnings represent its **economic reality**?
- In determining the quality of earnings over a long-term time frame, the portfolio manager wants to understand the **relationship of earnings to cash flow**.

Balance sheet

- How well Nestlé's balance sheet takes into account the company's **full rights and obligations**?
- Whether the **capital structure** of the company can support future operations and strategic plans?

To address the portfolio manager's concerns, the analyst develops a plan of analysis based on The financial statement analysis framework.

Continue next slide →

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

Framework	Application for Case study
1. Phase 1: <i>Define a Purpose for the Analysis</i>	- Identify the factors that have driven the company's financial success and assessing their sustainability. - Identify and understand the risks that may affect the sustainability of returns.
2. Phase 2: <i>Collect Input Data</i>	Financial statements can be collected on its website.
3. Phase 3: <i>Process Data</i>	3.1. Sources of earnings and return on equity. 3.2. Asset base. 3.3. Capital structure. 3.4. Capital allocation decisions.
3. Phase 4: <i>Analyze/Interpret Data</i>	3.5. Earnings quality and cash flow analysis. 3.6. Market value decomposition. 3.7. Anticipating changes in accounting standards.
4. Phase 5: <i>Develop conclusions and recommendations</i>	As a result of of the analyses performed, the analyst has gathered sufficient evidence and believes he is able to address the concerns.
5. Phase 6: <i>Follow-up</i>	The analyst concludes his findings and gives recommendation on Nestlé shares.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

1.

Phase 1 – Establish the objectives

The analyst states the purpose and context of the analysis as:

- Identify the factors that have driven the company's financial success and assessing their sustainability.
- Identify and understand the risks that may affect the sustainability of returns.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

2.

Phase 2 – Collect data

The analyst finds that Nestlé has an extensive collection of financial statements on its website. After gathering several years of annual reports, he is ready to begin processing the data.

Case study: General information



Nestlé S.A. is a global packaged food and beverage company headquartered in Switzerland.



L'ORÉAL

NESPRESSO.

DIESEL

YVES SAINT LAURENT



PURINA

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3. Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

Sources of earnings and return on equity can be decomposed using:



Recall: Five-Way DuPont Decomposition

$$\begin{aligned}
 \text{ROE} &= \frac{\text{Net income}}{\text{Equity}} \\
 &= \frac{\text{Net income}}{\text{EBT}} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Average TA}} \times \frac{\text{Average TA}}{\text{Average shareholders' equity}} \\
 &= \text{Tax burden} \times \text{Interest burden} \times \text{EBIT margin} \times \text{Total asset turnover} \times \text{Financial Leverage}
 \end{aligned}$$

- The **tax burden ratio** measures the proportion of its pretax profits that a company gets to keep. A *higher* tax burden ratio implies that the company can keep a *higher* percentage of its pretax profits.
- The **interest burden ratio** captures the effect of interest expense on ROE. High borrowing costs reduce ROE. As interest expense rises, EBT will fall as a percentage of EBIT, the interest burden ratio will fall, and ROE will also fall.
- The **EBIT margin** captures the effect of operating profitability on ROE.
- The **asset turnover** ratio is an indicator of the overall efficiency of the company, while the **leverage ratio** measures the total value of a company's assets relative to its equity capital.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3. Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

The analyst must consider the firm's sources of income and whether the income is generated internally from operations or externally:

Illustration 1: Extracted data from Nestlé S.A. Income statements, 2024 – 2012 (CHF millions)

	2014	2013	2012
Income statement data:			
Sales	91,612	92,158	89,721
Operating profit (EBIT)	10,905	13,068	13,388
Profit before taxes, associates and joint ventures (EBT)	10,268	12,437	12,683
Taxes	(3,367)	(3,256)	(3,259)
Income from associates and joint ventures	8,003	1,264	1,253
% of Profit	53.7%	12.10%	11.74%
Profit for the year	14,904	10,445	10,677

The analyst notes that Nestlé has a significant amount of “income from associates and joint ventures” in all three years, especially in 2014.

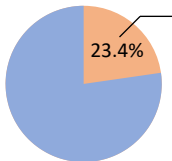
[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

From the analyst finding, in 2014, “income from associates and joint ventures” amounted to CHF8,003 million, or 53.7%, of Nestlé’s net income (referred to by Nestlé as “profit for the year”).



Nestlé has a significant equity position (23.4%) in L'Oréal (Paris exchange: OR), a French cosmetics company.

→ L'Oréal is accounted for in the financial statements as an investment in associates.

Equity position of L'Oréal

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

The returns earned by associates are not under the direct control of Nestlé's management → the analyst wants to remove the effects of the investments in associates from the balance sheet and income statement.

Illustration 2: Data needed for DuPont analysis (CHF millions)

Income statement data	2014	2013	2012
Sales	91,612	92,158	89,721
Operating profit (EBIT)	10,905	13,068	13,388
Profit before taxes, associates and joint ventures (EBT)	10,268	12,437	12,683
Profit for the year	14,904	10,445	10,677
(Less): Income from associates and joint ventures	(8,003)	(1,264)	(1,253)
Profit excluding associates and joint ventures	6,901	9,181	9,424

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

The returns earned by associates are not under the direct control of Nestlé's management → the analyst wants to remove the effects of the investments in associates from the balance sheet and income statement.

Illustration 2: Data needed for DuPont analysis (CHF millions)

Balance sheet data	2014	2013	2012
Total assets	133,450	120,442	125,877
(Less): Investment in associates and joint ventures	(8,649)	(12,315)	(11,586)
Total assets, excluding associates and joint ventures	124,801	108,127	114,291
Total equity	71,884	64,139	62,664
(Less): Investment in associates and joint ventures	8,649	12,315	11,586
Total equity, excluding associates and joint ventures	62,235	51,824	51,078

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

Let's continue with Nestlé S.A. case study, we use DuPont analysis based on data removed from the effects of the investments in associates:

Illustration 3: Expanded DuPont analysis

	2014	2013	2012
Tax burden (excl. associates)	67.21%	73.82%	74.30%
× Interest burden	94.16%	95.17%	94.73%
× EBIT margin	11.90%	14.18%	14.92%
= Net profit margin (excl. associates)	7.53%	9.96%	10.50%
× Associates' effect on net profit margin	216.07%	113.76%	113.33%
= Net profit margin	16.27%	11.33%	11.90%
Total asset turnover (excl. associates)	0.787	0.829	0.825
Effect of associates' investments on turnover	(0.065)	(0.081)	(0.075)
× Total asset turnover	0.722	0.748	0.750
= Return on assets	11.75%	8.47%	8.93%
× Leverage	1.87	1.94	1.98
= Return on equity (ROE)	21.97%	16.44%	17.67%

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

We compare ROE calculated *including and excluding* the effects of associates:

Illustration 4: ROE performance due to investment in associates

	2014 (%)	2013 (%)	2012 (%)
ROE including associates 	21.97	16.44	17.67
Less: Nestlé-only ROE	11.08	16.02	17.15
Associates' contribution to ROE	10.89	0.42	0.52

The trend in the ROE including associates, which shows a significant increase in 2014, is largely the result of the gains in 2014 from the transactions involving the investments in associates (*we have previously discussed*).

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

We compare ROE calculated *including and excluding* the effects of associates:

Illustration 4: ROE performance due to investment in associates

	2014 (%)	2013 (%)	2012 (%)
ROE including associates	21.97	16.44	17.67
Less: Nestlé-only ROE ↓	11.08	16.02	17.15
Associates' contribution to ROE	10.89	0.42	0.52

The analyst is particularly troubled by the sharp drop-off in the Nestlé-only ROE in 2014.



What happen with the ongoing operations of Nestlé?

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

Looking back in the Expanded DuPont analysis:

Illustration 3: Expanded DuPont analysis

	2014	2013	2012
Tax burden (excl. associates)	67.21%	73.82%	74.30%
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[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

Looking back in the Expanded DuPont analysis:



What happen with the ongoing operations of Nestlé?

The net profit margin excluding income from associates shows a disturbing trend: It decreased each year in the 2012–2014 period.

This finding prompts the analyst to try to identify a reason for the declining profitability of the Nestlé-only business. *(Further discussed in the next slides)*

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

Searching the income statements and notes in the annual reports:

Illustration 5: Profitability adjusted for provisions and impairment charges (CHF million)

	2014	2013	2012
Profit (excl. associates – <i>Illustration 2</i>)	6,901	9,181	9,424
Impairment of goodwill	1,908	114	14
Total provisions for restructuring, environmental, litigation, and other (not tax-affected)	920	862	618
Profit adjusted for unusual charges	9,729	10,157	10,056

The analyst notices that Nestlé has recorded goodwill impairments over the period under study, with a particularly large one, CHF1,908 million, occurring in 2014.

→ This impairment was related to Nestlé's acquisitions of ice cream and pizza businesses in the United States.

He also notices that Nestlé has recorded provisions each year for restructuring activities, environmental liabilities, litigation reserves, and other activities.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

To see how much these events affected the Nestlé-only profitability, he constructs the percentage of net profit margin with and without unusual charges:

Illustration 6: Profitability adjusted for provisions and impairment charges in percentage

		2014 (%)	2013 (%)	2012 (%)
Net profit margin (excl. associates, incl. all unusual charges)	↓	7.53	9.96	10.50
Net profit margin (excl. associates and unusual charges)	↓	10.62%	11.02%	11.21%
Profit margin consumed by unusual charges		3.09%	1.06%	0.71%

The analyst knows that there was an unusually large goodwill impairment charge in 2014 (CHF1,908 millions), which may explain the sudden decrease.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

To see the role played by such unusual charges in the ROE trend, he reworks the Nestlé-only ROE figures on the basis of revised net profit margins (excluding associates and unusual charges):

Illustration 7: Nestlé-only ROE, with unusual charges removed from pre-tax margins

	2014 (%)	2013 (%)	2012 (%)
Nestlé-only ROE ↓	15.63	17.73	18.31

Absent the unusual charges, the magnitude of the Nestlé-only ROE improved significantly in all three years, but the trend remained on a downward slope.

Illustration 4: ROE performance due to investment in associates

	2014 (%)	2013 (%)	2012 (%)
ROE including associates	21.97	16.44	17.67
Less Nestlé-only ROE ↓	11.08	16.02	17.15
Associates' contribution to ROE	10.89	0.42	0.52

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.1. Sources of earnings and return on equity

Returning to the DuPont analysis, the analyst now realizes the significance of the associates' earnings to the entire Nestlé entity

- The margin is greater in each year if the associates' earnings are included in net profit as opposed to looking at Nestlé alone. Consistently, the company's profit margins are smaller without the boost from associates' earnings.
- Asset turnover is consistently lower when assets include the investment in associates.
- The analyst assumes that there will be no change in the Nestlé capital structure and that a similar blend of debt and equity in the company's capital structure finances the investment in associates' assets and the Nestlé-only assets → The financial leverage ratio has not been adjusted by the analyst.



In conclusion:

- Nestlé's core business has deteriorated in profitability during the examined period.
- There was an unusually large goodwill impairment charge in 2014.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

3.

Phase 3 & 4 – Process and analyze data

3.2. Asset base

The analyst examines the composition of the balance sheet over time:

Illustration 8: Nestlé S.A. Balance sheets, 2014 – 2012 (CHF millions)

	2014	2013	2012
Current assets	33,961	30,066	34,020
<i>% of Total assets</i>	25.5%	24.9%	21.1%
PP&E, net	28,421	26,895	26,576
<i>% of Total assets</i>	21.3%	22.3%	21.1%
Goodwill	34,557	31,039	32,688
<i>% of Total assets</i>	25.9%	25.8%	26.0%
Intangible assets	19,800	12,673	13,018
<i>% of Total assets</i>	14.8%	10.5%	10.3%
Other non-current	16,711	19,769	19,575
<i>% of Total assets</i>	12.5%	16.4%	15.6%
Total assets	133,450	120,442	125,877

Given that Nestlé is a food manufacturer and marketer, he expected significant investments in current assets, inventory, and physical plant assets.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

3.

Phase 3 & 4 – Process and analyze data

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Intangible assets	19,800	12,673	13,018
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Other non-current	16,711	19,769	19,575
% of Total assets	12.5%	16.4%	15.6%
Total assets	133,450	120,442	125,877

However, he is surprised to see so much investment in intangible assets, indicating that Nestlé's success may depend, in part, on successful acquisitions.

→ The increases in Net profit margin and ROE incl. associates (*Illustration 3*) may be partially due to successful acquisitions.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

3.**Phase 3 & 4 – Process and analyze data**

3.2. Asset base

In conclusion:

There was a significant proportion of goodwill and intangible assets in asset structure of Nestlé – a food manufacturer and marketer. However, since goodwill is no longer amortized through the income statement, the analyst must consider the possibility of losses in the future if goodwill is determined to have been impaired.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

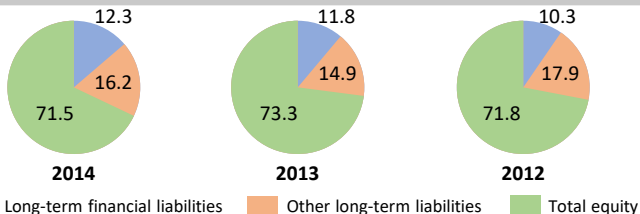
3. Phase 3 & 4 – Process and analyze data

3.3. Capital structure

From the DuPont analysis, the analyst understands that Nestlé's overall financial leverage was rather stable over the last three years. However, it does not reveal the true nature of the leverage, as some liabilities are more burdensome than others.

The analyst decides to investigate Nestlé's capital structure more deeply by constructing a chart on a common-size basis:

Illustration 9: Percentages of Long-term capital structure



Not only is the proportion of equity financing decreasing, but long-term financial liabilities have become a **significantly greater** part of the capital mix → Nestlé has been making its capital structure financially riskier.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

3.

Phase 3 & 4 – Process and analyze data

3.3. Capital structure

Given the increased leverage in the long-term capital structure, the analyst wonders whether there have also been changes in the company's working capital accounts:

Illustration 10: Nestlé working capital accounts and ratios, 2012 – 2014

		2014	2013	2012
Current ratio		1.03	0.91	0.88
Quick ratio		0.68	0.59	0.58
Defensive interval ratio		106.6	91.9	110.0
Days sales outstanding (DSO)		51.1	50.0	53.0
Days on hand of inventory (DOH)		67.4	65.7	69.3
Days payable outstanding (DPO)		(126.5)	(117.8)	(108.6)
Cash conversion cycle		(8.0)	(2.1)	13.7

- The current and quick ratios improved slightly.
- There was a modest improvement in the defensive interval ratio.
- The management of the working capital accounts has changed so much (mainly from the increase of DPO) that Nestlé now has a **negative 8 days** for its cash conversion cycle.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

3.

Phase 3 & 4 – Process and analyze data

3.3. Capital structure

In conclusion:

- Nestlé has the financial stability to fund growth in its existing operations and carry out its growth-by-acquisition strategy.
- The capital structure is capable of supporting future operations and strategic plans.

[LOS 6.d] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements and ratios

3. Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

The analyst identifies is that Nestlé reports segment information by management responsibility and geographic area, **not by segments based exclusively on geographic areas**:

Illustration 11: Revenue and EBIT by segment (CHF millions)

	Sales			EBIT		
	2014	2013	2012	2014	2013	2012
Europe	15,175	15,567	15,388	2,327	2,331	2,363
% of Total	16.6%	16.9%	17.2%	16.6%	16.6%	17.6%
Americas	27,277	28,358	28,613	5,117	5,162	5,346
% of Total	29.8%	30.8%	31.9%	36.5%	36.7%	39.7%
Asia, Oceania and Africa	18,272	18,851	18,875	3,408	3,562	3,579
% of Total	19.9%	20.5%	21.0%	24.3%	25.4%	26.6%
Nestlé Waters	7,390	7,257	7,174	714	665	640
% of Total	8.1%	7.9%	8.0%	5.1%	4.7%	4.8%
Nestlé Nutrition	9,614	9,826	7,858	1,997	1,961	1,509
% of Total	10.5%	10.7%	8.8%	14.2%	14.0%	11.2%
Other businesses	13,884	12,299	11,813	2,654	2,175	2,064
% of Total	15.2%	13.3%	13.2%	18.9%	15.5%	15.3%
Unallocated items	–	–	–	(2,198)	(1,809)	(2,037)
% of Total	–	–	–	(15.7%)	(12.9%)	(15.1%)
Total	91,612	92,158	89,721	14,019	14,047	13,464

[LOS 6.d] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements and ratios

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

In terms of contributing revenue and EBIT, the Americas segment is the largest segment while the Nestlé Waters segment is the smallest. Also, the percentage contribution to EBIT by the Americas segment declined from 31.9% to 29.8% over the period.

Illustration 11: Revenue and EBIT by segment – Extracted data (CHF millions)

	Sales			EBIT		
	2014	2013	2012	2014	2013	2012
Americas – <i>largest</i>	27,277	28,358	28,613	5,117	5,162	5,346
% of Total	29.8%	30.8%	31.9%	36.5%	36.7%	39.7%
Nestlé Waters – <i>smallest</i>	7,390	7,257	7,174	714	665	640
% of Total	8.1%	7.9%	8.0%	5.1%	4.7%	4.8%

[LOS 6.d] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements and ratios

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

The analyst remembers that Nestlé acquired the Wyeth Nutritionals business in 2012, which would explain the solid growth of Nestlé Nutrition segment.

Illustration 11: Revenue and EBIT by segment – Extracted data (CHF millions)

	Sales			EBIT		
	2014	2013	2012	2014	2013	2012
Nestlé Nutrition	9,614	9,826	7,858	1,997	1,961	1,509
% of Total	10.5%	10.7%	8.8%	14.2%	14.0%	11.2%
Other businesses	13,884	12,299	11,813	2,654	2,175	2,064
% of Total	15.2%	13.3%	13.2%	18.9%	15.5%	15.3%

“Other businesses,” which is a collectively large group of disparate businesses (*mainly includes Nespresso, Nestlé Professional, Nestlé Health Science, and Nestlé Skin Health*), also increased in importance between 2012 and 2014.

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

The analyst moves on with his segment analysis and evaluation of capital allocation:

Illustration 12: Assets and Capital expenditures by Segment

	Assets			Capital expenditures		
	2014	2013	2012	2014	2013	2012
Europe	11,308	11,779	11,804	747	964	1,019
% of Total	11.4%	13.8%	13.4%	20.0%	20.5%	20.2%
Americas	20,915	21,243	22,485	1,039	1,019	1,073
% of Total	21.1%	24.9%	25.5%	27.9%	21.6%	21.3%
Asia, Oceania and Africa	15,095	14,165	14,329	697	1,280	1,564
% of Total	15.2%	16.6%	16.2%	18.7%	27.2%	31.0%
Nestlé Waters	6,202	6,046	6,369	308	377	407
% of Total	6.2%	7.1%	7.2%	8.3%	8.0%	8.1%
Nestlé Nutrition	24,448	22,517	24,279	363	430	426
% of Total	24.6%	26.4%	27.5%	9.7%	9.1%	8.5%
Other businesses	21,345	9,564	9,081	573	642	550
% of Total	21.5%	11.2%	10.3%	15.4%	13.6%	10.9%
Total	99,313	85,314	88,347	3,727	4,712	5,039

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3. Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

Using the information from Illustration 11 to calculate EBIT margins, as well as the information about the asset and capital expenditure distribution from Illustration 12:

Illustration 13: EBIT margin and CAPEX % to Assets % by Segment – Ranked by EBIT margin

	EBIT margin (%)			CAPEX%/Assets%		
	2014	2013	2012	2014	2013	2012
Nestlé Nutrition	20.77	19.96	19.20	0.39	0.34	0.31
Other businesses	19.12	17.68	17.47	0.72	1.21	1.06
Americas	18.76	18.20	18.68	1.32	0.87	0.84
Asia, Oceania and Africa	18.65	18.90	18.96	1.23	1.64	1.91
Europe	15.33	14.97	15.36	1.75	1.49	1.51
Nestlé Waters	9.66	9.16	8.92	1.34	1.13	1.13

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.
Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

Ratios of CAPEX% to Total assets%:

$$\text{Ratio} = \frac{\text{CAPEX\%}}{\text{Total assets\%}} = \frac{\text{CAPEX of segment/Total CAPEX}}{\text{Total assets of segment/Total assets}}$$

Implication:

- Ratio > 1: The segment is receiving a “growth allocation” of capital spending → Comparing the ratio with the EBIT margin percentage gives the analyst an idea of whether **the company is investing its capital in the most profitable segments.**
- Ratio < 1: The segment is allocated a smaller percentage of its capital expenditures to a segment than its proportion of total assets. If these trends continue, the segments will represent a more or less significant proportion of the firm over time.

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3. Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

Illustration 13: EBIT margin and CAPEX % to Assets % by Segment – Ranked by EBIT margin

	EBIT margin (%)			CAPEX%/Assets%		
	2014	2013	2012	2014	2013	2012
Nestlé Nutrition	20.77	19.96	19.20	0.39	0.34	0.31
Other businesses	19.12	<i>Highest</i>	17.47	0.72	<i>Lowest</i>	1.06
Americas	18.76	18.20	18.68	1.32	0.87	0.84
Asia, Oceania and Africa	18.65	18.90	18.96	1.23	1.64	1.91
Europe	15.33	14.97	15.36	1.75	1.49	1.51
Nestlé Waters	9.66	9.16	8.92	1.34	1.13	1.13

- The most profitable segment is Nestlé Nutrition, but over the last three years, it has received the lowest proportion of capital expenditures.
- The company has invested in the nutrition segment by acquisition, such as the Wyeth Nutritionals business in 2012.

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

**Illustration 13: EBIT margin and CAPEX % to Assets % by Segment
– Ranked by EBIT margin**

	EBIT margin (%)			CAPEX%/Assets%		
	2014	2013	2012	2014	2013	2012
Nestlé Nutrition	20.77	19.96	19.20	0.39	0.34	0.31
Other businesses	19.12	17.68	17.47	0.72	1.21	1.06
Americas	18.76	18.20	18.68	1.32	0.87	0.84
Asia, Oceania and Africa	18.65	18.90	18.96	1.23	1.64	1.91
Europe	15.33	14.97	15.36	1.75	1.49	1.51
Nestlé Waters	9.66	9.16	8.92	1.34	1.13	1.13

- The Other businesses segment is the next most profitable segment in EBIT margin terms. Although it did not receive a “growth allocation” of capital expenditures in 2014, it received a growth allocation in the previous two years.
- Given their profitability levels and substantial operations, the analyst is encouraged to see that Americas segment and the Asia, Oceania, and Africa segment are receiving “growth allocations” of capital spending.

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

Illustration 13: EBIT margin and CAPEX % to Assets % by Segment – Ranked by EBIT margin

	EBIT margin (%)			CAPEX%/Assets%		
	2014	2013	2012	2014	2013	2012
Nestlé Nutrition	20.77	19.96	19.20	0.39	0.34	0.31
Other businesses	19.12	17.68	17.47	0.72	1.21	1.06
Americas	18.76	18.20	18.68	1.32	0.87	0.84
Asia, Oceania and Africa	18.65	18.90	18.96	1.23	1.64	1.91
Europe	15.33	14.97	15.36	1.75	1.49	1.51
Nestlé Waters	9.66	9.16	8.92	1.34	1.13	1.13

- Although European segment and Nestlé Waters segment have high proportional allocation of capital spending, they have had the lowest profit margins.
- In a worst-case scenario, if the company were to continue making growth allocations of capital toward the lowest-margined businesses, such as Europe and Nestlé Waters, the overall Nestlé-only returns might be affected negatively.

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

Accrual-based measures such as EBIT may not be a good indicator of an entity's ability to generate cash flow. The analyst would rather evaluate segmental capital allocation decisions based on cash flows generated by each segment:

As segment cash flow data is generally not reported, the analyst can approximate cash flow as:

$$\text{Estimated cash flow} = \text{EBIT} + \text{Depreciation and amortization}$$

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

Illustration 14: Depreciation and amortization by Segment

	2014	2013	2012
Europe	473	517	533
Americas	681	769	899
Asia, Oceania and Africa	510	520	553
Nestlé Waters	403	442	491
Nestlé Nutrition	330	337	176
Other businesses	525	437	295

Illustration 15: Segment cash generation and Average assets (CHF millions)

	Segment cash generation			Average total assets		
	2014	2013	2012	2014	2013	2012
Europe	2,800	2,848	2,896	11,544	11,792	11,683
Americas	5,798	5,931	6,245	21,079	21,864	22,783
Asia, Oceania and Africa	3,918	4,082	4,132	14,630	14,247	14,068
Nestlé Waters	1,117	1,107	1,131	6,124	6,208	6,486
Nestlé Nutrition	2,327	2,298	1,685	23,483	23,398	18,564
Other businesses	3,179	2,612	2,359	15,455	9,323	10,009

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3. Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

The analyst computes each segment's cash operating return on total assets (Illustration 15) and compares the results with the 2014 ranking of capital expenditures (Illustration 13) as well as the EBIT margins.

**Illustration 16: Segment cash operating return on assets
– Ranked by CAPEX%/Asset%**

	2014		Segmental CF/Aver. TA		
	CAPEX%/Asset%	EBIT%	2014 (%)	2013 (%)	2012 (%)
Europe	1.75	15.3	24.3	24.2	24.8
Nestlé Waters	1.34	9.7	18.2	17.8	17.4
Americas	1.32	18.8	27.5	27.1	27.4
Asia, Oceania and Africa	1.23	18.7	26.8	28.7	29.4
Other businesses	0.72	19.1	20.6	28.0	23.6
Nestlé Nutrition	0.39	20.8	9.9	9.8	9.1

(Lighter shading indicates the highest numbers. Darker shading indicates the lowest numbers)

It can be seen that the Nestlé Nutrition segment—while continuing to produce superior operating margins—has fallen behind in cash generation in the latest year.

[LOS 6.e] Analyze and interpret how modifications affect a company's FS, financial ratios, and overall financial condition

3.

Phase 3 & 4 – Process and analyze data

3.4. Capital allocation decisions

In conclusion:

- The acquisition activities in the Nutrition and Health Science product group do not appear to build on the company's traditional strengths.
- Although the Nestlé Nutrition segment show excellent EBIT margins, they rank very low in return on assets.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3. Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Earnings quality refers to the persistence and sustainability of a firm's earnings. **Earnings that are closer to operating cash flow are considered higher quality.**

Earnings analysis

Earnings are subject to accrual accounting events that require numerous judgments and estimates. Accruals ratio is measured based on Net operating assets (NOA) in two approaches:

Balance sheet approach

Accruals ratio for time t

$$= \frac{NOA_t - NOA_{t-1}}{(NOA_t + NOA_{t-1})/2}$$

Cash flow statement approach

Accruals ratio for time t

$$= \frac{[NI_t - (CFO_t + CFI_t)]}{(NOA_t + NOA_{t-1})/2}$$

Although both accruals ratios are conceptually equivalent (both measure the degree of accruals present in a firm's earnings), their results can differ because of acquisitions and divestitures, exchange rate gains and losses, and inconsistent treatment of specific items on the balance sheet and on the cash flow statement.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

The analyst decides to examine the balance-sheet-based accruals and the cash-flow-based accruals over the last few years:

Illustration 17: Selected information from BS and SOCF (CHF millions)

Balance sheet Accrual info	2014	2013	2012
Total assets	133,450	120,442	125,877
Cash and short-term investments	8,881	7,053	9,296
Operating assets (1)	124,569	113,389	116,581
Total liabilities	61,566	56,303	63,213
Long-term debt	12,396	10,363	9,008
Debt in current liabilities	8,810	11,380	18,408
Operating liabilities (2)	40,360	34,560	35,797
Net operating assets [(1) – (2)]	84,209	78,829	80,784
Balance-sheet-based aggregate accruals (NOA_t – NOA_{t-1})	5,380	(1,955)	8,114
Average NOA	81,519	79,807	76,727

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

The analyst decides to examine the balance-sheet-based accruals and the cash-flow-based accruals over the last few years:

Illustration 17: Selected information from BS and SOCF (CHF millions)

Statement of Cash flow accrual info	2014	2013	2012
Profit from continuing operations (NI)	14,904	10,445	10,677
Operating cash flow	(14,700)	(14,992)	(15,668)
Investing cash flow	3,072	1,606	14,491
Cash-flow-based aggregate accruals $[NI_t - (CFO_t + CFI_t)]$	3,276	(2,941)	9,500

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Using the data contained in Illustration 17, the analyst calculate the balance sheet and cash flow statement accruals ratios:

Illustration 18: Accruals ratios (CHF millions)

	2014	2013	2012
<i>Balance sheet approach</i>			
Balance-sheet-based accruals $(NOA_t - NOA_{t-1})$	5,380	(1,955)	8,114
Divided by: Average NOA	81,519	79,807	76,727
Balance-sheet-based accruals ratio	6.6%	-2.4%	10.6%
<i>Cash flow approach</i>			
Cash-flow-based accruals $[NI_t - (CFO_t + CFI_t)]$	3,276	(2,941)	9,500
Divided by: Average NOA	81,519	79,807	76,727
Cash-flow-based accruals ratio	4.0%	-3.7%	12.4%

Under both approaches, the accruals ratio has fluctuated widely over the period. Wide fluctuations may be an indication of earnings manipulation.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Because of the potential for earnings manipulation by increasing accruals, the analyst decide to compare Nestlé's cash flow to its operating income.

Cash flow analysis

In order to compare the two measures, it is necessary to eliminate cash paid for interest and taxes from operating cash flow by adding them back. This adjusted figure is the cash generated from operations (CGO).

$$\text{CGO} = \text{CFO} + \text{Cash paid for Interest expense and Tax expense}$$

or

$$\text{CGO} = \text{EBIT} + \text{Non-cash charges} - \text{Increase in working capital}$$

Implication: If the cash flow generated by operations was **significantly or consistently less than** operating profit, one would have reason to be suspicious about the quality of the operating profit.

Note: Firms that follow IFRS have the choice of reporting cash paid for interest as an operating cash flow or as a financing cash flow
→ If a firm reports the interest as a financing cash flow, no interest adjustment is necessary.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

To convince himself of the representativeness of the Nestlé earnings, the analyst compares the cash generated by operations with the operating profit:

Illustration 19: Operating cash flow to Operating profit (CHF millions)

	2014	2013	2012
Cash generated from operations	17,199	18,206	18,525
Operating profit	10,905	13,068	13,388
Cash generated from operations/Operating profit	1.58	1.39	1.38

The ratio confirms that cash generated from operations has exceeded operating income over the past three years.

→ The results of this analysis reduce our earlier concerns of potential earnings manipulation from our accruals analysis.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

In order to evaluate Nestlé's recent acquisitions, the analyst examines the cash return on total assets:

Illustration 20: Operating cash flow to Total assets (CHF millions)

	2014	2013	2012
Cash generated from operations	17,199	18,206	18,525
Average total assets	126,946	123,160	119,659
Cash return on total assets 	13.5%	14.8%	15.5%

- Although the 13.5% cash return on total assets is a high return on investment, the trend is declining.
- The analyst thinks back to the 2014 goodwill impairment and the poor EBIT return on assets in the Nutrition and Health Science product group, in which acquisitions have been occurring lately.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Given the negative trend in asset returns, the analyst looks at Nestlé's liquidity and funding ability relative to cash flow. He decides to compare cash flow with reinvestment, debt, and debt-servicing capacity:

Illustration 21: Selected cash flow ratios (CHF millions)

<i>Cash flow to reinvestment</i>	2014	2013	2012
Cash generated from operations (CGO)	17,199	18,206	18,525
CAPEX (incl. Expenditure on intangible assets)	4,423	5,330	5,598
CGO/Total CAPEX	3.89	3.42	3.31
<i>Cash flow to total debt</i>			
Cash generated from operations (CGO)	17,199	18,206	18,525
Total debt	21,963	22,124	27,839
CGO/Total debt	78.3%	82.3%	66.5%
<i>Cash flow interest coverage</i>			
Cash generated from operations (CGO)	17,199	18,206	18,525
Cash interest paid	518	505	559
Cash flow interest coverage	33.2	36.1	33.1

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Cash flow to reinvestment: The ratio indicates the amount of cash flow that management reinvests in a business.

Illustration 21: Selected cash flow ratios – Extracted data (CHF millions)

<i>Cash flow to reinvestment</i>	2014	2013	2012
Cash generated from operations (CGO)	17,199	18,206	18,525
CAPEX (incl. Expenditure on intangible assets)	4,423	5,330	5,598
CGO/Total CAPEX	3.89	3.42	3.31

The analyst sees that reinvestment needs have been covered by cash flow by a factor of 3.89 in 2014, 3.42 in 2013, and 3.31 in 2012. Even better, the trend is improving.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Cash flow to total debt: The ratio indicates that additional borrowing could be arranged should an investment opportunity arise.

Illustration 21: Selected cash flow ratios – Extracted data (CHF millions)

<i>Cash flow to total debt</i>	2014	2013	2012
Cash generated from operations (CGO)	17,199	18,206	18,525
Total debt	21,963	22,124	27,839
CGO/Total debt	78.3%	82.3%	66.5%

- Based on the relationship of cash flow to total debt, the company is not highly leveraged, with cash generated from operations at 78.3% of total debt at the end of 2014.
- The analyst notes that Nestlé has the capacity to pay off its debt in approximately two years even while maintaining its current reinvestment policy $[21,963 / (17,199 - 4,423)]$.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

Cash flow interest coverage: The ratio indicates that the company has sufficient financial capacity to add more debt if there is an investment opportunity.

Illustration 21: Selected cash flow ratios – Extracted data (CHF millions)

<i>Cash flow interest coverage</i>	2014	2013	2012
Cash generated from operations (CGO)	17,199	18,206	18,525
Cash interest paid	518	505	559
Cash flow interest coverage	33.2	36.1	33.1

The cash flow interest coverage ratio indicates more than satisfactory financial strength in the current year, with cash flow 33.2 times the interest paid.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.5. Earnings quality and cash flow analysis

In conclusion:

- The ratio of operating cash to operating profit has been consistently favorable, providing confidence in the quality of the earnings.
- The negative trend also shows in the cash returns on total assets. They have decreased each year since 2012.
- Measures comparing cash flow with reinvestment, debt, and debt-servicing capacity indicate strength in financial capacity.

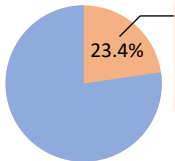
[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.6. Market value decomposition

When a parent company has an ownership interest in an associate (subsidiary or affiliate), it may be beneficial to determine the *standalone* value of the parent; that is, the implied value of the parent without regard to the value of the associate.



Nestlé has a significant equity position (23.4%) in L'Oréal (Paris exchange: OR), a French cosmetics company.

→ L'Oréal is accounted for in the financial statements as an investment in associates.

Equity position of L'Oréal

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3. Phase 3 & 4 – Process and analyze data

3.6. Market value decomposition

The implied value of the parent = The parent's market value – The parent's pro rata share of the associate's market value.

Illustration 22: Nestlé implied value as of 31 December 2014 (Currency in millions, except share prices)

L'Oréal value

31 Dec 2014 share price	€139.30
Shares held by Nestlé (millions)	129.881
L'Oréal holding value (in €)	€18,092
31 Dec CHF/EUR rate	1.202
L'Oréal holding value (in CHF)	CHF21,747

Nestlé market value, with and without L'Oréal:

Nestlé 29 Dec 2014 share price	Pro rata MV:	CHF72.95
Shares outstanding (millions)	L'Oréal 9.4%	3,168.400
Nestlé market capitalization	Nestlé 90.6%	CHF231,135
Value of L'Oréal holding		(21,747)
Implied value of Nestlé operations		CHF209,388

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.6. Market value decomposition

The analyst now wants to remove the earnings of L'Oréal from the earnings of the combined entity to make a price-to-earnings comparison for Nestlé earnings alone:

Illustration 23: Calculation of Nestlé standalone earnings as of 31 Dec 2014

	2014
Nestlé consolidated earnings	14,904
Less: L'Oréal earnings (Nestlé discloses in its annual report)	(934)
Nestlé standalone earnings	13,970
Less: Non-controlling interests	(448)
Nestlé standalone earnings to shareholders	13,522

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.6. Market value decomposition

Illustration 24: Comparison of Decomposed Nestlé earnings and P/E ratios

	Market value	Earnings	P/E
L'Oréal	21,747	934	23.3
% of Total actual earnings available to Nestlé	9.4%	6.5%	
Implied Nestlé-only	209,388	13,522	15.5
% of Total actual earnings available to Nestlé	90.6%	93.5%	
Actual earnings available to Nestlé parent company shareholders	231,135	14,456	16.0

At the time of the analysis (early 2015), Nestlé's common stock traded at a price-to-earnings multiple of 16.0, a discount of 20% to the price-to-earnings multiple of 19.9 for the S&P 500 Index at year-end 2014. The shares of the "Nestlé-only" company are selling at a slightly higher discount (15.5 times earnings).

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.6. Market value decomposition

In conclusion:

Given that the company has consistently demonstrated meaningful cash flows and earnings and possesses low financial leverage, the analyst concludes that Nestlé shares may be discounted by the market because investors may be developing a skeptical attitude toward the company.

[LOS 6.b,c] Identify financial reporting choices and evaluate the quality of financial data and recommend appropriate adjustments

3.

Phase 3 & 4 – Process and analyze data

3.7. Anticipating changes in accounting standards

Throughout the analysis, we must always keep in mind of the difference in accounting standards when making adjustment and for comparison purposes.

Application for case study:

- To keep the DuPont analysis as logically consistent as possible throughout all the periods of study, the analyst revises the 2011 balance sheet (from that reported in the 2012 Consolidated Financial Statements of the Nestlé Group) for the effects of implementing IFRS 11 and IAS 19 revised.
- Implied Nestlé-only P/E is a crude measure because of the potential differences in accounting methods used by the two firms.

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

4.

Phase 5 – Develop conclusions and recommendations

As a result of the analyses performed, the analyst has gathered sufficient evidence, that summarize his findings as follows:

Support for investment in Nestlé shares

- Nestlé has the financial stability to fund growth in its existing operations and carry out its growth-by-acquisition strategy – [Refer to 3.3 and 3.5.](#)
- The operating cash flows have consistently exceeded the operating earnings. Measures comparing cash flow with reinvestment, debt, and debt-servicing capacity indicate strength in financial capacity – [Refer to 3.5.](#)
- Decomposing earnings into Nestlé-only and L'Oréal and considering the respective price-to-earnings ratios, it appears that the implied Nestlé-only portion is undervalued – [Refer to 3.6.](#)

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

4.

Phase 5 – Develop conclusions and recommendations

As a result of the analyses performed, the analyst has gathered sufficient evidence, that summarize his findings as follows:

Causes for concern

- Although Nestlé has significant, world-class brands and global reach, its core business has deteriorated in profitability in the last several years, as shown by the decomposition of the ROE – *Refer to 3.1.*
- The company's write-down of goodwill from earlier acquisitions may signal ineffective allocation of capital – *Refer to 3.1.*
- The negative trend also shows in the cash returns on total assets – *Refer to 3.5.*
- The company's priorities in the allocation of capital in making acquisitions are of some concern. The acquisition activities in the Nutrition and Health Science product group do not appear to build on the company's traditional strengths. – *Refer to 3.4.*

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

4.**Phase 5 – Develop conclusions and recommendations**

The analyst concludes that Nestlé is not clearly a good investment opportunity at this time and recommends waiting to see whether a further discount makes it more attractive, or the operations improve.

5.**Phase 6 – Follow-up**

The analyst is asked to update his findings in the initial research report at each reporting period, emphasizing the quality measures expressed by the accruals tests and the cash flow support of earnings, with particular regard to return on assets.



Publication 505
2018-19

Tax Withholding and Estimated

Capital Gains and Losses

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- 7.b.** Compare “growth relative to GDP growth” and “market growth and market share” approaches to forecasting revenue
- 7.c.** Evaluate whether economies of scale are present in an industry by analyzing operating margins and sales levels
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LEARNING OUTCOMES

- 7.i.** Explain how competitive factors affect prices and costs
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- 7.m.** Explain considerations in the choice of an explicit forecast horizon
- 7.n.** Explain an analyst's choices in developing projections beyond the short-term forecast horizon

OVERVIEW

Financial statement modeling is a key step in the process of valuing companies and the securities they have issued.

Financial statement modeling

Forecast horizon (LOS 7.m,n)

Cash flow statement

1. CFO forecast
2. CFI forecast
3. CFF forecast

**Cash flow
statement**
(LOS 7.f,g)

Income statement

1. Revenue forecast
2. Operating expense forecast
3. EBIT forecast
4. Non-operating forecast

**Income
statement**
(LOS 7.a-e)

Balance sheet
(LOS 7.f,g)

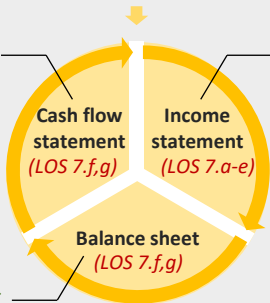
Balance sheet

1. CAPEX forecast
2. Depreciation & Amortization forecast
3. Working capital forecast

Behavioral finance
(LOS 7.h)

Competitor factors
(LOS 7.i,j,k)

**Technological
developments**
(LOS 7.l)



[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy Cointreau Group (Rémy) case study: Company overview

Rémy Cointreau Group (Rémy), a French company that sells primarily spirits, whose reporting year ends 31 March, operates and reports three business segments:

1

Cognac. This division, composed primarily of Rémy Martin brand cognac, represented approximately 73% of FY2021 (year-end 31 March 2021) revenue and 94% of total current operating profit. Current operating profit is a non-IFRS measure reported by Rémy equal to IFRS operating profit excluding items related to discontinued brands or items deemed infrequent or immaterial, such as impairment or litigation provisions.

2

Liqueurs & Spirits. A diverse portfolio of spirits brands, the main brands in this segment are Cointreau, Metaxa, St-Rémy, Mount Gay, Bruichladdich, and The Botanist. The segment represented approximately 25% of FY2021 revenue and 14% of current operating profits.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy Cointreau Group (Rémy) case study: Company overview

Rémy Cointreau Group (Rémy), a French company that sells primarily spirits, whose reporting year ends 31 March, operates and reports three business segments:

3

Partner Brands. This segment includes other companies' brands that are marketed through Rémy's distribution network. They represented approximately 3% of FY2021 revenue and just under 0% of current operating profit, earning a slight operating loss in FY2021 of –EUR0.8 million. This division's importance has declined significantly over time as the company discontinued distribution ("partner brand") contracts.

Data sources for this example include the company's fiscal year ended 31 March 2021 and 2020 annual reports, the company's interim reports, and corresponding investor presentations for additional information on the underlying results of the respective divisions.

After providing a brief overview of the company, we will focus primarily on the mechanics of constructing pro forma income statements, statements of cash flows, and balance sheets.

[LOS 7.a] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models

1.
Income statement forecast
1
Revenue forecast
2
Operating expense forecast
COGS forecast
SG&A forecast
3
EBIT forecast
Segments forecast check
4
Non-operating expense forecast
Interest expense forecast
**Income tax expense
forecast**
**Share outstanding
forecast**

[LOS 7.a] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models

1.

Income statement forecast

1.1. Revenue forecast

Analysts must decide whether to use a top-down, bottom-up, or hybrid approach to projecting future revenue:

1. Top-down approach

Begin at the level of the overall economy.

Example: sector, industry, and market for a specific product.

2. Bottom-up approach

Begin at the level of the individual company or a unit within company.

Example: individual product lines, locations, or business segments.

3. Hybrid approach

Combine elements of both top-down and bottom-up analysis.

[LOS 7.b] Compare “growth relative to GDP growth” and “market growth and market share” approaches to forecasting revenue

1. Income statement forecast

1.1. Revenue forecast – Top-down approach

Top-down analysis starts with expectations about a macro variable

Growth relative to GDP growth

Forecast the relationship between nominal GDP and company sales, or use the real GDP growth rate to forecast quantity and an inflation forecast to estimate prices.

Example:

If an analyst forecasts that nominal GDP will grow at 5% and believes a company's revenue will grow at a 20% faster rate, he will project the company's sales to increase by:

$$5\% \times (1 + 0.20) = 6\%$$

Market growth and market share

Begin with an estimate of industry sales (market growth), then estimates company revenue as a percentage of industry sales based on the company's expected market share.

Example:

A company that currently has \$12 million in sales, a 12% share of industry sales of \$100 million.

If an analyst expects the company to increase its market share next year to 13%, and forecasts industry sales to grow to \$106 million (6% growth rate), the company is forecasted to have in sales:

$$13\% \times \$106 \text{ million} = \$13.78 \text{ million}$$

[LOS 7.a] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models

1.

Income statement forecast

1.1. Revenue forecast – Bottom-up approaches

Bottom-up analysis starts with an individual company or its reportable segments

Time series

Forecast based on historical growth rates or time-series analysis.

Returns-based measure

Forecast based on balance sheet accounts. For example, interest revenue for a bank can be calculated as loans multiplied by the average interest rate.

Capacity-based measure

Forecast, for example, in retailing, based on same-store sales growth (for stores that have been open for at least 12 months) and sales related to new stores.

[LOS 7.a] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models

1.

Income statement forecast

1.1. Revenue forecast – Hybrid approaches

Hybrid approaches combine elements of both top-down and bottom-up analysis, and in practice, they are the most common approaches.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

Segment financial information is summarized in Illustration 1:

Illustration 1: Analysis of Rémy's turnover and operating profit

Revenue (€ millions)	2019	2020	2021
Cognac	774	736	735
Liqueurs & Spirits	264	262	248
Partner Brands	87	28	27
Total revenues	1,126	1,025	1,010
<i>Current operating profit (€ millions)</i>			
Cognac	236	200	221
Liqueurs & Spirits	39	38	33
Partner Brands	5	−2	−1
Holding/Corporate-level costs	−15	−20	−17
Total current operating profit	264	215	236

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

Segment financial information is summarized in Illustration 1:

Illustration 1: Analysis of Rémy's turnover and operating profit

Revenue (€ millions)	2019	2020	2021
<i>Current operating profit margins</i>			
Cognac	30.4%	27.1%	30.1%
Liqueurs & Spirits	14.7%	14.3%	13.3%
Partner Brands	5.6%	−6.2%	−3.0%
Holding/Corporate-level costs (% of total revenue)	−1.3%	−2.0%	−1.7%
Total current operating profit	23.5%	21.0%	23.4%

Source: Based on information in consolidated financial statements of Rémy Cointreau Group for year ended 31 March 2021 and 2020.

As shown, the company's largest business segment is also its most profitable segment: The Cognac segment earned a current operating profit margin of approximately 30% (= EUR221 million/EUR735 million) in fiscal year 2021.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

With this case study, we use hybrid approach for revenue forecast:

Overall cognac and spirits market development

COVID-19

Top-down approach

Revenue

Bottom-up approach

Quantity (volume)

Price

Revenue of Cognac segment

Revenue of Liqueurs & Spirits segment

Revenue of Partner Brands segment

In this section, let's focus on forecast Revenue of Cognac segment.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

In terms of volume growth forecast

- Volume growth is expected to remain robust but be slower than the 9.1% achieved in 2021 as the global recovery from the COVID-19 pandemic and associated recession fades.
 - The growing number of affluent Asian consumers will likely keep demand high, while developed market consumption is likely to be rather flat.
- **The assumption is for 7% volume growth in 2022, declining to 6% in 2023 and 2024.**

In terms of price growth forecast

- Favorable structure of the industry.
 - The company's efforts to increase the share of revenues accounted for by what it calls "exceptional spirits".
- **A 4% price/mix contribution to revenue growth is assumed in 2022, with the trend maintained into 2023 and 2024.**

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

Changes in revenue attributable to volume or price/mix are organic growth and are shown separately from the impact of acquisitions and divestitures (scope change) and foreign exchange (forex impact in the model).

The combined projections for volume growth and price/mix impact results in overall organic revenue growth, calculated as:

$$\text{Organic growth \%} = [(1 + \text{Volume growth \%}) \times (1 + \text{Price/mix \%})] - 1.$$

Illustration 2: Organic growth calculations

	2018	2019	2020	2021	2022E	2023E	2024E
Volume growth %	6.0%	5.9%	−10.1%	9.1%	7.0%	6.0%	6.0%
Price/mix %	7.2%	6.0%	2.6%	−5.4%	4.0%	4.0%	4.0%
Organic growth %	13.6%	12.3%	−7.8%	3.2%	11.3%	10.2%	10.2%

$$\text{Organic growth \% in 2022} = [(1 + 0.07) \times (1 + 0.04)] - 1 = 11.3\%$$

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

In terms of other factors forecast

Rémy's revenues are also affected by movements in exchange rates. However, the model forecasts no foreign currency impact on revenue in the 2022–24 forecast period.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

Illustration 3 summarizes historical and projected information for the Cognac segment's revenue:

Illustration 3: Historical and projected information for Cognac segment revenue

	2018	2019	2020	2021	2022E	2023E	2024E
Cognac segment revenues (€ millions)	760	774	736	735	818	902	994
YoY%	7.4%	1.9%	−5.0%	−0.1%	11.3%	10.2%	10.2%
Volume growth %	6.0%	5.9%	−10.1%	9.1%	7.0%	6.0%	6.0%
Price/mix %	7.2%	6.0%	2.6%	−5.4%	4.0%	4.0%	4.0%
Organic growth %	13.6%	12.3%	−7.8%	3.2%	11.3%	10.2%	10.2%
Forex impact and scope change %	−5.8%	−4.0%	2.5%	−3.8%	0.0%	0.0%	0.0%
Effect of IFRS15 adoption	0.0%	−6.0%	0.0%	0.0%	0.0%	0.0%	0.0%
YoY%	7.8%	2.3%	−5.3%	−0.6%	11.3%	10.2%	10.2%

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Revenue forecast

A similar analysis can be performed to project revenue for the other segments. Then, the amounts can be summed to derive projected consolidated revenue:

Illustration 4: Historical and projected revenue by segment

	2018	2019	2020	2021	2022E	2023E	2024E
Cognac	760	774	736	735	818	902	994
Liqueurs & Spirits	267	264	262	248	251	253	256
Partner Brands	100	87	28	27	26	26	26
Total revenues	1,127	1,126	1,025	1,010	1,095	1,181	1,275

[LOS 7.c] Evaluate whether economies of scale are present in an industry by analyzing operating margins and sales levels

1. Income statement forecast

1.2. Operating costs

Variable costs are directly linked to revenue, and they might be best modeled as a percentage of revenue or as projected unit volume multiplied by unit variable costs.

Fixed costs are not directly related to revenue; rather, they are related to investment in property, plant, and equipment (PP&E) and to total capacity.

When estimating costs, analysts should pay particular attention to fixed costs.

Analysts should determine whether, at its current level of output, the subject company has **economies of scale**.

A company with economies of scale will have **higher operating margins** (because of lower average cost) as production volume increases, and **sales volume and margins will tend to be positively correlated**.

[LOS 7.d] Demonstrate methods to forecast cost of goods sold and operating expenses

1.

Income statement forecast

1.2. Operating costs – COGS and Gross margins

- Because COGS is closely related to revenue, future COGS is usually estimated as a *percentage of future revenue*:

$$\text{Forecast COGS} = (\text{historical COGS} / \text{revenue}) \times (\text{estimate of future revenue})$$

or

$$\text{Forecast COGS} = (1 - \text{gross margin}) \times (\text{estimate of future revenue})$$

- Future COGS can be estimated as a *percentage of gross margin*.
 - If a company is losing market share to new substitute products with lower prices, gross margins are likely to decline.
 - If the company is gaining market share because it has introduced differentiated products—especially if it has done so in combination with achieving cost advantages—gross margins are likely to rise.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Operating costs – COGS and Gross margins forecast

Back to the Rémy case study, the company gross margin has remained roughly flat from FY2018 (67.5%) to FY2021 (67.3%) as total sales have decreased modestly.

Illustration 5: COGS and Gross margins forecast

	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
COGS	366	415	348	330	347	362	379
Gross profit	761	711	677	680	748	819	897
Gross margin	67.5%	63.1%	66.1%	67.3%	68.3%	69.3%	70.3%
Change in gross margin	0.8%	-4.4%	2.9%	1.3%	1.0%	1.0%	1.0%

Going forward, we project gross margin to increase by 100 bps in each of the next three years based on increasing total revenues, particularly from price/mix, which is strongly accretive to gross margin.

[LOS 7.d] Demonstrate methods to forecast cost of goods sold and operating expenses

1.

Income statement forecast

1.2. Operating costs – SG&A Expenses

- Compared to COGS, selling, general, and administrative (SG&A) operating expenses are **less sensitive** to changes in sales volume.
 - **Selling and distribution expenses** often have a large variable component and can be modeled as a percentage of sales.
 - **General and administrative costs** are more fixed and might be better modeled using a fixed growth rate based on expected wage inflation.
- Segment disclosures are unlikely to provide specific line items such as COGS and SG&A by segment. Therefore, an analyst creating segment forecasts can only rely on summary information such as operating margin by segment.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Operating costs – SG&A Expenses forecast

Distribution costs forecast

- In particular, the setup of Rémy's distribution network in Asia increased the cost base.
- Rémy is very committed to its brand building and is also diversifying geographically.

→ We estimate modest increases in distribution costs as a percentage of revenue, of 20 bps per year.

Illustration 6: Distribution costs forecast

	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
Distribution costs	433	346	355	342	373	404	439
<i>Distribution costs as % of sales</i>	38.4%	30.7%	34.6%	33.8%	34.0%	34.2%	34.4%

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Operating costs – SG&A Expenses forecast

Administrative costs forecast

- Administrative costs as a percentage of revenue have increased from 8.1% to 10.1% as revenues have fallen, owing to the COVID-19 pandemic.
- However, the growth in absolute euro amounts has been modest, with costs of approximately EUR100 million in FY2019–FY2021.

→ We expect 1% growth in administrative costs per year through FY2024E.

Illustration 7: Administrative costs forecast

	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
Administrative costs	92	101	107	103	104	105	106
<i>Administrative costs as % of sales</i>	8.1%	8.9%	10.4%	10.1%	9.5%	8.9%	8.3%

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Operating costs – SG&A Expenses forecast

Other operating expense (income) forecast

- Other operating expense (income), composed primarily of provisions for impairments of intangible assets, restructurings, and divestiture gains, has fluctuated from –EUR2 million to EUR20 million from FY2018 to FY2021.
- Because we do not anticipate any transactions that would result in other operating expenses or income, we forecast zero for this line in the model.

Illustration 8: Other operating expense (income) forecast

	2018	2019	2020	2021	2022E	2023E	2024E
Other operating expense (income)	13	–2	20	0	0	0	0

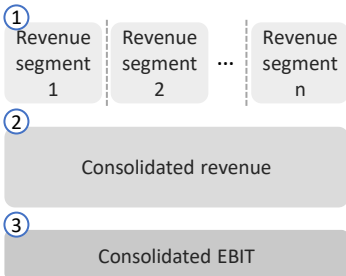
[LOS 7.d] Demonstrate methods to forecast cost of goods sold and operating expenses

1. Income statement forecast

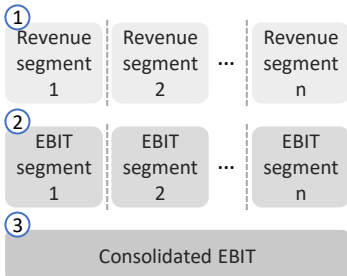
1.3. Operating profit (EBIT)

Operating profit and margin can be estimated using two approaches:

Approach 1: Consolidated approach



Approach 2: Segment approach



While a segment approach can be used instead of a consolidated approach to forecasting revenue and operating profit, it is also commonly used as a “check” on the consolidated forecasts.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Operating profit forecast

Approach 1: Consolidated approach

Illustration 9 provides a consolidated income statement for Rémy through the EBIT and EBIT margin line, in which components we have estimated above:

Illustration 9: Consolidated historical and projected EBIT							
	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
COGS	366	415	348	330	347	362	379
Gross profit	761	711	677	680	748	819	897
Distribution costs	433	346	355	342	373	404	439
Administrative costs	92	101	107	103	104	105	106
Other operating expense (income)	13	−2	20	0	0	0	0
EBIT	223	266	196	236	272	310	352
EBIT margin	19.8%	23.6%	19.1%	23.3%	24.8%	26.2%	27.6%

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Operating profit forecast

Approach 2: Segment approach

According to step 1.1, we have performed revenue forecast for each segment. Illustration 4 summarizes historical and projected information for the each segment revenue:

Illustration 4: Historical and projected revenue by segment

	2018	2019	2020	2021	2022E	2023E	2024E
Cognac	760	774	736	735	818	902	994
Liqueurs & Spirits	267	264	262	248	251	253	256
Partner Brands	100	87	28	27	26	26	26
Total revenues	1,127	1,126	1,025	1,010	1,095	1,181	1,275

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Operating profit forecast

Approach 2: Segment approach

Rémy discloses current operating profit for each of its segments as well as an operating cost at the corporate or holding company level.

The forecast of EBIT by segment is based on operating margin expectation:

Illustration 10: Historical and projected operating profit margins (%) by segment

	2018	2019	2020	2021	2022E	2023E	2024E
Cognac	26.9%	30.4%	27.1%	30.1%	31.2%	32.3%	33.4%
Liqueurs & Spirits	16.0%	14.7%	14.3%	13.3%			
Partner Brands	5.3%	5.6%	−6.2%	−3.0%			
Holding/Corporate-level costs	−1.4%	−1.3%	−2.0%	−1.7%			
Total current operating profit margins	21.0%	23.5%	21.0%	23.4%			

For the Cognac segment, the expectation is based on:

- The forecast of higher revenue growth and higher distribution costs.
- Comparison with financial results of another cognac brand (Hennessy).

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Operating profit forecast

Approach 2: Segment approach

For the other segments, there is not much upside:

- In the Liqueurs & Spirits division, we assume operating margin to increase modestly to 13.6%.
- In Partner Brands segment, the operating margin is projected to remain flat from the year ended FY2021 level as the company discontinued distribution (“partner brand”) contracts.

Illustration 10: Historical and projected operating profit margins (%) by segment

	2018	2019	2020	2021	2022E	2023E	2024E
Cognac	26.9%	30.4%	27.1%	30.1%	31.2%	32.3%	33.4%
Liqueurs & Spirits	16.0%	14.7%	14.3%	13.3%	13.4%	13.5%	13.6%
Partner Brands	5.3%	5.6%	−6.2%	−3.0%	−3.0%	−3.0%	−3.0%
Holding/Corporate-level costs	−1.4%	−1.3%	−2.0%	−1.7%			
Total current operating profit margins	21.0%	23.5%	21.0%	23.4%			

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Operating profit forecast

Approach 2: Segment approach

In total, Rémy Cointreau Group's consolidated operating margin is forecast to improve from 23.4% in FY2021 to 27.6% in FY2024, largely because of growth and margin improvement in the Cognac segment, the most profitable division, and leverage from that sales growth on corporate-level costs.

Illustration 10: Historical and projected operating profit margins (%) by segment

	2018	2019	2020	2021	2022E	2023E	2024E
Cognac	26.9%	30.4%	27.1%	30.1%	31.2%	32.3%	33.4%
Liqueurs & Spirits	16.0%	14.7%	14.3%	13.3%	13.4%	13.5%	13.6%
Partner Brands	5.3%	5.6%	-6.2%	-3.0%	-3.0%	-3.0%	-3.0%
Holding/Corporate-level costs	-1.4%	-1.3%	-2.0%	-1.7%	-1.5%	-1.3%	-1.1%
Total current operating profit margins	21.0%	23.5%	21.0%	23.4%	24.8%	26.2%	27.6%

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Operating profit forecast

Approach 2: Segment approach

Using the information from Illustration 10 as well as the information about the projected revenue by segment from Illustration 4, we have EBIT by segment as follows:

Illustration 11: Historical and projected operating profit by segment for Rémy							
	2018	2019	2020	2021	2022E	2023E	2024E
Cognac	204	236	200	221	255	291	332
Liqueurs & Spirits	43	39	38	33	34	34	35
Partner Brands	5	5	−2	−1	−1	−1	−1
Holding/Corporate-level costs	−16	−15	−20	−17	−16	−15	−14
Total current operating profit	237	264	215	236	271	309	352

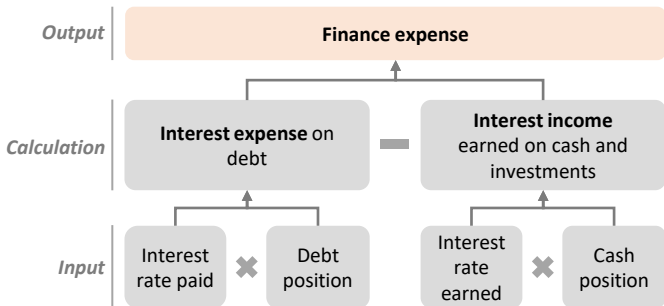
[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

1. Income statement forecast

1.4. Non-operating costs

Three types of non-operating line items are included in the model: **finance expenses (i.e., interest expenses), income taxes, and shares outstanding.**

a. Financing expenses



Therefore, forecasting net finance cost requires estimating the debt and cash positions as well as interest rates paid and earned.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Non-operating costs – Financing expenses forecast

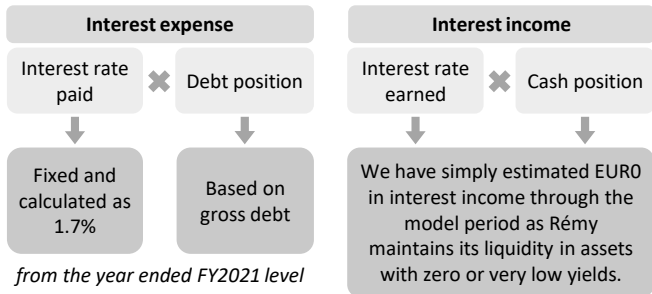


Illustration 12: Debt position and financial costs and income for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Gross debt (incl. long-term and short-term debt)	470	522	720	515	515	515	515
Interest expense	14.5	13.7	12.9	12.1	8.7	8.7	8.7
Interest rate		2.9%	2.5%	1.7%	1.7%	1.7%	1.7%
Interest income	0.0	0.0	0.1	0.2	0.0	0.0	0.0
Net finance cost	14.5	13.7	12.8	11.9	8.7	8.7	8.7

[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

1. Income statement forecast

1.4. Non-operating costs

b. Corporate income tax

Income taxes are primarily determined by the geographic composition of profits and the tax rates in each geography but can also be influenced by the nature of a business.

Primary tax rates

Statutory rate

The percentage tax charged in the country where the firm is domiciled

Effective tax rate

Income tax expense as a percentage of pretax income on the income statement

Cash tax rate

Cash taxes paid as a percentage of pretax income

Differences arise from:

- *Tax credits*
- *Withholding tax on dividends*
- *Adjustments to previous years*
- *Non-deductible expenses*

The difference



Changes in Deferred tax items

$$\text{Income tax expense} = \text{Cash tax due} + \Delta \text{DTL} = \Delta \text{DTA}$$

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Non-operating costs – Corporate income tax forecast

- The French statutory tax rate at the time of analysis is 32%.
 - Rémy Cointreau Group's effective tax rate has, over the longer run, been close to the statutory rate.
- An estimated 32% effective tax rate is used in the forecast period.

[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

1.
Income statement forecast

1.4. Non-operating costs

c. Shares outstanding

Typically, the two major factors that affect shares outstanding over time are:

- ✓ Share issuance related to equity-based compensation of employees (increases shares outstanding).
- ✓ Share repurchases (decreases shares outstanding).

Less common but sometimes significant transactions that also affect shares outstanding include:

- ✓ Acquisitions financed with stock.
- ✓ Secondary issuance.
- ✓ Conversions of preferred stock or other instruments to common stock.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Non-operating costs – Shares outstanding forecast

Illustration 13 shows beginning and ending basic shares outstanding for the past six fiscal years as well as the annual net amount of share repurchases and issuance:

Illustration 13: Shares outstanding for Rémy (€ millions)

	2016	2017	2018	2019	2020	2021
Beginning basic shares outstanding	48.6	48.6	49.6	50.0	49.8	49.8
Share repurchases	−0.0	0.0	−0.3	−1.0	−0.0	0.0
Share issuance	0.0	1.0	0.7	0.8	0.1	0.4
Ending basic shares outstanding	48.6	49.6	50.0	49.8	49.8	50.3
Weighted average basic shares	48.6	49.1	49.8	50.1	49.8	50.1
Dilutive securities	0.1	2.7	2.6	2.6	2.6	2.6
Weighted average diluted shares	48.7	51.8	52.4	52.7	52.4	53.1

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Non-operating costs – Shares outstanding forecast

- Shares outstanding for Rémy have not changed materially in six years because the company does not pay significant share-based compensation nor has it repurchased shares.
- Management has not disclosed an intention to repurchase shares in the near term.

→ **The model assumes that weighted average basic and diluted shares outstanding on the income statement remain flat at the FY2021 level.**

Illustration 14: Projected shares outstanding for Rémy (€ millions)

	2018	2019	2020	2021	2022E	2023E	2024E
Weighted average basic shares	49.8	50.1	49.8	50.1	50.1	50.1	50.1
Weighted average diluted shares	52.4	52.7	52.4	53.1	53.1	53.1	53.1

[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

1.
Income statement forecast

1.5. Other items

Dividends

Can be estimated based on historical data, using a constant growth rate or a constant payout ratio.

Unusual charges

- Almost impossible to predict → Typically excluded from forecasts.
- If certain recurring costs are classified as unusual → Should consider some normalized level of charges in the model.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Income statement

Lines labeled “other” are aggregated and zeroed out going forward because they are immaterial, difficult to forecast, or both.

Illustration 15: Consolidated historical and projected Income statement for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
COGS	366	415	348	330	347	362	379
Gross profit	761	711	677	680	748	819	897
Distribution costs	433	346	355	342	373	404	439
Administrative costs	92	101	107	103	104	105	106
Other operating expense (income)	13	−2	20	0	0	0	0
EBIT	223	266	196	236	272	310	352
Net finance costs	15	14	13	12	9	9	9
Other financial expenses	8	19	15	3	0	0	0
Total financial expenses	22	33	28	15	9	9	9
Profit before tax	201	233	167	221	263	301	344

Continue next slide →

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Income statement

Lines labeled “other” are aggregated and zeroed out going forward because they are immaterial, difficult to forecast, or both.

Illustration 15: Consolidated historical and projected Income statement for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Profit before tax	201	233	167	221	263	301	344
Income tax	54	68	61	78	84	96	110
<i>Effective tax rate</i>	26.6%	29.0%	36.4%	35.1%	32.0%	32.0%	32.0%
Income from associates	1	−7	0	1	0	0	0
Profit from continuing operations	148	159	107	144	179	205	234
Profit from discontinued operations	0	0	6	0	0	0	0
Net profit for the year	148	159	113	144	179	205	234
YoY%		8%	−29%	27%	24%	14%	14%

Continue next slide →

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Income statement

Lines labeled “other” are aggregated and zeroed out going forward because they are immaterial, difficult to forecast, or both.

Illustration 15: Consolidated historical and projected Income statement for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
EPS basic continuing operations	2.97	3.18	2.14	2.88	3.58	4.09	4.67
EPS diluted continuing operations	2.82	3.02	2.04	2.74	3.40	3.89	4.44
EPS basic total	2.97	3.18	2.27	2.88	3.58	4.09	4.67
EPS diluted total	2.82	3.02	2.16	2.74	3.40	3.89	4.44
Average number of shares, basic, mlns	49.8	50.1	49.8	50.1	50.1	50.1	50.1
Average number of shares, diluted, mlns	52.4	52.7	52.4	53.1	53.1	53.1	53.1

[LOS 7.f] Describe approaches to balance sheet modeling

2. Balance sheet forecast

2.1. Working capital

Three balance sheet items comprise working capital forecasts—accounts receivable, inventories, and accounts payable.

$$\text{Inventories turnover} = \frac{\text{COGS}}{\text{Avg. inventories}}$$

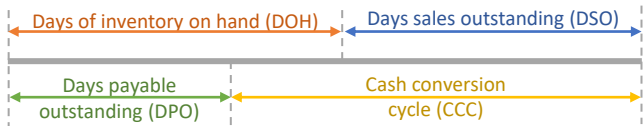
$$\text{DOH} = \frac{365}{\text{Inventories turnover}}$$

$$\text{Receivables turnover} = \frac{\text{Revenue}}{\text{Avg. receivables}}$$

$$\text{DSO} = \frac{365}{\text{Receivable turnover}}$$

Purchases

Sales



Pay cash

Receipt cash

$$\text{Payables turnover} = \frac{\text{Purchase}}{\text{Average payables}}$$

$$\text{DPO} = \frac{365}{\text{Payables turnover}}$$

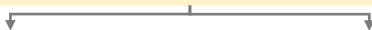
$$\text{Cash conversion cycle} = \text{DOH} + \text{DSO} - \text{DPO}$$

[LOS 7.f] Describe approaches to balance sheet modeling

2. Balance sheet forecast

2.2. Capital investments

Forecasting capital investments in tangible and intangible assets requires an analyst to use the cash flow statement to determine acquisitions and dispositions, and the income statement to determine depreciation and amortization expense.



Capital expenditures (CAPEX)

- Maintenance CAPEX:
 - Necessary to **sustain** the current business.
 - Forecasts are based on historical depreciation and amortization expenses.
- Growth CAPEX:
 - Necessary to **expand** the business.
 - Forecasts are based on management's expansion plans and revenue growth.

Depreciation and amortization

Forecasts are based on:

- Net PP&E and intangibles on the balance sheet.
- Useful lives as assumed by management's accounting policies.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

3.

Statement of cash flows forecast

Net income (Income statement)

Share based compensation

Working capital

Depreciation & amortization

CAPEX

Cash flows from operating activities

Cash flows from investing activities

Share repurchase and issuance

Dividends

Debt issuance and repayment

Cash flows from financing activities



[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Working capital forecast

Illustration 16: Working capital development for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
COGS	366	415	348	330	347	362	379
Average inventories	1,170	1,246	1,364	1,493			
Average account receivables	210	271	199	158			
Average account payables	517	544	534	586			
Working capital, net	863	973	1,029	1,065			
% of sales	77%	86%	100%	105%			
Change in working capital		-110	-56	-36			
DOH	1,166	1,095	1,431	1,650			
DSO	68	88	71	57			
DPO	515	478	561	648			

The largest working capital component is inventory because much of Rémy's cognac requires years of aging.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Working capital forecast

The model assumes that:

- The inventory days are expected to decline through FY2024 as the inventory increase that occurred during the COVID-19 pandemic is worked through.
- The days sales outstanding are expected to remain at FY2021 levels.
- Days payable outstanding are expected to decline back to an average level, again reflecting a normalization after the COVID-19 pandemic.

Illustration 16: Working capital development for Rémy – Extracted data

		2018	2019	2020	2021	2022E	2023E	2024E
DOH		1,166	1,095	1,431	1,650	1,500	1,350	1,200
DSO		68	88	71	57	57	57	57
DPO		515	478	561	648	628	608	588

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Working capital forecast

As a result of the decrease in inventory days, the model projects a net positive contribution from working capital to the reconciliation of net income to cash flows from operations on the statement of cash flows, which is in stark contrast to prior years' negative contribution.

Illustration 16: Working capital development for Rémy – Extracted data

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
COGS	366	415	348	330	347	362	379
Average inventories	1,170	1,246	1,364	1,493	1,426	1,340	1,245
Average account receivables	210	271	199	158	171	185	200
Average account payables	517	544	534	586	597	604	610
Working capital, net	863	973	1,029	1,065	1,000	922	835
% of sales	77%	86%	100%	105%	91%	78%	65%
Change in working capital		-110	-56	-36	64	79	87

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: CAPEX and Depreciation forecast

Given the healthy volume growth prospects, we expect CAPEX to remain at a modestly above historical average level of 5.0% of sales through FY2024.

Illustration 17: CAPEX and Depreciation/Amortization breakdowns

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
CAPEX, PP&E and Intangibles	34	45	65	54	55	59	64
<i>% of sales</i>	3.0%	4.0%	6.3%	5.3%	5.0%	5.0%	5.0%
Deprecation and Amortization (D&A)	22	30	33	34	36	36	37
<i>% of prior year fixed assets</i>		4.0%	4.3%	4.2%	4.2%	4.2%	4.2%
CAPEX/D&A ratio	1.6	1.5	1.9	1.6	1.5	1.6	1.7

With Rémy's growing fixed asset base, it is logical that depreciation will increase. The model assumes that depreciation and amortization (D&A) is equal to 4.2% of prior year fixed assets, the average of the past three years.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Statement of cash flows

Note: Analysts normally have a choice of whether to focus on the balance sheet or cash flow statement; the third financial statement will naturally result from the construction of the other two.

Here, we focus on the Statement of cash flows.

- Net income, D&A, change in working capital, capex, and debt estimates are already in place (*as illustrated in previous steps*).
- The model assumes flat share-based compensation, no share repurchases or issuance, and dividends paid equal to the FY2021 level through FY2024.
- Lines labeled “other” are aggregated and zeroed out going forward because they are immaterial, difficult to forecast, or both.

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Statement of cash flows

Illustration 18: Projected Statement of cash flow for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Net profit for the year	148	159	113	144	179	205	234
Deprecation and Amortization (D&A)	22	30	33	34	36	36	37
Share-based compensation	3	3	4	2	2	2	2
Investment in working capital	-7	-162	-72	-13	64	79	87
Other non-cash amounts	20	22	3	10	0	0	0
Cash flow from operations	185	53	81	177	281	322	360
CAPEX	-34	-45	-65	-54	-55	-59	-64
Other investing activities	2	92	12	62	0	0	0
Cash flow from investments	-32	47	-53	8	-55	-59	-64
Debt issuance (repayments)	0	11	196	-246	0	0	0
Share issuance (repurchases)	-27	-104	-2	2	0	0	0
Dividend paid	-25	-9	-132	-10	-10	-10	-10
Cash flow from financing	-52	-102	62	-253	-10	-10	-10

Continue next slide →

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Statement of cash flows

Illustration 18: Projected Statement of cash flow for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Cash flow from operations	185	53	81	177	281	322	360
Cash flow from investments	−32	47	−53	8	−55	−59	−64
Cash flow from financing	−52	−102	62	−253	−10	−10	−10
FX translation effects	8	−6	1	−1	0	0	0
Net change in cash	109	−8	91	−68	217	254	287
<i>Cash and equivalents, beginning</i>	<i>78</i>	<i>187</i>	<i>179</i>	<i>269</i>	<i>201</i>	<i>418</i>	<i>671</i>
<i>Cash and equivalents, end</i>	<i>187</i>	<i>179</i>	<i>269</i>	<i>201</i>	<i>418</i>	<i>671</i>	<i>958</i>

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Balance sheet

The forecasted balance sheet is based on the combination of the projected income statement (Illustration 15), the projected statement of cash flows (Illustration 18), and the historical starting balance sheet.

Illustration 19: Projected Balance sheet for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Cash and equivalents	187	179	269	201	418	671	958
Account payables	517	544	534	586	597	604	610
Inventories	1,170	1,246	1,364	1,493	1,426	1,340	1,245
Other current assets	16	5	16	10	10	10	10
Total current assets	1,583	1,700	1,848	1,861	2,025	2,206	2,412

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model



Rémy case study: Pro forma Balance sheet

Net PP&E for each year = net PP&E prior period + CAPEX – Depreciation expense

$$\text{Net PP\&E in 2022} = 845 + 55 - 36 = 864$$

Illustration 19: Projected Balance sheet for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
PP&E, intangibles, goodwill, net	752	785	808	845	864	887	913
Investment in associates	20	1	1	2	2	2	2
Other non-current assets	186	139	131	73	73	73	73
Total assets	2,542	2,625	2,789	2,781	2,964	3,168	3,400

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Rémy case study: Pro forma Balance sheet

Illustration 19: Projected Balance sheet for Rémy

(in € millions)	2018	2019	2020	2021	2022E	2023E	2024E
Short-term/current debt	73	98	268	92	92	92	92
Accounts payable	5517	544	534	586	597	604	610
Other current liabilities and accrued expenses	26	31	39	42	42	42	42
Total current liabilities	616	673	842	720	731	737	744
Long-term/non-current debt	397	424	452	424	424	424	424
Other non-current liabilities	121	102	92	88	88	88	88
Total common equity	1,407	1,425	1,403	1,548	1,720	1,918	2,144
NCI	1	1	1	1	1	1	1
Total equity and liabilities	2,542	2,625	2,789	2,781	2,964	3,168	3,400

[LOS 7.h] Explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases

	Definition	Consequence	Mitigation
Overconfidence bias	Having <i>too much faith</i> in one's own work	May underestimate forecasting errors → narrower confidence interval of forecasts	Share forecasts, critiques, evaluate past forecasts, scenario analysis
Illusion of control bias	A false sense of security in one's forecasts	Overfitted models → perform poorly out of sample, conceal outdated assumption	Focus on variables with known explanatory power, seeking outside opinions
Conservatism bias	Making <i>small adjustments</i> to prior forecasts when new info is available	Reluctance to incorporate negative news, lags in incorporating positive news	Periodic evaluation of forecasting errors, flexible models
Representativeness bias	Classifying data based on past info and known classifications	Incorrect understanding that biases all future thinking on the info	Start with base rate and determining differentiating factors
Confirmation bias	Pay attention to data that affirms their earlier beliefs	Pursue only positive news or certain criteria, narrow scope	Read and seek researches with contradicting views and opinions

**[LOS 7.i,j] Explain how competitive factors affect prices and costs;
Evaluate the competitive position based on Porter's five forces analysis**

How factors of Porter's Five Forces Framework affect prices and costs

1. Power of suppliers

- Company prospects for earnings growth are lower when the bargaining power of suppliers is high.
- If suppliers are few, they may be able to extract a larger portion of any value added.

2. Power of buyers

- Companies have less pricing power when the bargaining power of customers is higher, especially in a circumstance where a small number of customers are responsible for a large proportion of a firm's sales and also when switching costs are low.

3. Threat of substitutes

- Companies have less (more) pricing power when the threat of substitute products is high (low) and switching costs are low (high).

4. Threat of entry

- Companies have more pricing power and better prospects for earnings growth when the threat of new entrants is low.
- Significant barriers to entry into an industry make it possible for existing companies to maintain high returns on invested capital.

[LOS 7.i,j] Explain how competitive factors affect prices and costs; Evaluate the competitive position based on Porter's five forces analysis

How factors of Porter's Five Forces Framework affect prices and costs

5. Intensity of rivalry

- Companies have less (more) pricing power when the intensity of industry rivalry is high (low).
- Pricing power is low when industry concentration is less, when fixed costs and exit barriers are high, when industry growth is slow or negative, and when products are not differentiated to a significant degree.

[LOS 7.k] Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation

Inflation and deflation (i.e., the overall increase and decrease in the prices of goods and services) can significantly affect the accuracy of forecasts for a company's future revenue, profit, and cash flow.

The relation between Inflation/Deflation and:

Industry sales

- Industry structure.

Example: The industry structure of beer market in the US is an oligopoly, which has likely contributed to increases in beer prices roughly in line with the US CPI.

- Price elasticity of demand.

Example: If the demand is price elastic, a company's efforts to pass on inflation through higher prices can have a negative impact on volume (if cheaper substitutes are available).

Company sales

- Price elasticity of demand.
- Pricing strategy and market position.

Example: Sysco Company (the largest food distributor to restaurants and institutions in North America) has sometimes not passed on food price increases in recessionary conditions out of concern of not financially weakening already recession-affected customers.

[LOS 7.k] Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation

The relation between Inflation/Deflation and:

Industry costs

- Purchasing characteristics of an industry.

Example: Long-term price-fixed forward contracts and hedges can delay the impact of price increases.

- Drivers of input prices.

Example: weather conditions can have a dramatic impact on the price of agricultural products and consequently on the cost base of industries.

- Competitive environment.

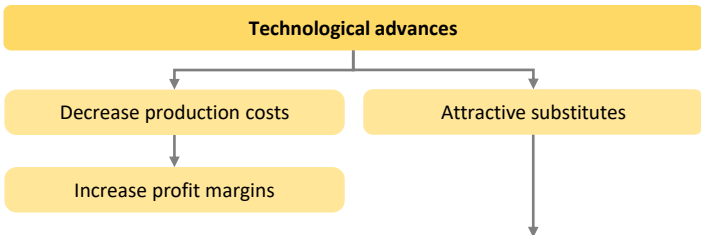
Example: If the participants within the industry have access to alternative inputs or are vertically integrated, the impact of volatility in input costs can be mitigated.

Company costs

Segment the cost structure by category and geography.

This assessment should take into account the company's ability to substitute cheaper alternatives for expensive inputs or to increase efficiency to offset the impact of increases in input prices.

[LOS 7.I] Evaluate the effects of technological developments on demand, selling prices, costs, and margins



One way for an analyst to model the introduction of new substitutes for a company's products is to estimate a **cannibalization factor**.

Cannibalization factor is the percentage of new product sales that will replace existing product sales.

$$\text{Cannibalization rate} = \frac{\text{new product sales that } \textit{replace} \text{ the existing product sales}}{\text{total new product sales}}$$

[LOS 7.I] Evaluate the effects of technological developments on demand, selling prices, costs, and margins



Example: Cannibalization

Alpha, Inc., manufactures LED lightbulbs for sale to businesses as well as to households. Seventy-five percent of Alpha's unit sales are to business customers. Technological advances have enabled bulb manufacturers to produce a new bulb that is more energy efficient, and Alpha is planning to introduce a bulb next year that uses this new technology. Projected sales for the new bulb are shown in the following table.

Market	Units
Businesses	87,000
Households	11,000
Total units	98,000

Calculate the lost unit sales of current product due to cannibalization, assuming a cannibalization factor of 50% for businesses and 20% for households.

[LOS 7.I] Evaluate the effects of technological developments on demand, selling prices, costs, and margins



Example: Cannibalization

Answer:

$$\text{Cannibalization rate} = \frac{\text{new product sales that replace the existing product sales}}{\text{total new product sales}}$$

New product sales that *replace* the existing product sales = The lost unit sales of current product = Cannibalization $\times$ Total new product sales.

- For businesses:

$$\text{The lost unit sale} = 50\% \times 87,000 = 43,500 \text{ units}$$

- For households:

$$\text{The lost unit sale} = 20\% \times 11,000 = 2,200 \text{ units}$$

$$\text{Total lost unit sales} = 45,700 \text{ units}$$

[LOS 7.m] Explain considerations in the choice of an explicit forecast horizon

The choice of the forecast time horizon is determined by:

- The **investment strategy** for which the security is being considered.
- The **cyclicity** of the industry: long enough to include the midpoint of a business cycle.
- **Company-specific** factors (*e.g., acquisitions, mergers, or restructurings*): long enough to allow the benefits of the changes to be measurable.
- The analyst's employer's preferences.



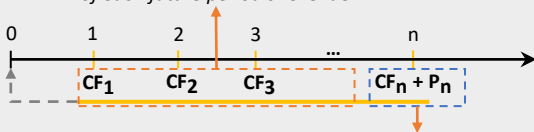
Although the underlying principles remain the same if one extends the time horizon beyond the explicit short-term period, certain considerations and choices are available to the analyst when developing longer-term projections.

Longer-term projections should focus on adjusting certain input factors

[LOS 7.m] Explain considerations in the choice of an explicit forecast horizon

We will consider various alternatives for two aspects of long-term forecasting: revenue forecasts and terminal value.

Pro forma financial results can be estimated based on the projection of each future period's revenue



Terminal value requires inputs outside the explicit period, which are adjusted for future forecasts

Revenue forecasts: methods were covered in LOS7.a,b.

Terminal value: can be estimated using either (1) relative approach or (2) DCF approach.

(1) Relative approach: must ensure that the multiple used is consistent with the estimate of the company's growth rate and required rate of return.

(2) DCF approach:

- ✓ Cash flow or earnings measure
- ✓ Expected future growth rate

[LOS 7.m] Explain considerations in the choice of an explicit forecast horizon

Challenges in determining terminal value

Inflection points – those instances when the future will not be like the past, due to change in a company's or an industry's competitive environment or to changes in the overall economy.

Inflection points occur due to changes in:

Overall economic environment

Example: 2008 financial crisis, COVID-19...

Business cycle stage

Example: If the analyst is examining a cyclical company, using a boom year as the starting point for the perpetuity could result in a grossly overstated valuation.

Government regulations

Example: Pharmaceutical manufacturers are heavily exposed to both regulation and technological advances.

Technology

Example: Semiconductor manufacturers must constantly keep up with new technology but experience relatively light regulation.

Lecturing Slide

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